HACKberry

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Introduction
01HACKberryThe world we aim for

[1] What is HACKberry?

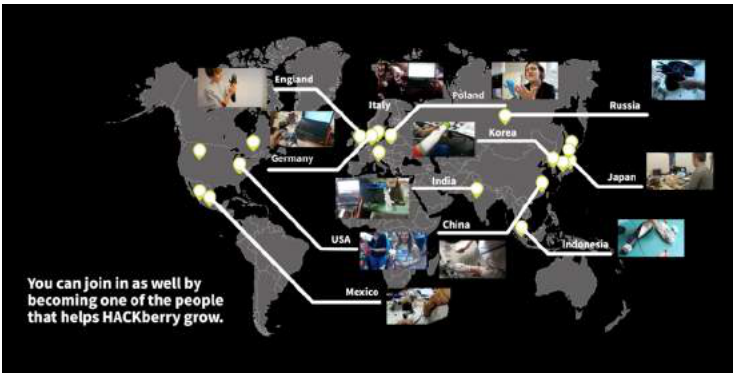
HACKberry is an electric prosthetic hand designed by exiii Inc. and released in 2015. Its housing is 3D printed, and all design data is available online as open source. Since September 2016, it has been taken over by the nonprofit organization Mission ARM Japan, which continues to improve and promote the device.

An electric prosthetic hand is an electrically operated prosthetic hand that is operated via electrical signals from muscles. An electromyographic sensor is placed on the surface of the skin, and the user's intentions are read from the measured signal patterns, and the motor built into the prosthetic hand moves accordingly. The user puts a little pressure on the arm.

By simply applying force, the prosthetic arm can be moved intuitively, just like a normal person.

The development of electric prosthetic hands has a long history, with some already being commercially available in the 1960s. However, traditional manufacturing and distribution systems make it difficult to lower the price, and they are still very expensive, costing over 1.5 million yen. As a result, only a select few people in developed countries can afford electric prosthetic hands.

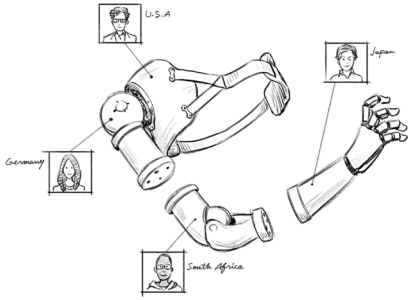
HACKberry aims to make powered prosthetic hands an "easy option" by combining 3D printing with open source. 3D printing does not require initial costs such as mold costs, and by accessing open source data via the web, you can start making them right away. HACKberry



HACKberry's global expansion

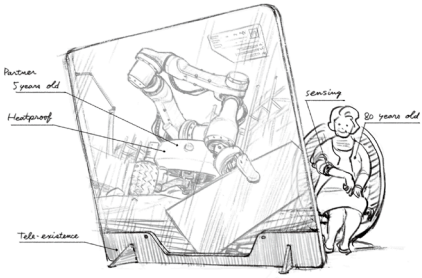
[2] HACKbery's Vision I'm starting to make HACKberries.

In this way, we can reduce manufacturing and distribution costs,
If you pay about 50,000 yen for materials, you can make it anywhere in the world.
We have also created a system that allows people to make their own electric prosthetic arms.
It has not yet reached the practical quality of commercially available products.
However, many people around the world
Customizing prosthetic hands HACK berry's vision is "easy choice"
Until now, prosthetic hands have been used as welfare devices to replace the natural hand.
It is also important to broaden the scope of design.
Our hobbies and lifestyles vary widely.
While designs that resemble the original have been common, if we think of prosthetic arms as a part of fashion, like shoes, clothes, glasses, etc., the design



possibilities for prosthetic arms expand.
Developed by an open source community. This is due to two factors: price and design.
As mentioned above, the current situation
The electric prosthetic arm is very expensive, and if the price does not come down,
It will not be widely adopted. However, it will be possible to

One example of this is HACKberry, which has a stylish design that is deliberately not skin-colored. Also, in order to make prosthetic arms a "casual option," we must focus not only on the "user" but also on the "maker." Prosthetic arm users and makers are two sides of the same coin. In order to increase the opportunities to use prosthetic arms, we must increase the opportunities to make them. And in order to increase the opportunities to make prosthetic arms, we believe it is important to convey the "joy of making." For example, when cooking, we need to consider the lifestyle of those who use prosthetic arms.



Applications other than prosthetic hands

We aim to create a world where prosthetic limbs of various designs are created one after another using ingenuity and creativity, while imagining what they will look like.

The name H A C K berry was chosen based on this vision. Hackberry is the English name for the Enoki tree, a broad-leaved tree in the elm family. Enoki trees spread their branches widely and produce many leaves and fruit. With this in mind, we hope to grow H A C K berry into a large tree that will bear fruit in the form of various prosthetic limbs all over the world, with the hope that "by improving (HACKing) we can broaden the ways in which we can enjoy our products, and reach users and bear fruit (berry)."

Pinching leaves HACKberry

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[1] How HACKberry works

First, the standard specifications of the HACK bery are shown in the table on the next page. The dimensions and weight are for the hand part and do not include the socket part.

The HACK berry uses a single simple pressure sensor. This sensor is made by layering a 10mm thick sponge with a hollow interior on top of an inexpensive distance sensor known as a photoreflector. By placing the sensor on the user's arm (preferably in a muscled area) and securing it around the arm with a non-elastic band, it is possible to physically read the movement of the user's arm muscles. (This technology is based on that developed by Professor Masahiro Yoshikawa of the Osaka Institute of Technology.)
The angles of the three servo motors change depending on the value read by the simple pressure sensor. A microcomputer processes this information. When the user applies force to the arm, the fingers close, and when the force is released, the fingers open. The stronger the force applied, the faster the fingers close. The program released as open source is set up like this, but it is also possible to rewrite the program to set different settings (for example, to reverse the opening and closing of the fingers).

By adopting a link mechanism like the one shown on the page, it is possible to combine the functions of "pinching" small objects and "grasping" large objects. With this link mechanism, when an object hits the middle of the finger, the rotational force of the joint at the base is sent to the joint at the tip, so with just one motor, it is possible to passively switch between pinching small objects and grasping large objects.

(2) Ease of assembly
Prosthetic hands have more complex mechanisms than typical electrical appliances. They are also used in a variety of environments, which increases
Yes, it is possible.

Considering the convenience in daily life, the risk of breakdown. Furthermore, H A C K berry is open source and is intended to be assembled by third parties. For these reasons, it is important that H A C K berry be easy to assemble and reassemble. All of H A C K berry's components are available globally. Furthermore, gears were used instead of wires to drive the finger joints. Wires are easy to achieve soft movements similar to tendons, but they are prone to loosening and require frequent maintenance. Gears, on the other hand, result in stiff movements, but they do not loosen and are easy to repair in the event of a breakdown.

(3) Wrist mechanism

[2] HACKberry design features

With current technology, it is difficult to perfectly reproduce the dexterity of a natural hand, so when designing a prosthetic hand, careful consideration must be given to functionality. HACK berry has been designed with practicality in daily life in mind, and has been given the following features:

(1) Achieving both "pinch" and "grasp"
When people pick up something, they notice the shape of the object.

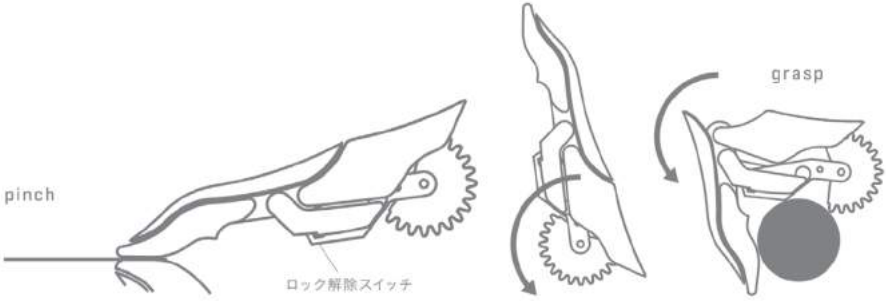
Not only the movement of the tip but also the movement of the wrist cannot be overlooked. The angle of the wrist of the H A C K berry can be adjusted manually. By pressing the button located on the wrist, the wrist can be rotated or tilted. In addition, the hand and wrist can be detached and the position can be adjusted in 90 degree increments. By combining these, the three degrees of freedom of the wrist (pronation/supination, palmar flexion/dorsiflexion, radial flexion/ulnar flexion) can be reproduced. For more information,

Please see "3. Have more fun with HACKberry" (p. 145).

By skillfully adjusting the position of your fingers and wrists to suit your
However, in the case of prosthetic hands, the number of motors
Because of the limited range of movement, it is not possible to adjust freely like with a natural hand.
Therefore, HACKberry is the next index finger.

inch Law	225 x 150 x 60 (mm) *When the thumbs are in a parallel position
Heavy quantity	450-500g *Varies depending on ingredients
S h s m a m	Simple pressure sensor x1
motor	Servo motor (large) x1 *index finger flexion Servo motor (small) x2 *Thumb abduction, middle/ring/pinky finger flexion
battery	Li-ion battery (7.2V, 2200mAh)
Microcomputer	Mk1: Arduino micro Mk2: Arduino nano interchangeable

HACKberry specifications



HACKberry index finger link mechanism

Introduction
03 License type

If you use HACKberry as open source, you must comply with the following terms and conditions established by exiiii Inc.

====
HACKberry Open Source Project Policy

In this policy, Ixy Inc. (hereinafter referred to as "our company") defines the content and terms of use of the source code and data related to "HACKberry" (hereinafter referred to as "this project") that will be made public.

1. Purpose
1.1. The purpose of this Project is to promote the research, development, and popularization of prosthetic limbs by making the source code and data available to all users of this Project (hereinafter referred to as "Users") in accordance with this Policy and by granting licenses to our inventions.

1.2. The use of source code and data released under this Policy, or inventions licensed under this Policy, for the purposes set out in the following items is prohibited:

1.2.1. Military Use
1.2.2. Use for purposes contrary to public order and morals
1.2.3. Use for the purpose of damaging the reputation or credibility of our company or HACKberry. 1.2.4. Use for the purpose that our company determines may fall under any of the above items.

2. Software Source Code
The source code of software related to HACKberry is released under the following license.

If you wish to use the source code under conditions different from each license (commercial use, etc.), please contact us at the address below.

info@exiii.jp

Source Code License
2.1. A rd u in o sketch (meaning the source code to be written to the A rd u in o Micro built into the prosthetic hand). GNU General Public License version 3

3. Hardware Data
All data related to HACKberry hardware is released under the following licenses or conditions. If you wish to use the data under conditions different from the licenses (commercial use, etc.), please contact us at the address below.

info@exiii.jp

Data License
3.1. 3D data (shape data of each part used in the prosthetic hand) Creative Commons License BY-NC-SA 4.0 International
3.2. Hand PCB data (referring to the circuit board for connecting the microcontroller, motor, sensor and battery. Creative Commons License BY-NC-SA 4.0 International
3.3. Sensor board data (This refers to the circuit board for mounting the reflective optical sensor (photoreflector) for detecting muscle bulges.) Creative Commons License BY-NC-SA 4.0 International
3.4. Battery board data (This refers to the circuit board built into the battery box.) Creative Commons License BY-NC-SA 4.0 International

4. Patents
4.1. The Company grants the User a non-exclusive, royalty-free, unlimited territorial and time-limited license (excluding sublicenses) to practice inventions for which the Company has applied for patents or for which the Company has registered patents, solely to the extent necessary to create a prosthetic arm using the source code or data in accordance with the licenses set forth in 2. and 3., and only to the extent that the license does not violate 1.2.

4.2. The invention set forth in 4.1. relates to a joint mechanism that enables movement similar to that of a human hand with a simple structure. A patent application is currently pending, and as the application process progresses, information regarding the specifics of the invention will be announced in an appropriate manner, such as through this policy or forums.

5. Trademarks and Content
5.1. The following trademarks are registered trademarks of our company in Japan.

Trademark: exiii
Registration number: 5786270

5.2. The following trademarks are pending in Japan:

Trademark: HACKberry
Application number: 2016-042844, 2016-042845

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5.4. If a user commits or uploads source code or data made public pursuant to this policy to Git Hub or the forum (http://exiii-hackberry.com/forums/), and that content is a work under copyright law, the user's copyright will be transferred to the Company upon committing. In addition, in this case, the user shall not exercise moral rights against the Company or third parties authorized by the Company.

5.5. Trademarks, etc. and Content may not be used in any way without our prior written permission.

5.6. If you wish to use the Trademarks or Content, please contact us at the address below.

<Contact> info@exiii.jp

6. Individual License Prevails
In the event that there is a difference between the provisions of this Policy and the license granted to the source code or data stipulated in paragraphs 2 or 3 with respect to the same matter, the provisions of each license shall take precedence unless otherwise specified.

7. Disclaimer
We make no express or implied guarantee that this Website, the source code and data provided on this Website, and the licenses set forth in this Policy are free from factual or legal defects (including defects related to safety, reliability, accuracy, completeness, validity, fitness for a particular purpose, security, etc., errors, bugs, infringement of rights, etc.). We are not obligated to provide users with HACKberry services without removing such defects.

7.2. We shall not be liable for any disputes or damages of any kind arising between Users or between Users and third parties in connection with the use of source code and data provided by us or the interpretation and application of each license set forth in this Policy.

7.3 We do not guarantee that the source code and data will always be available on this Website. We shall not be liable for any damages incurred by users or third parties due to the inability to provide the source code and data on this Website, whether due to system maintenance or inspection or other reasons.

8. Changes to the method and content of provision
8.1. We may, at any time without prior notice to users, change the method or content of providing source code or data or the content of the license for inventions set forth in this Policy, or discontinue the provision or license (hereinafter referred to as "Changes, etc.").

8.2. We shall not be liable for any disputes or damages incurred by users or third parties in connection with the changes, etc. set forth in the preceding paragraph.

9. Changes to this Policy
9.1. We may change the contents of this policy at any time without prior notice to users.

By continuing to obtain or use source code or data provided by the Company or by continuing to use the Company's inventions after any change to this Policy, you will be deemed to have validly and irrevocably agreed to the changed Policy, except for the content of the license applicable to source code or data obtained by you under the previous Policy before the change.

10. Governing Law and Jurisdiction
10.1. The official text of this Policy is in Japanese, and the governing law shall be Japanese law.

10.2. Any dispute arising out of or relating to this Policy shall be submitted to the exclusive jurisdiction of the Tokyo District Court or the Tokyo Summary Court, depending on the amount of the claim.

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These are the terms and conditions that must be followed when using HACK berry as open source. Furthermore, any content related to HACK berry uploaded to the forums on exiiii-hackberry.com or to Git Hub, including but not limited to program code, bug reports, opinions, ideas, and other content, whether for commercial or non-commercial use, may be used in the development of HACK berry and may be distributed not only under the Creative Commons license but also under a closed source license established by exiiii Inc.

HACK berry uses 3D printers and open source to make it easy for anyone in the world to make their own electric prosthetic arm. The reason for providing HACK berry in this way is to increase opportunities for people who use prosthetic arms to try them out and for people who make them to experience the joy of making them.

However, the HACK berry does not have the same level of safety assurance as commercially available electric prosthetic hands, and is not sold as a finished product with any quality guarantee. When using the HACK berry as a prosthetic hand beyond the scope of research or trial use, you must fully understand the performance and precision of the HACK berry and do so at your own risk. eXIII Inc. and the nonprofit organization Mission ARM Japan assume no responsibility for any disputes or damages incurred by users or third parties as a result of using the HACK berry.

2. Let's assemble it

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05 Uploading the program ····· P052

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08 Assembling the index finger ····· P080

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14 Initial Position Setting P134

Assembling
01 WhichHackberryWould you like to assemble

There are several configurations of HACKberry, so first decide which HACKberry you want to build.

A. Right hand

This is the most basic type of HACKberry. When using it as a prosthetic arm, you will need to have a prosthetics manufacturer create a socket that fits your actual arm. In addition to this image, you will need an external battery case to share power with the sensors used to operate HACKberry.

B. Right hand (with dummy socket)

This is a type of HACKberry A with a 3D printed temporary socket. The battery is stored inside the socket, so an external battery case is not required. A sensor separate from this image is required to operate the HACKberry.

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C. Left hand

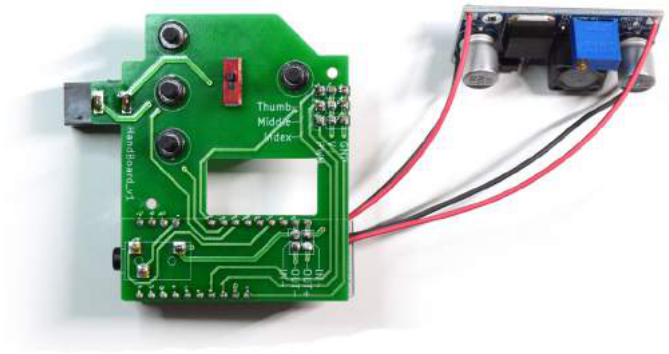
This is the most basic type of HACKberry for the left hand. When using it as a prosthetic arm, you will need to have a prosthetics manufacturer create a socket that fits your actual arm. There is basically no difference between this and a right hand. In addition to this image, you will need an external battery case to share the power and sensors to operate HACKberry.

D. Left hand (with dummy socket)

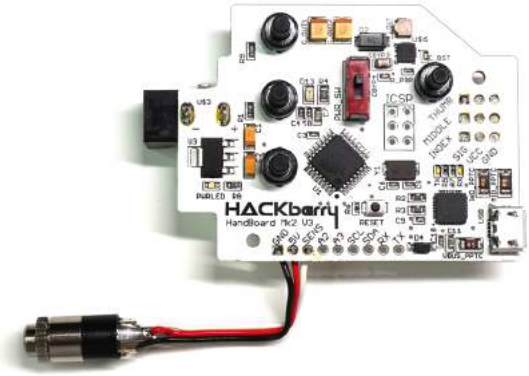
This is a C HACKberry with a 3D printed temporary socket. The battery is stored inside the socket, so an external battery case is not required. A sensor separate from this image is required to operate the HACKberry.

There are two types of HACKberry boards: Mk1 and Mk2. Both boards will work with HACKberry, but the Mk2 is easier to implement and is cheaper. At the same time, the board itself is smaller and has ports for expanding functionality, making it highly customizable.

・HACKberry hand board Mk1



・HACKberry hand board Mk2



Assembling
02 How to obtain parts

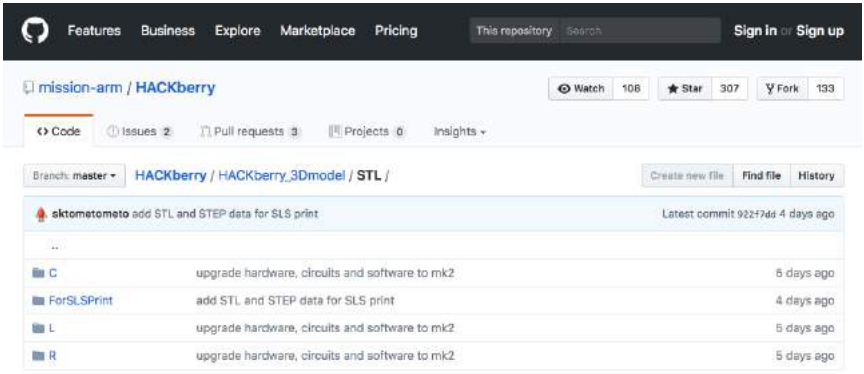
[1] Printing 3D printed parts

Data for 3D printed parts can be obtained from the internet. Open the link below and click on DATA(GITHUB) on the page that appears.

The Github site will open.

From there
HACKberry_3Dmodel / STL / ...

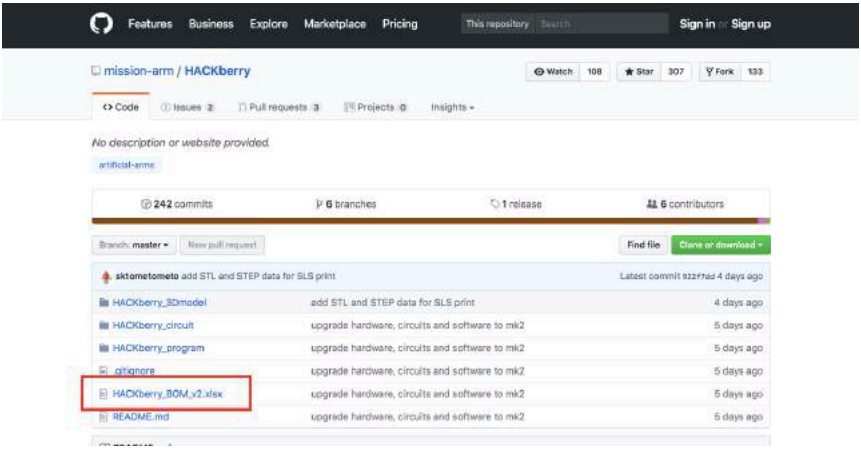
and download the desired STL data.


Features
Business
Explore
Marketplace
Pricing
This repository
Search
Sign in or Sign up
mission-arm / HACKberry
Watch 108
Star 307
Fork 133
Code
Issues 2
Pull requests 3
Projects 0
Insights
Branch: master
HACKberry / HACKberry_3Dmodel / STL /
Create new file
Find file
History
sktometomato add STL and STEP data for SLS print
Latest commit 922f7dd 4 days ago
..
C
upgrade hardware, circuits and software to mk2
6 days ago
ForSLSPrint
add STL and STEP data for SLS print
4 days ago
L
upgrade hardware, circuits and software to mk2
5 days ago
R
upgrade hardware, circuits and software to mk2
5 days ago

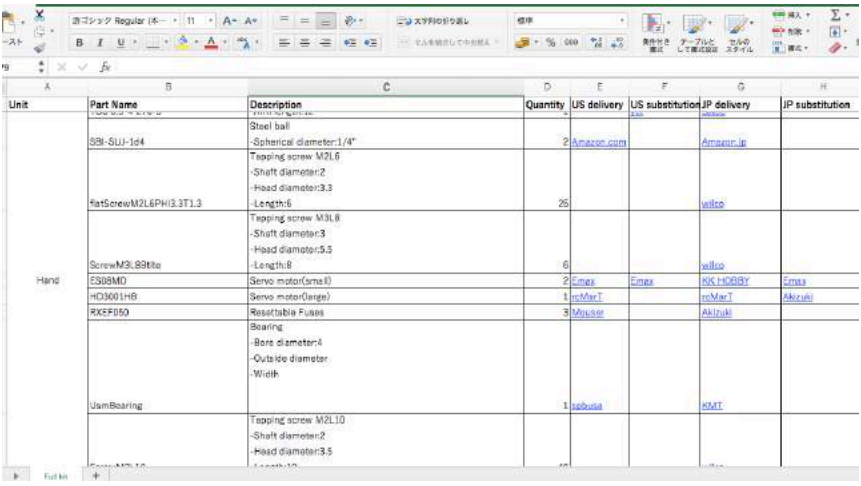
[2] How to purchase mechanical and electronic parts other than 3D printed parts

First, please refer to the previous section and access the Github page.

Download "HACKberry_BOM_v2.xlsx" from that page. (The BOM version may be updated in the future.)

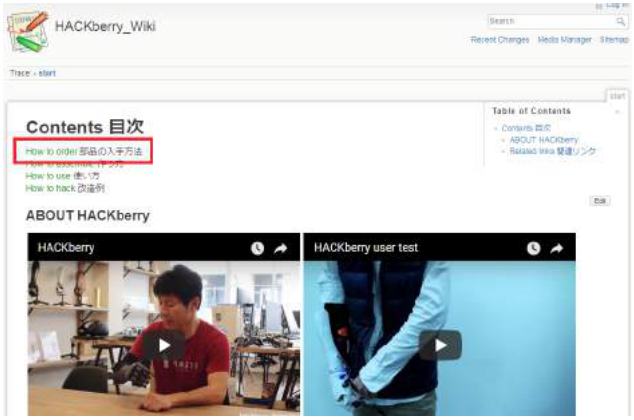


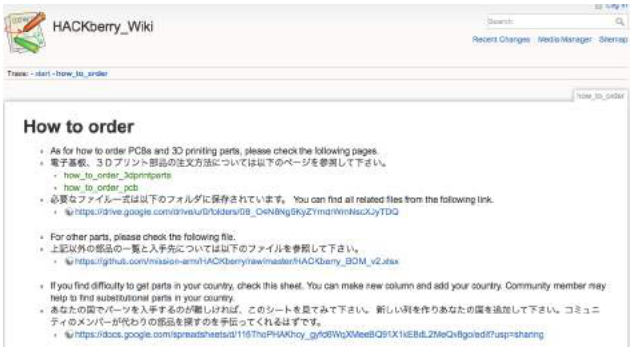
Open the downloaded file using Excel or a compatible software.



You can open the supplier's website from the link on the right side of the opened file screen, so please purchase the part by referring to the part name and required quantity.

To find out how to obtain the circuit board to use with HACKberry, please contact Mission ARM Japan. (You can also create your own circuit board by referring to the circuit diagram without going through Mission ARM Japan, or use a printed circuit board manufacturing service such as Seeed Studio.) For details, access the URL below, click TUTORIAL, and refer to the page that opens.




HACKberry_Wiki
Search
Recent Changes Media Manager Sitemap
Trace: » start » how_to_order
how_to_order
How to order
As for how to order PCBs and 3D printing parts, please check the following pages.
電子基板、３Ｄプリント部品の注文方法については以下のページを参照して下さい。
how_to_order_3dprintparts
how_to_order_pcb
必要なファイル一式は以下のフォルダに保存されています。 You can find all related files from the following link.
https://drive.google.com/drive/u/0/folders/0B_OeN8Ng5KyZYmdrWmNscXJyTDQ
For other parts, please check the following file.
上記以外の部品の一覧と入手先については以下のファイルを参照して下さい。
https://github.com/mistlon-arm/HACKberry/raw/master/HACKberry_BOM_v2.xlsx
If you find difficulty to get parts in your country, check this sheet. You can make new column and add your country. Community member may help to find substitutional parts in your country.
あなたの国でパーツを入手するのが難しければ、このシートを見てみて下さい。 新しい列を作りあなたの国を追加して下さい。コミュニティのメンバーが代わりの部品を探すのを手伝ってくれるはずです。
https://docs.google.com/spreadsheets/d/1I6ThqPHAKhoy_gyfd6WqXMeeBQ91X1kEBdL2MeQvBgo/edit?usp=sharing

Assembling
03 Before assembly

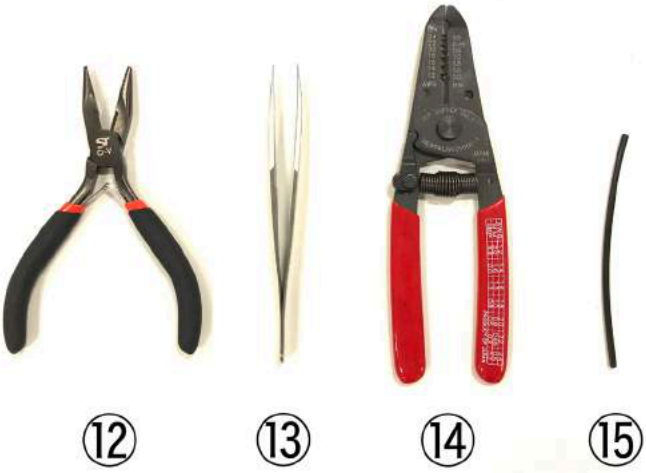
Precautions
- Please assemble in the order given in the manual.
・The following explanation will mainly be for the right hand. If you are assembling the left hand, please flip the image.
- Depending on the version, there may be slight differences between the image in this manual and the part you have.

-Essential Tools-

1. Soldering iron and soldering iron holder
2. Solder wire
3. PC
4. Micro USB cable
5. Nippers
6. Driver
7. Pin vise
8. Drill (φ1.6mm, φ2.2mm)
9. Major
10. Tester
11. Vinyl tape

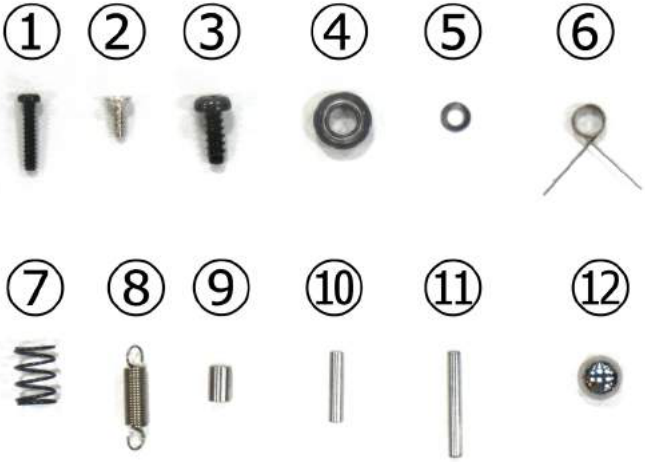
- Useful tools and equipment -

12. Needle-nose pliers 13. Tweezers 14. Wire stripper
15. Heat shrink tubing



- General-purpose parts -

1. Screw (M2L10) 2. Screw (M2L6) 3. Screw (M3L8)
4. Bearings 5. Washer 6. Torsion spring
7. Coil springs 8. Tension spring 9. Spacer
10. Shaft (10mm) 11. Shaft (15mm) 12.steel ball



-How to read part numbers-

This explains the naming convention for 3D printed parts. Characters are separated by hyphens (-).

The first item is the major classification.

R	Right hand parts
L	Left hand parts
C	Left and right common parts

The second item is the classification of the unit.

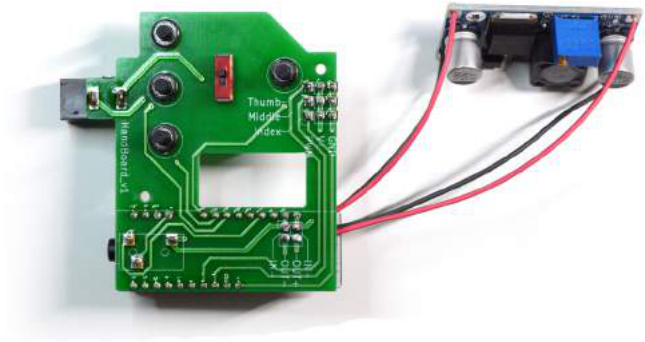
SO	Socket Unit
T	Thumb unit
I	Index finger unit
The	Three-finger unit (middle finger, ring finger, little finger)
H	Palm Unit
IN	Wrist Unit
B	Battery case unit
SE	Sensor Unit

The third item is the number assigned within that unit.

For example, R-SO-01 is the first part of a right-handed socket unit.

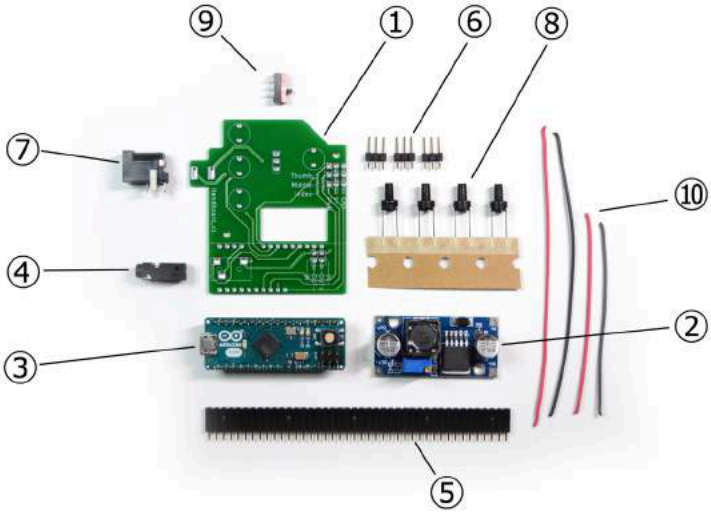
Assembling
04 PCB soldering

Soldering HACKberry hand board Mk1



-Parts used-

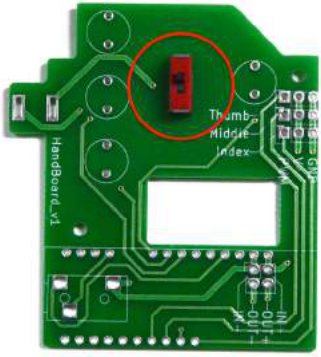
1. HACKberry hand board Mk1
2. DC-DC converter
3. Arduino Micro
4. Stereo jack
5. Pin socket
6. Pin headers
7. DC jack
8. Tactile Switch
9. Slide Switch
10. Lead wire


⑨
①
⑥
⑧
⑦
⑩
④
②
③
⑤

*When soldering, make sure to insert the electrical element (component) firmly all the way in and take care not to solder it while it is floating or at an angle.
* Small electrical elements on the board may be damaged by static electricity. Before soldering, use an anti-static pad or similar to remove as much static electricity as possible.
*Touching electrical components with wet hands can cause a short circuit and damage the components. Be sure to wipe off any moisture before soldering.

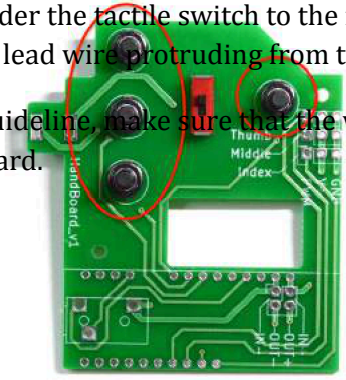
(1) Solder the slide switch to the front side of the board.The side of the board with white letters printed on it is the front.

Also, the slide switch has no polarity (direction).



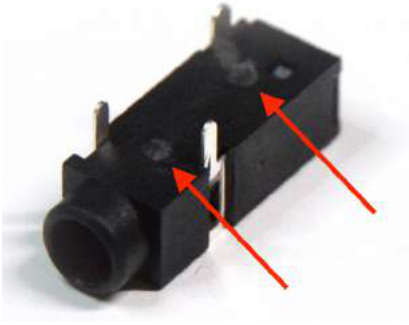
[2] Solder the tactile switch to the front side of the board. Use pliers to secure the excess lead wire protruding from the back side.

As a guideline, make sure that the wire does not protrude more than 2 mm from the board.



back side

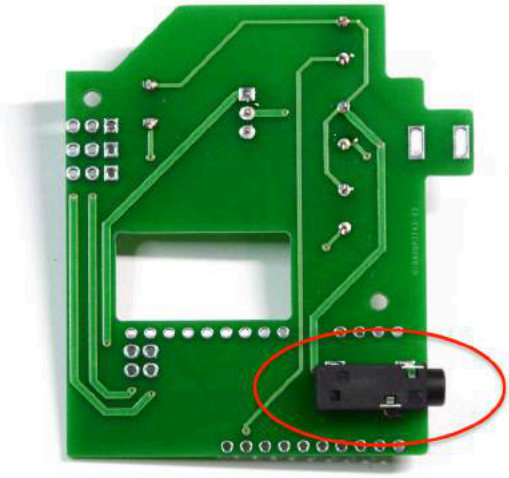
[3] Next, solder the back side. If you are soldering a right-handed board, cut off the protrusion on the bottom of the stereo jack with a cutter or similar tool before creating the circuit.



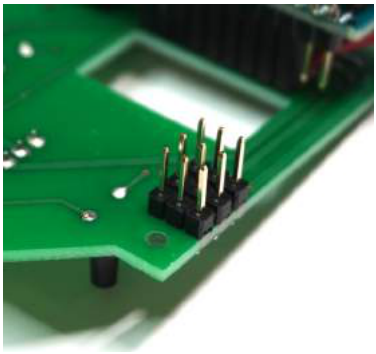
The protrusion has been cut off

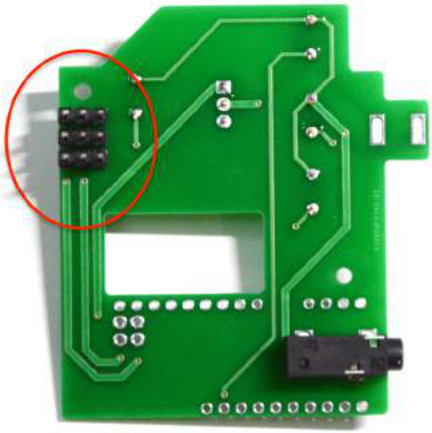
No protrusions cut off

[4] Next, solder the stereo jack in the position shown in the photo.

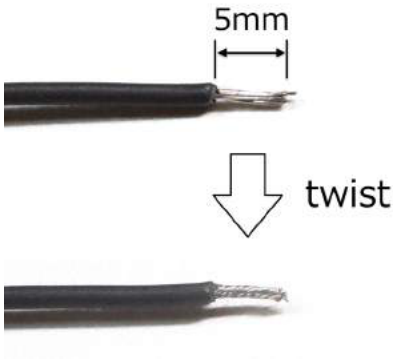


[5] Pin headerThe shorter metal partInsert it into the board and solder it.Make sure the pins are not at an angleI'll be careful.

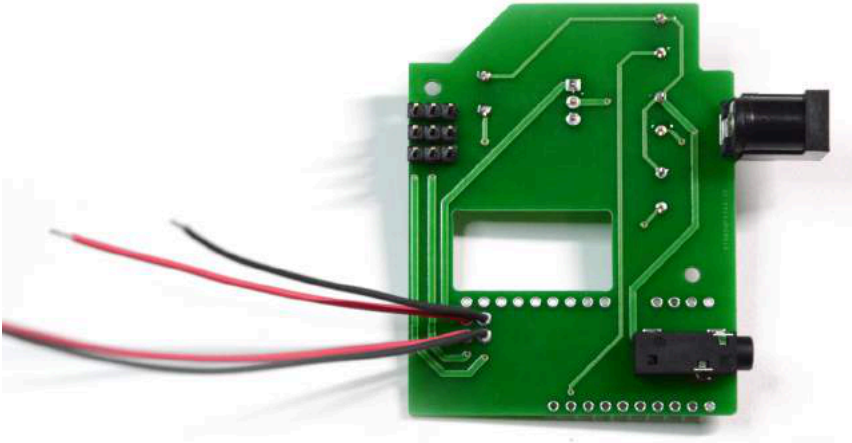


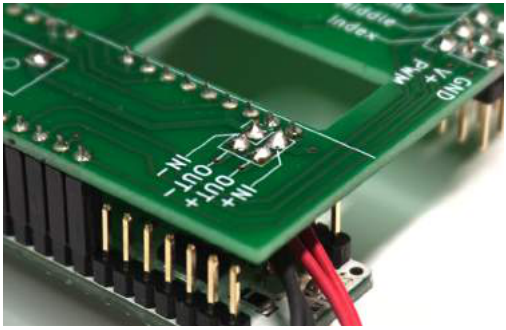


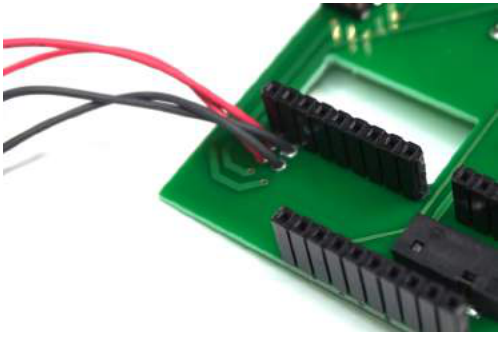
[6] Use a wire stripper or similar tool to strip 5 mm of the insulation from the end of the lead wire, and twist the wires together. If you do not have a wire stripper, use pliers to pinch the insulation and strip it carefully, being careful not to cut the core wire.


5mm
twist

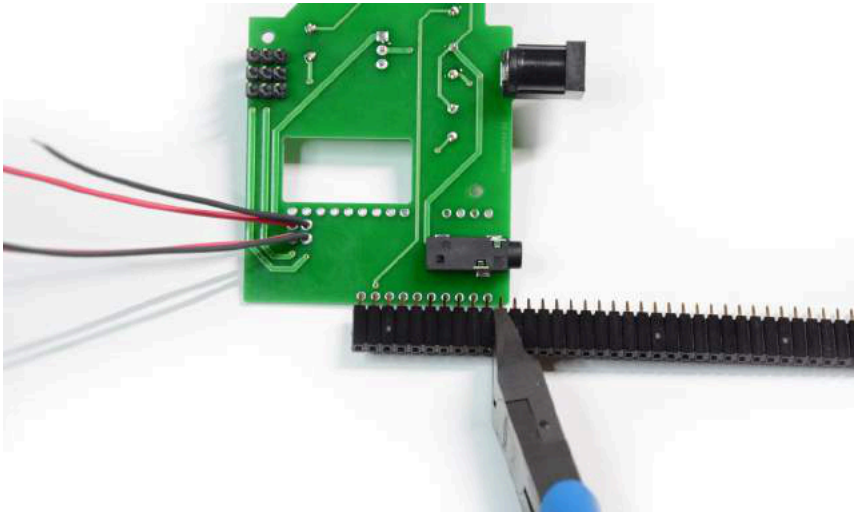
[7] The longer end is the OUTPUT and the shorter end is the INPUT. Solder them to the circuit board so that the red wire connects to the + terminal and the black wire connects to the - terminal. Solder the DC jack as well.

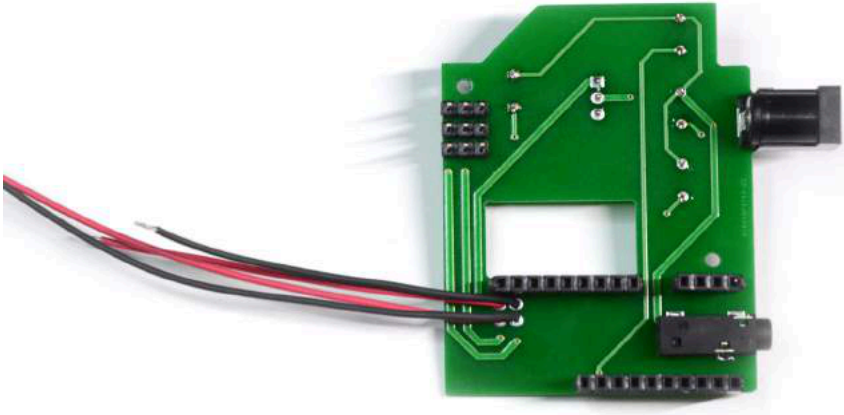




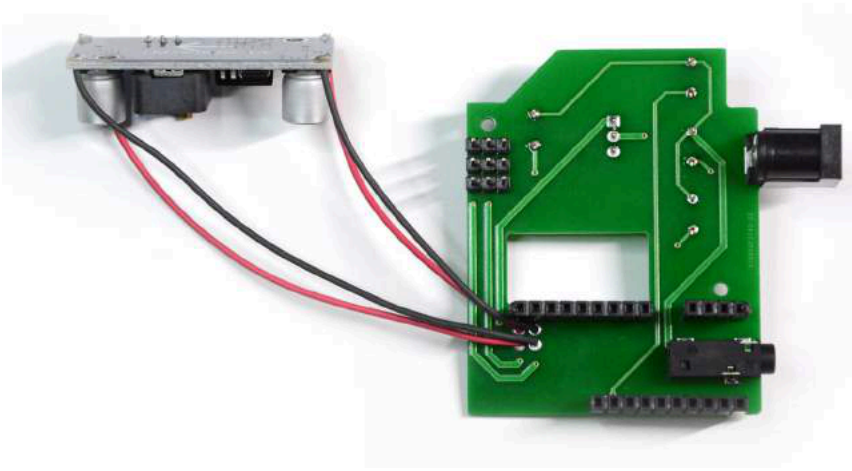


[8] Use pliers to cut the pin sockets to fit the holes in the board, then solder them in three places.

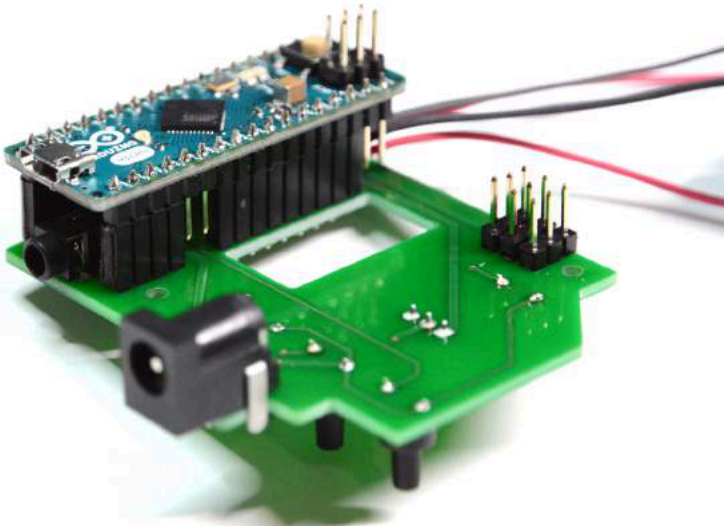




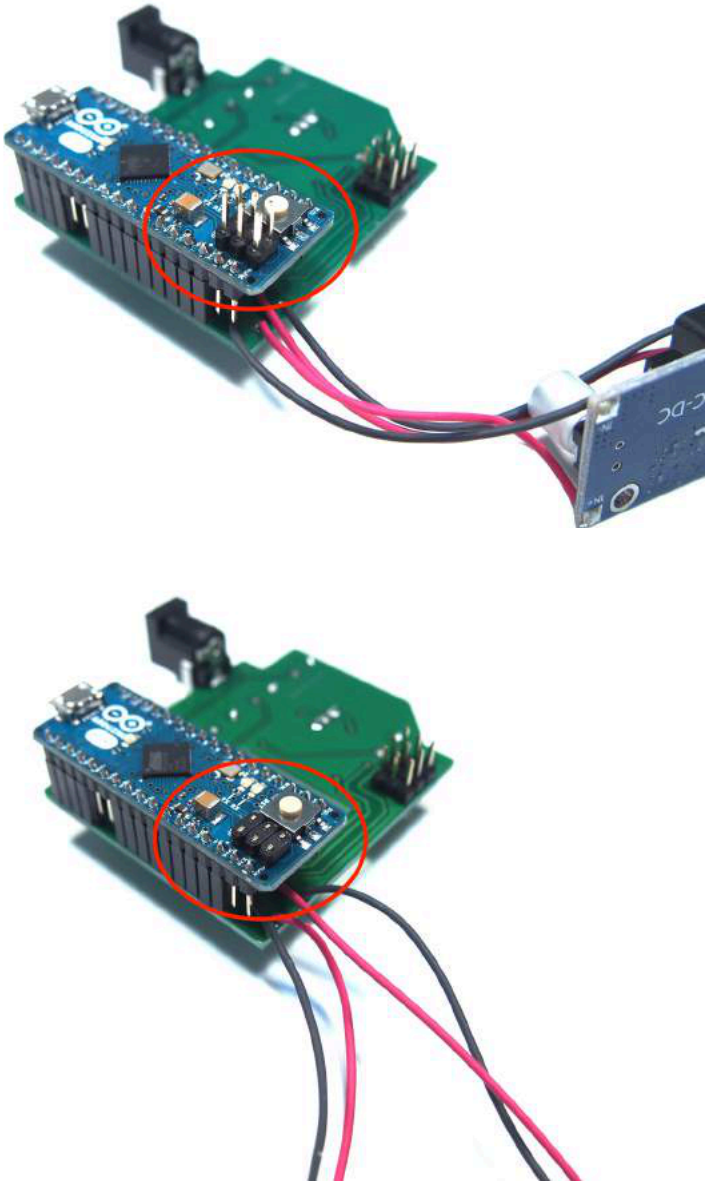
[9] Solder the DC-DC converter to the OUT+, OUT-, IN+, and IN- terminals on the HACKberry hand board Mk1.



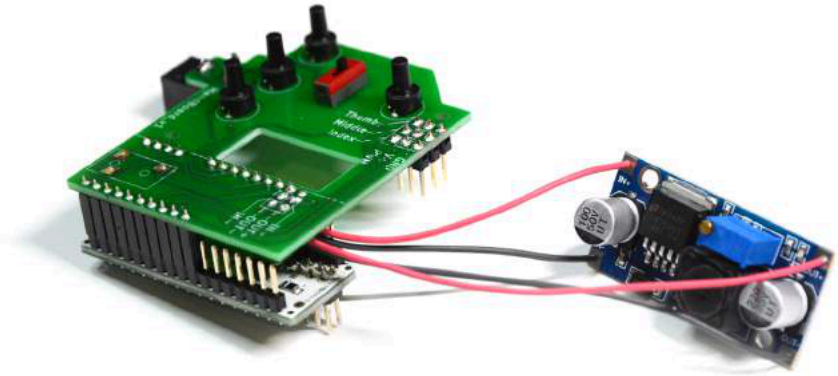
[1 0] Insert the Arduino Micro all the way into the pin header, being careful not to get the hole in the wrong position.

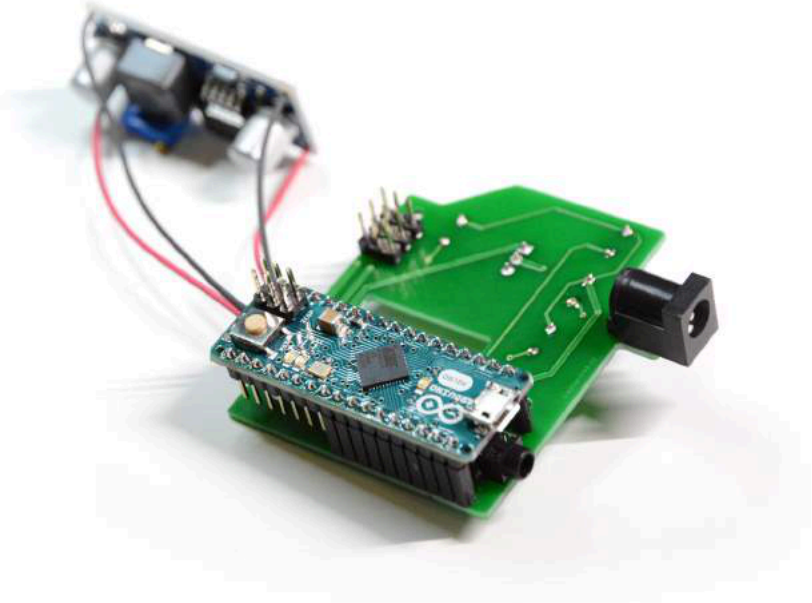


[1 1] For left-handed boards, use pliers to cut the pins in the red frame in the diagram as shown in the photo.



[1 2] The HACKberry hand board Mk1 is now complete.



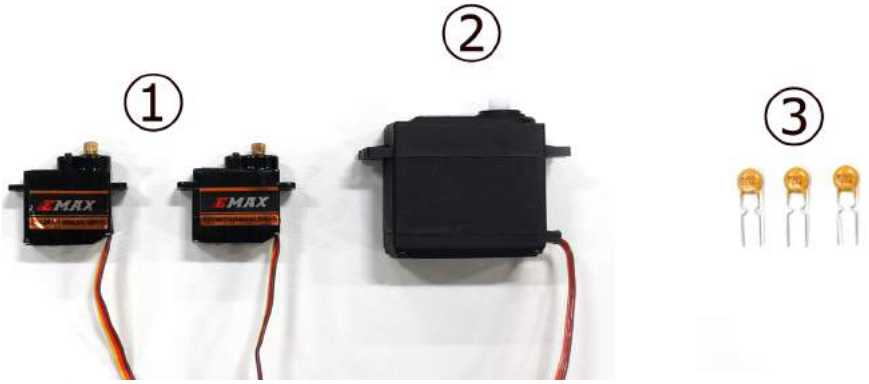


[1 3] Next, insert the polyswitch inside the servo motor.

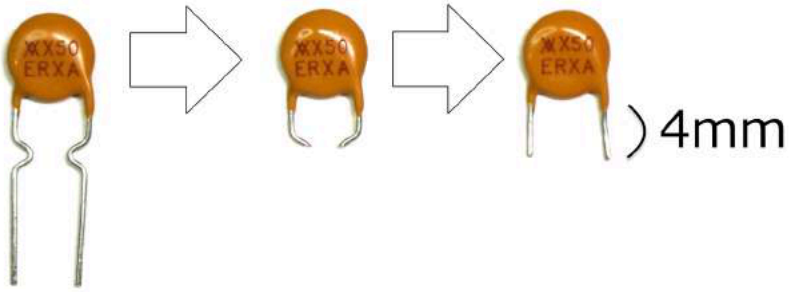
*It is possible to operate the servo motor without this step, but installing a polyswitch will prevent damage to the motor when excessive load is applied.

*This step is not necessary on Mk2 boards as they already have polyswitches implemented on the board itself.

1. Small servo motor 2. Large servo motor 3. PolySwitch

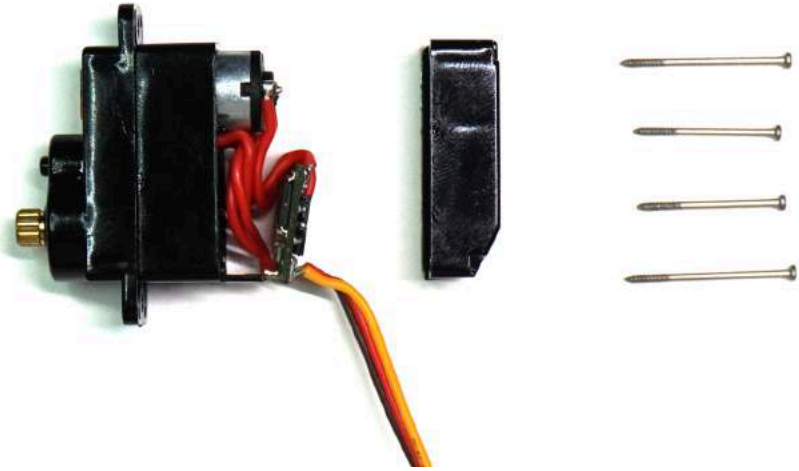


[1 4] First, cut the lead wire of the PolySwitch leaving about 4 mm, and then use pliers to straighten it out. Do this for three pieces.



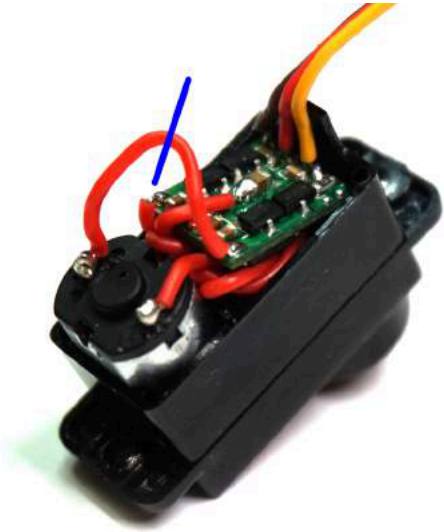
35

[1 5] Loosen the screws on the small servo motor and remove the cover. There is a manufacturer's sticker on the side, so peel it off or cut it at the joint of the part.

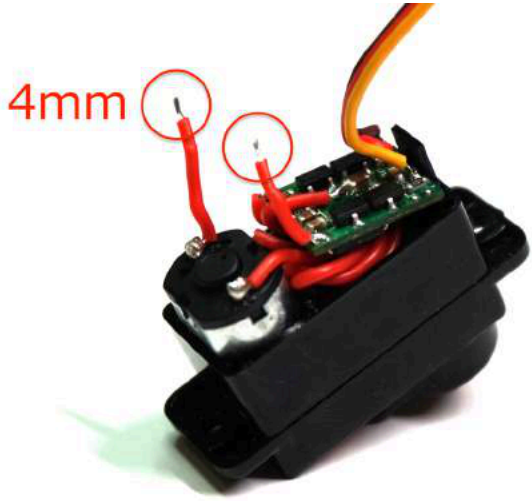


[1 6] Cut one of the two cables soldered to the DC motor inside at the blue line in the photo.

cut

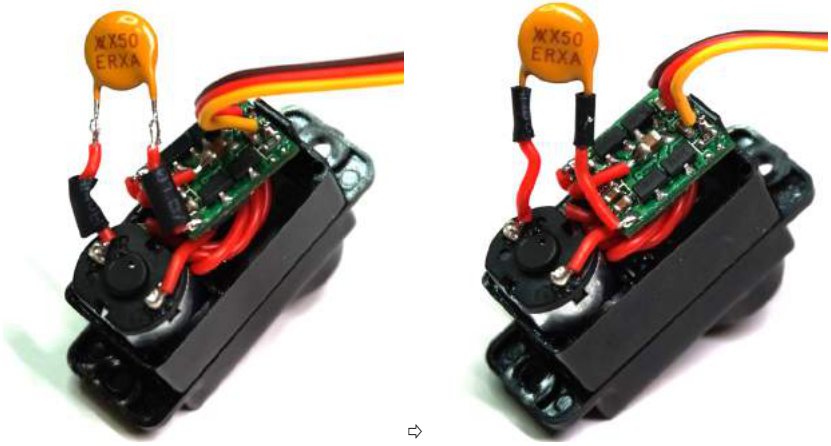


[1 7] Use a wire stripper or similar tool to strip about 4 mm of the insulation from the cut cable and twist the core wire.

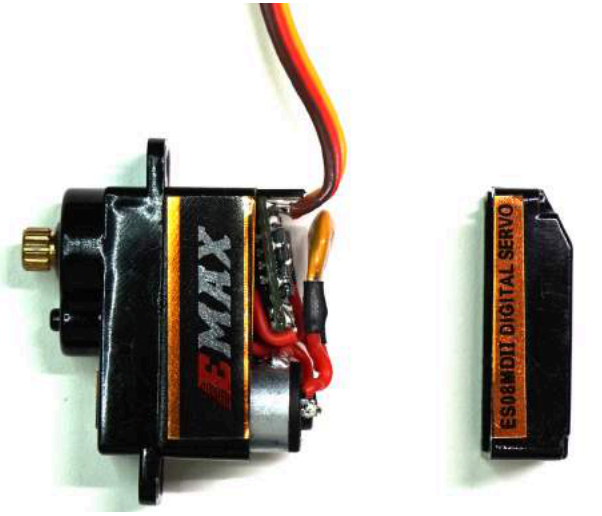


[1 8] Solder a polyswitch to the cut cable and cover it with vinyl tape.It is dangerous to expose the metal parts, so make sure to cover the entire metal part.

*If you have heat shrink tubing, it is safer to use it. *PolySwitch has no polarity.



[1 9] Fold the cable as shown in the figure, close the cover, and tighten the screws.At this time, be careful not to pinch the cable.

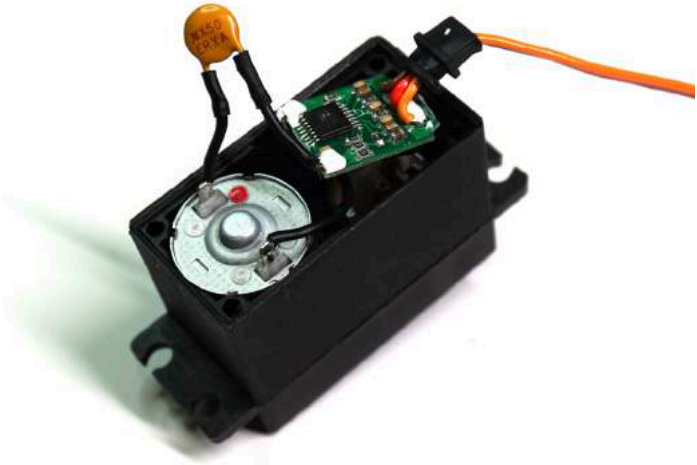


[2 0] Loosen the servo motor's large screw and remove the cover. A waterproof rubber gasket will appear, but it will not be necessary when you finally close the cover.

[2 1] Cut one of the two cables soldered to the DC motor inside as shown in the photo, strip about 4 mm of the insulation using a wire stripper, and twist the core wires together.

[2 2] Solder a polyswitch to the cut cable and cover it with vinyl tape.It is dangerous to expose the metal parts, so make sure to cover the entire metal part.

*If you have heat shrink tubing, it is safer to use it. *PolySwitch has no polarity.

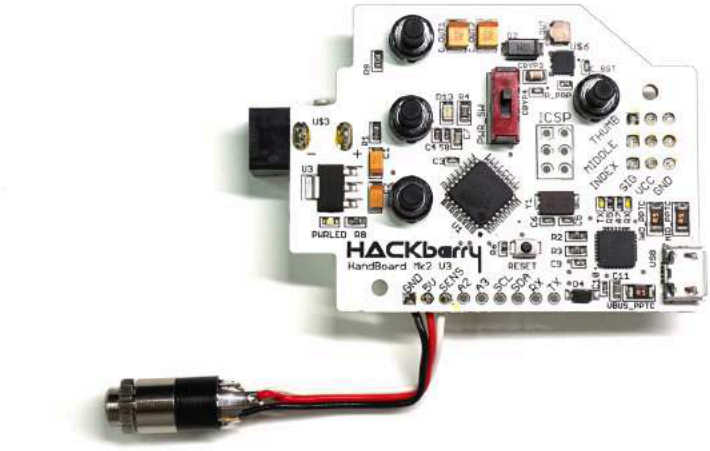


[2 3] Close the cover and tighten the screws. This completes the PolySwitch insertion.

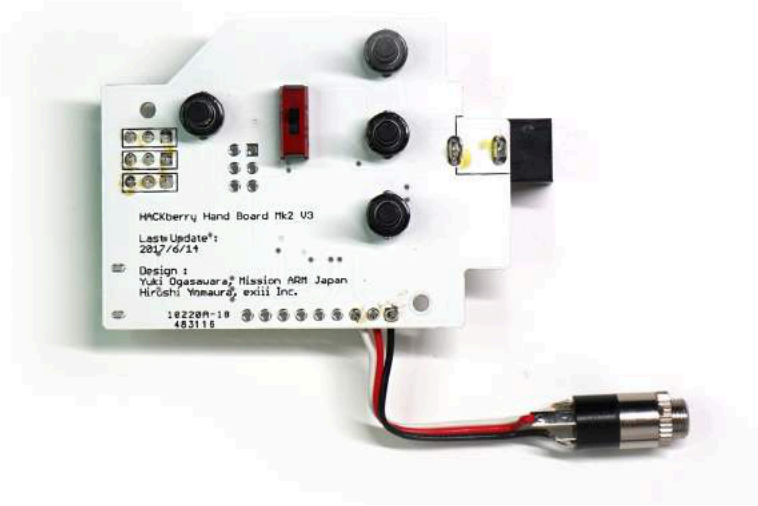
Mk2 board soldering

The HACKberry hand board Mk2 is a control board that can be used on both the left and right sides.The front and back will be reversed depending on whether you use it on the left or right side.Please be careful not to make a mistake as the soldering instructions for left-handed use are listed after the instructions for right-handed use.

Mk2 board completed image (right-handed)



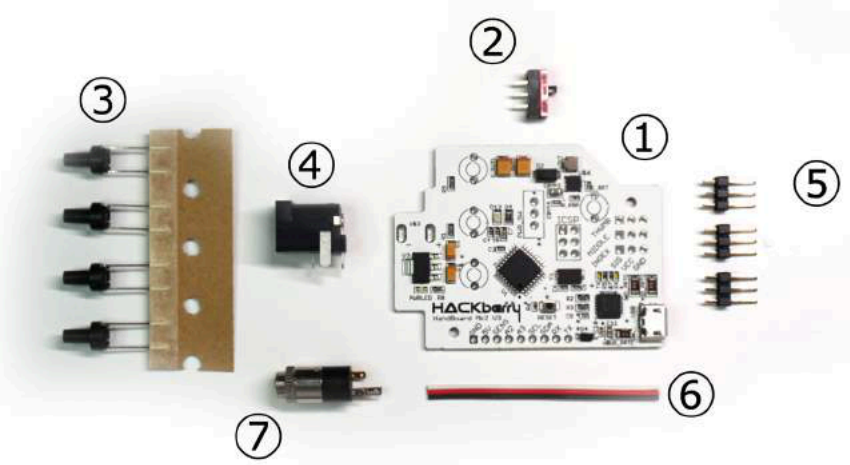
Mk2 board completed image (left-handed)



*The appearance may differ slightly from the photo above depending on the version of the board.

-Parts used-

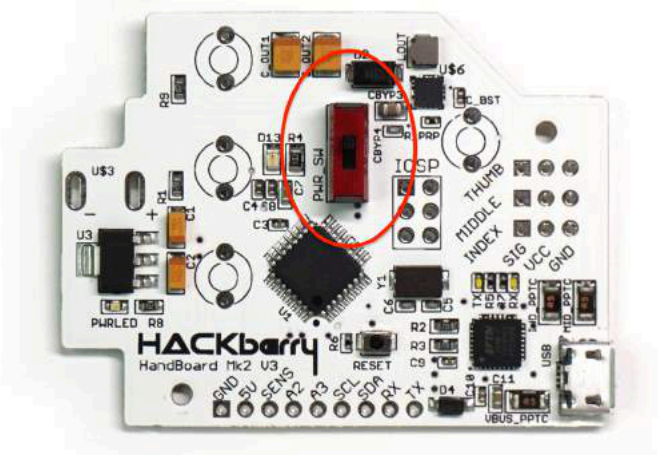
1. HACKberry hand board Mk2
2. Slide switch
3. Tactile switches
4. DC jack
5. Pin headers
6. Flat Cable
7. Stereo jack



*When soldering, make sure to insert the electrical element (component) firmly all the way in and take care not to solder it while it is floating or at an angle.
* Small electrical elements on the board may be damaged by static electricity. Before soldering, use an anti-static pad or similar to remove as much static electricity as possible.
*Touching electrical components with wet hands can cause a short circuit and damage the components. Be sure to wipe off any moisture before soldering.

Right-handed soldering

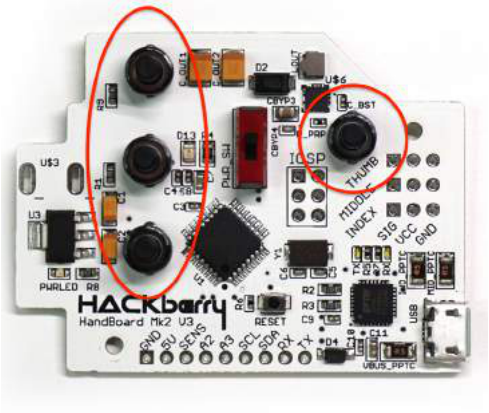
[1] Slide the slide switch to theFront sideSolder it to.The side of the board with the HACKberry logo printed on it is the front.Also, the slide switch has no polarity.



[2] Solder the tactile switch to the front side of the board. Cut off any excess lead wires sticking out from the back side. As a guideline, cut them within 2 mm of the board surface.

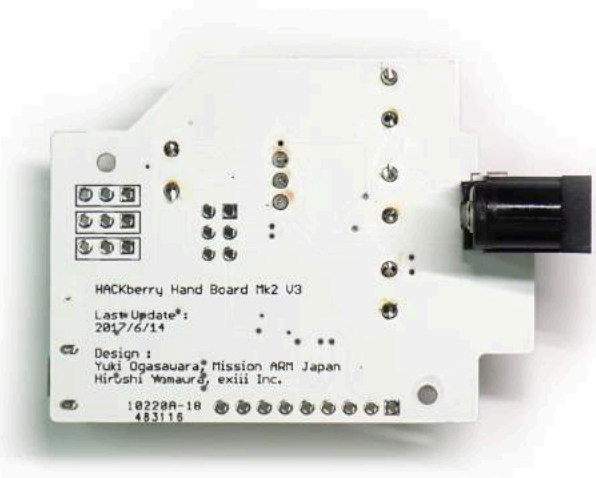
Front side

Back side



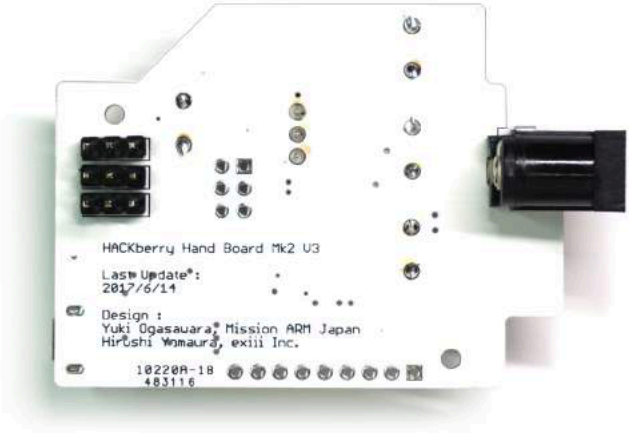
43

(3) Solder the DC jack as shown in the diagram.

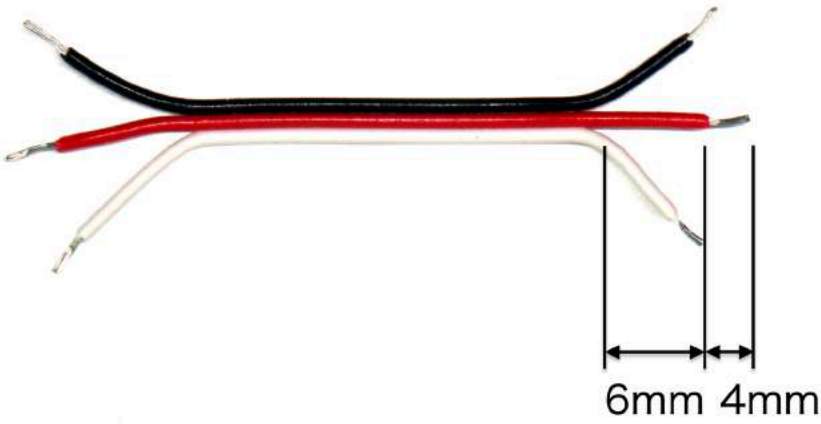


[4] Pin headerThe shorter metal partInsert it into the board and solder it.Easy to tilt diagonallySo be careful

Please put it on.

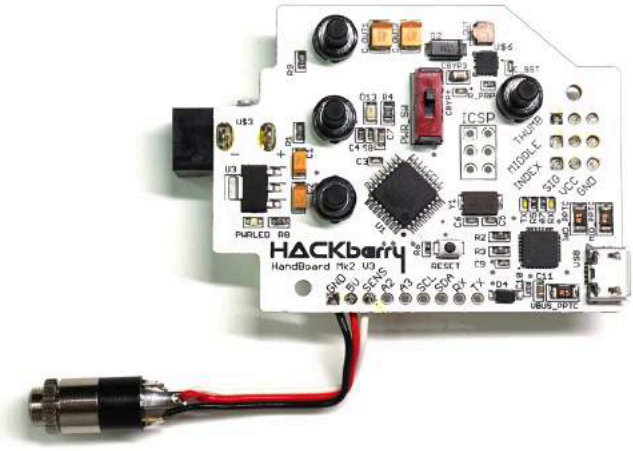


[5] Split each end of the flat cable by about 10 mm, and then strip about 4 mm of the insulation using a wire stripper or similar tool. Twist the core wires together by hand.



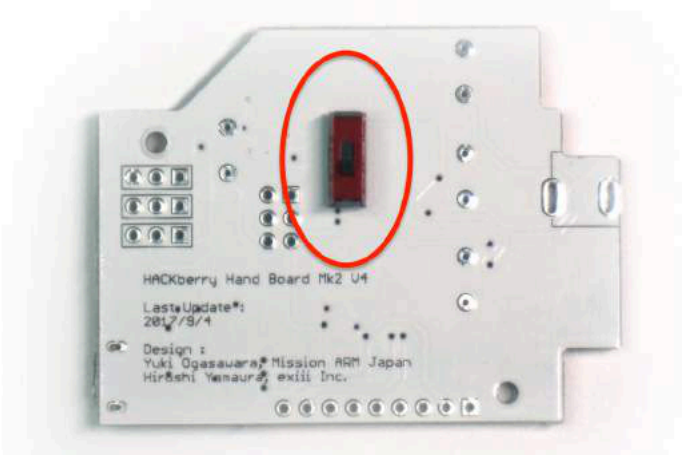
[6] Solder the flat cable to the stereo jack terminal.White cable to the top right terminal, red cable to the bottom terminal, black cable to the top left terminalSolder carefully to avoid any mistakes.

[7] Solder the other end of the flat cable to the circuit board.White cable is SENS, red is 5V, black is GNDThis completes the soldering of the HACKberry hand board Mk2.



Soldering for left-handed use

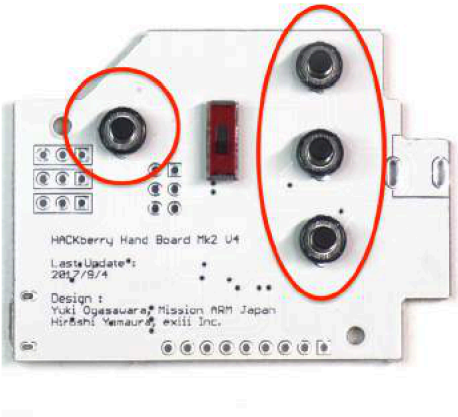
[1] Slide the slide switch to theBack sideThe board is soldered toThe side without the HACKberry logo is the back.Also, the slide switch has no polarity (direction).

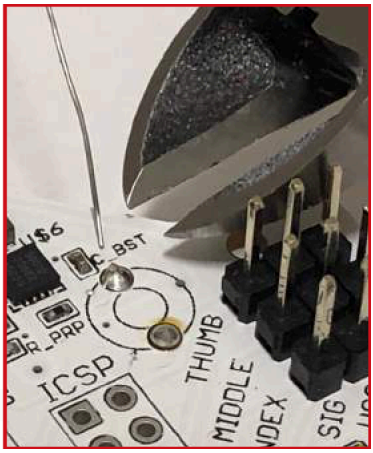


[2] Solder the tactile switch to the front side of the board. Cut off any excess lead wires sticking out from the back side. As a guideline, cut them within 2 mm of the board surface.

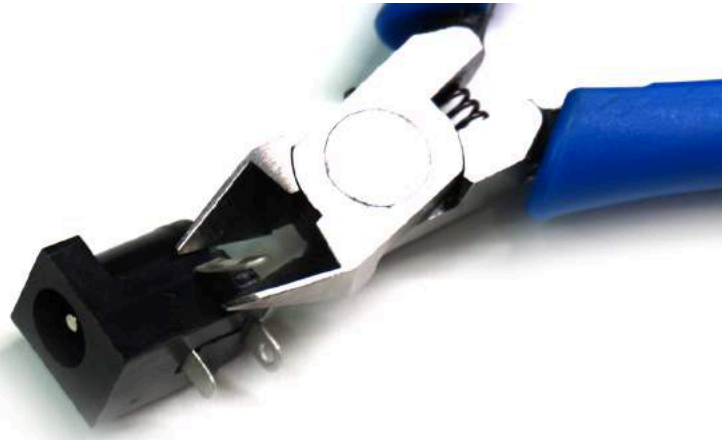
Back side

Front side

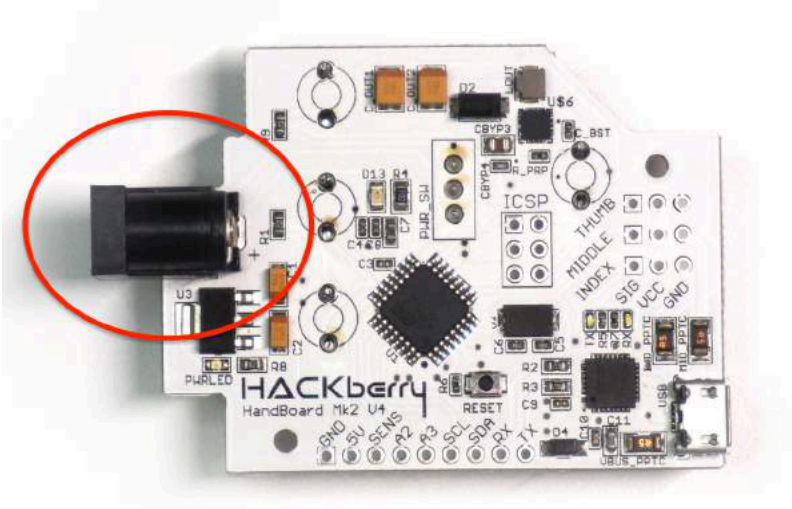

HACKberry Hand Board Mk2 V4
Last Update*:
2017/9/4
Design :
Yuki Ogasawara, Mission ARM Japan
Hirohashi Yamaura, exiii Inc.


ICSP
THUMB
MIDDLE
SIG

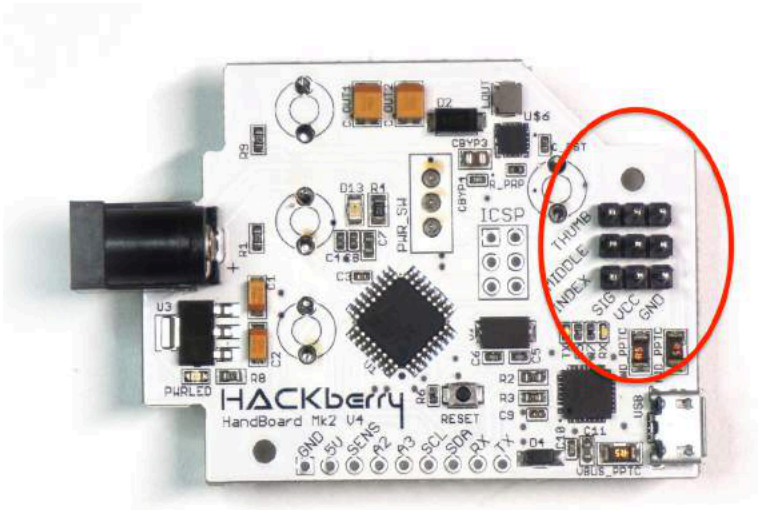
[3] Next, solder the front side. Use pliers to cut off the metal terminals protruding from the side of the DC jack.



(4) Solder the DC jack as shown in the diagram.

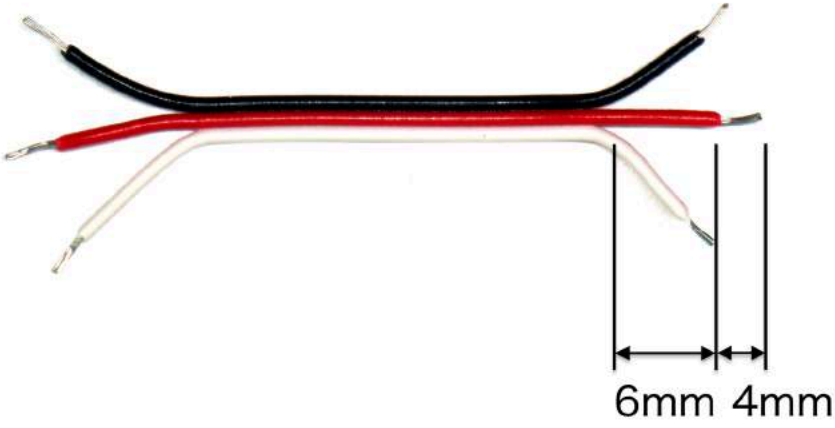


[5] Insert the short end of the pin header into the board and solder it.Easy to tilt diagonallySo please be careful.



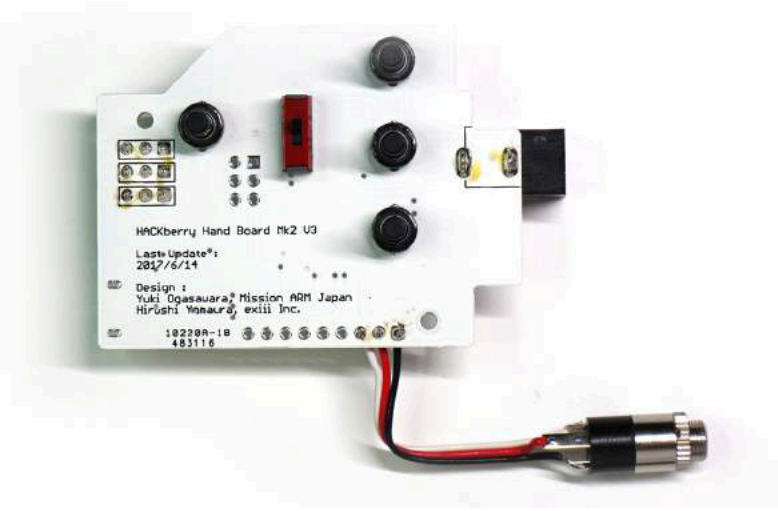
[6] Split both ends of the flat cable by about 10 mm, and then remove the coating by about 4 mm from each end.

Strip the wire using a ripper or similar tool. Twist the core wire together by hand.



[7] Solder the flat cable to the stereo jack terminal.White cable to the top right terminal, red cable to the bottom terminal, black cable to the top left terminalSolder carefully to avoid any mistakes.

(8) Solder the other end of the flat cable to the circuit board.White cable is SENS, red is 5V, black is GNDThis completes the soldering of the HACKberry hand board Mk2 board.

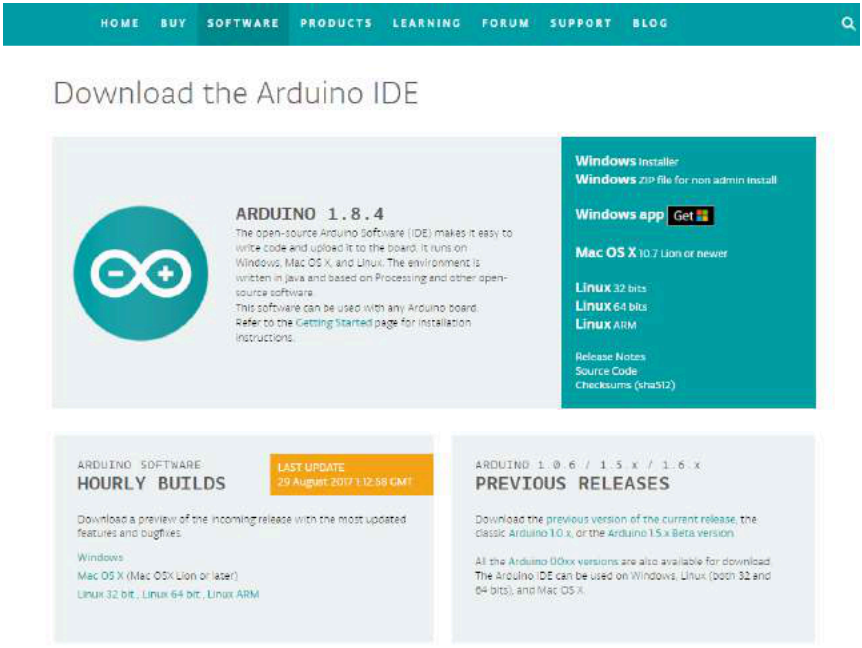


How to change the program for different board types and left and right

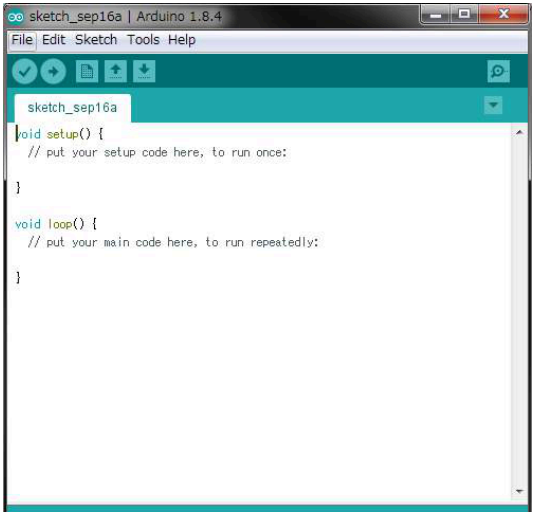
This section explains how to upload a program using Windows.

[1] Visit the official Arduino website and download the latest version of the Arduino IDE software.

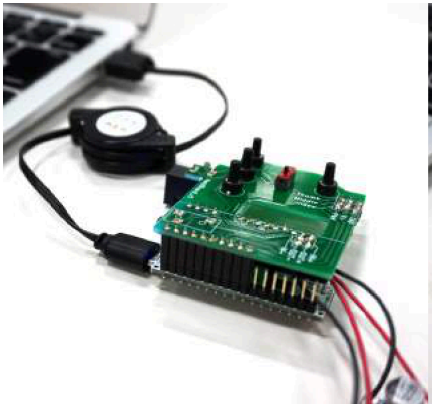
*The website design and version may differ from the time this book was written.

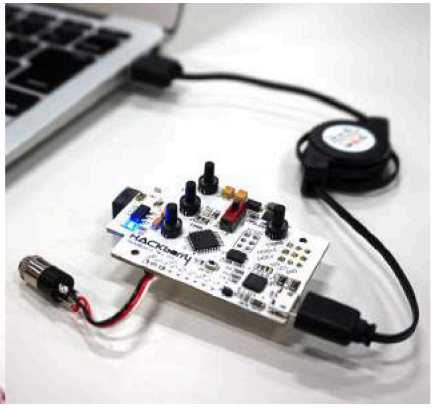


(2) Run the Arduino installation file and follow the instructions to install Arduino on your computer. Once the installation is complete, launch the Arduino IDE.



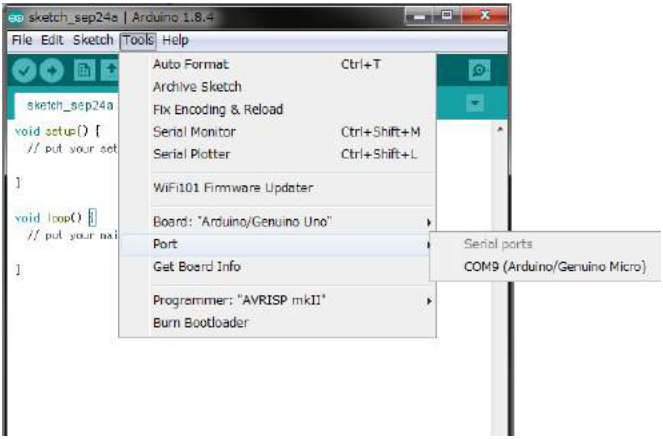
[3] If you are using an Mk1 board, connect the Arduino Micro to your PC. If you are using an Mk2 board, connect the Mk2 board to your PC using a micro USB cable.





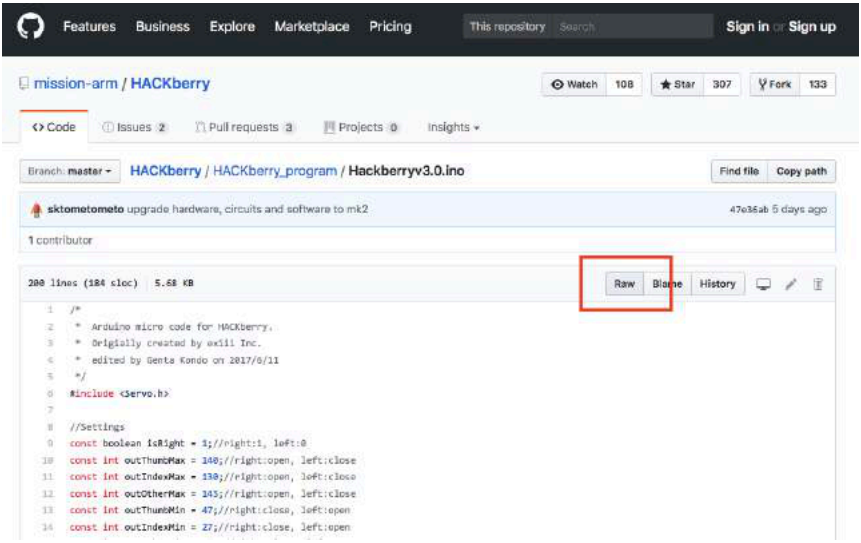
[4] The driver installation will begin automatically. Once the installation is complete, a new COM port will appear in the Ports field of the Tools tab in the Arduino IDE. Select it.

*If it does not appear, unplug the micro USB cable from the Arduino Micro or Mk2 board and then plug it back in.

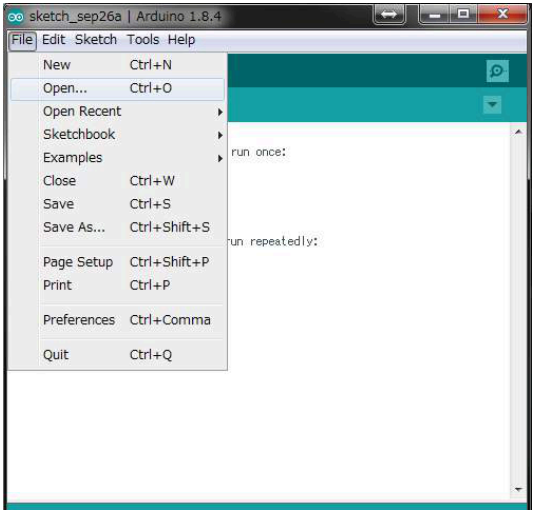


[5] Github's HACKberry repositoryHACKberry/HACKberry_programAccess /Hackberryv3.0.ino and if you have a HACKberry hand board Mk1,Hackberryv3.0"of,
If you have a HACKberry hand board Mk2HackberryMk2v1.0", right-click the "RAW" button, and select "Save link as."

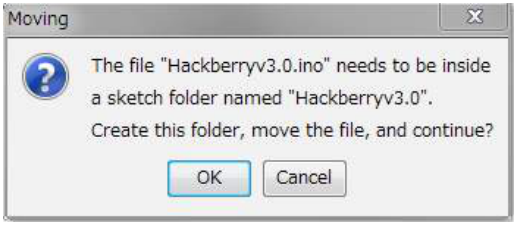
*The program version may have changed since this book was written.


Features Business Explore Marketplace Pricing
This repository Search
Sign in or Sign up
mission-arm / HACKberry
Watch 108
Star 307
Fork 133
Code
Issues 2
Pull requests 3
Projects 0
Insights
Branch: master
HACKberry / HACKberry_program / Hackberryv3.0.ino
Find file
Copy path
sktometometo upgrade hardware, circuits and software to mk2
47e36ab 5 days ago
1 contributor
200 lines (184 sloc) 5.68 KB
Raw
Blame
History
/*
* Arduino micro code for HACKberry.
* Orignally created by exiii Inc.
* edited by Genta Kondo on 2017/6/11
*/
#include <Servo.h>
//Settings
const boolean isRight = 1;//right:1, left:0
const int outThumbMax = 140;//right:open, left:close
const int outIndexMax = 130;//right:open, left:close
const int outOtherMax = 145;//right:open, left:close
const int outThumbMin = 47;//right:close, left:open
const int outIndexMin = 27;//right:close, left:open

[6] In the Arduino IDE, go to “File → Open” and select the file you just downloaded.

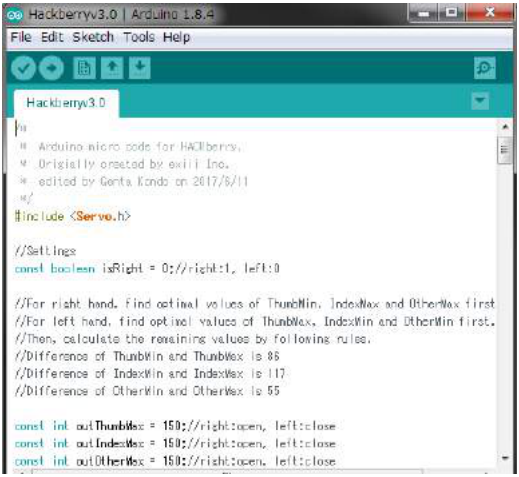


[7] The following message may appear during the process, but click OK to proceed.



[8] Next, select the appropriate microcontroller board from "Tools → Microcontroller Board." Finally, press the "➡" button in the upper left corner of the screen to upload the program.

*For the microcontroller board, select A r du ino Micro if you are using an M k 1 board, and A r du ino Nano if you are using an M k 2 board.



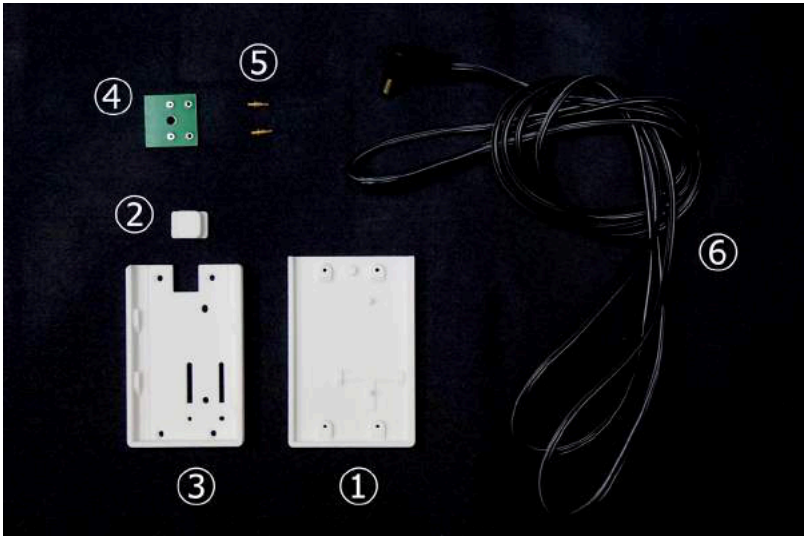
Assembling
06 Assembling the battery case and socket

Assembling the battery case

This section will show you how to assemble a battery case for a HACKberry that does not include a dummy socket. If you are assembling a HACKberry with a dummy socket, please start from page 62.

-Parts used-

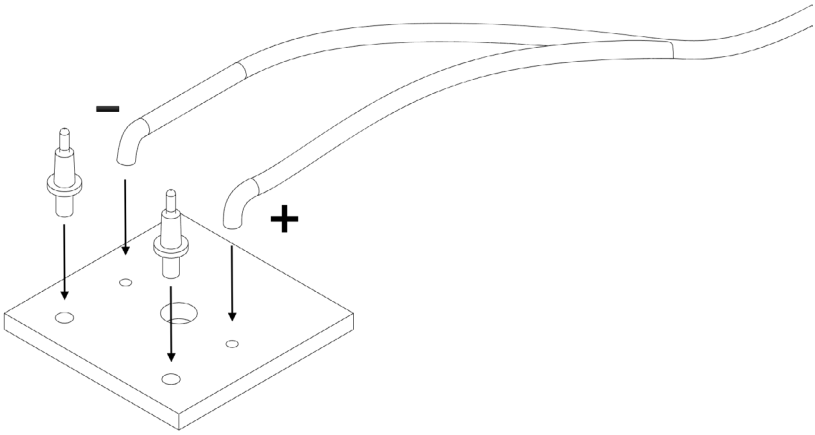
1. C-B-01
2. C-B-02
3. C-B-03
4. Battery board
5. Battery Probe
6. DC plug

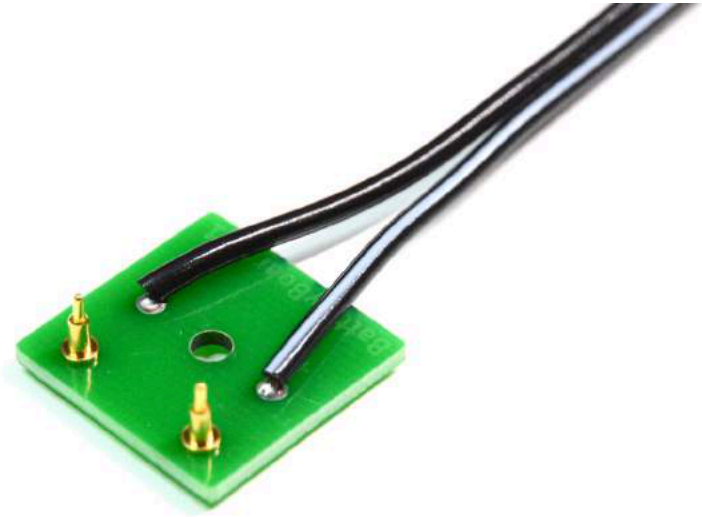


-General-purpose parts used-

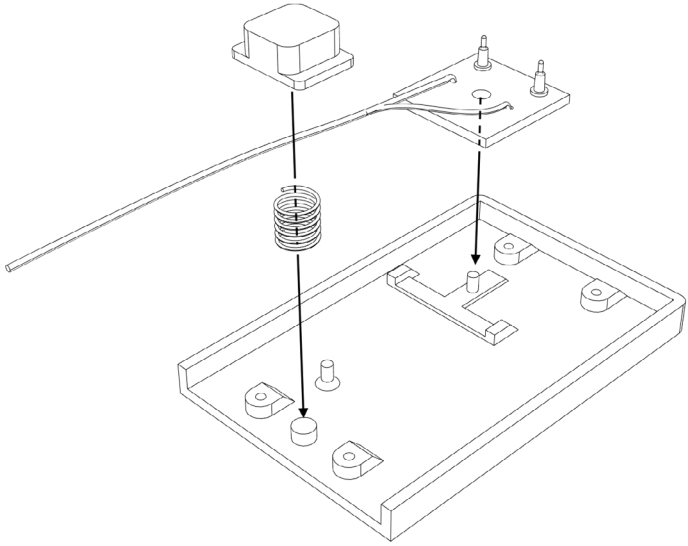
58

(1) Solder the battery probe and DC plug to the battery board as shown in the diagram below. Make sure to solder the cable with the white line to the + terminal.

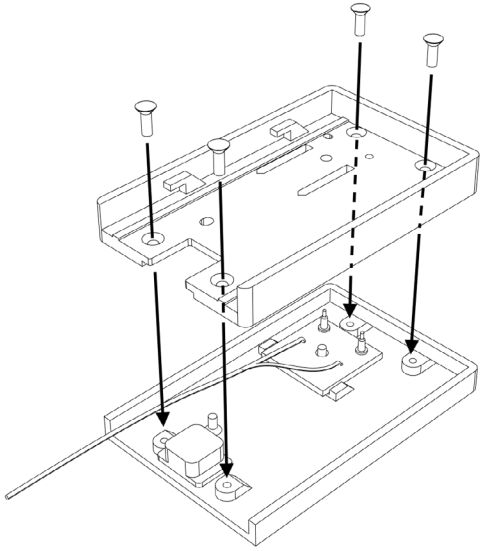




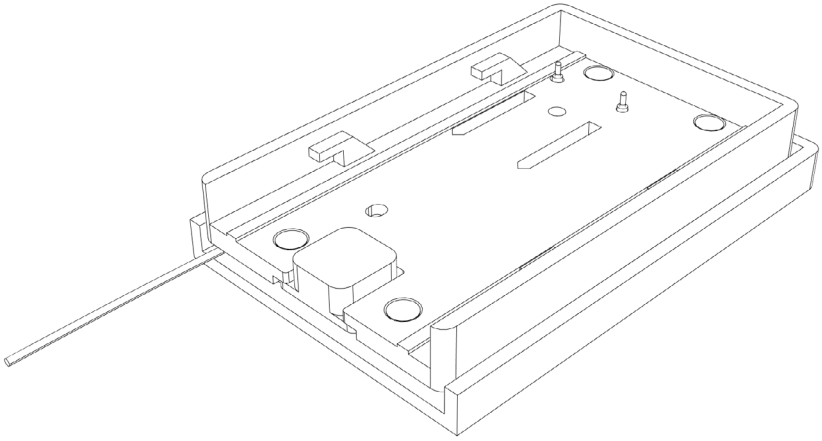
(2) Place the coil spring and C-B-02 on C-B-01 as shown in the diagram below.



(3) Secure C-B-03 with four screws (M2L6) as shown in the diagram below.



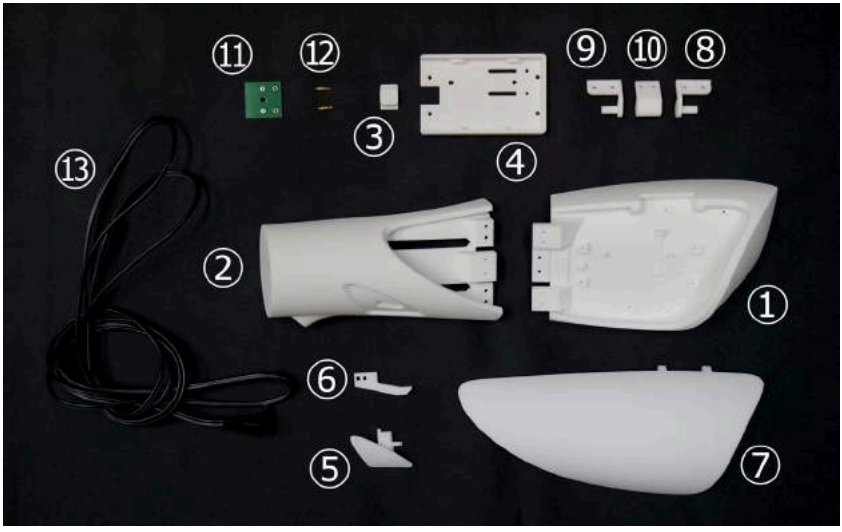
[4] This completes the assembly of the battery case.



Assembling the dummy socket

-Parts used-

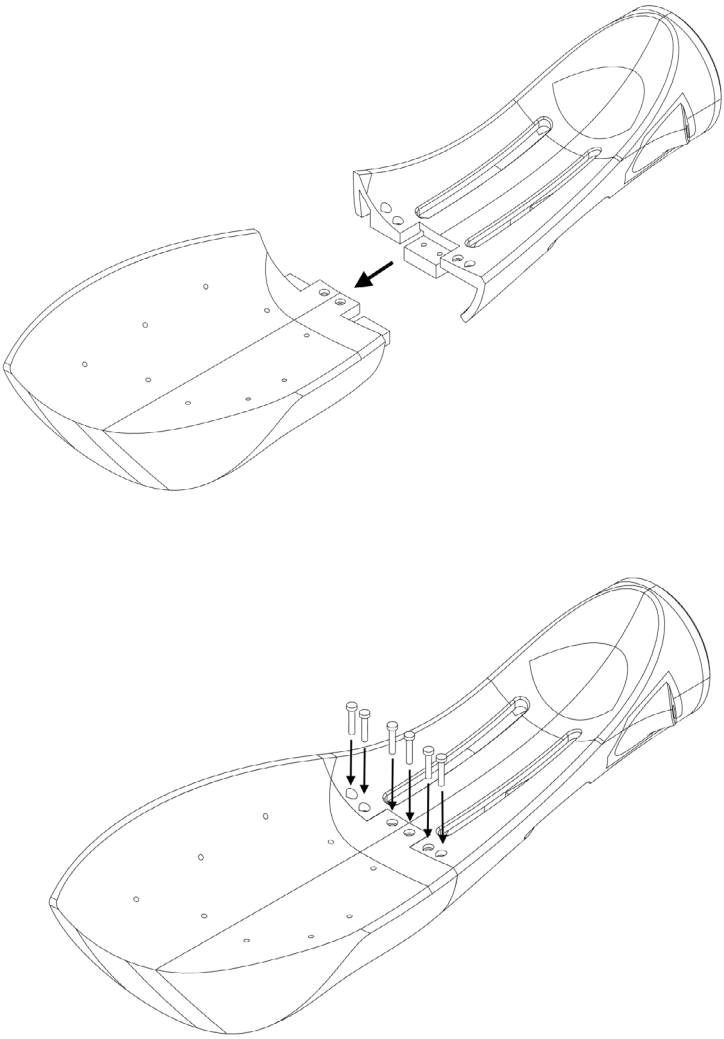
1. R-SO-01
2. R-SO-02
3. R-SO-03
4. R-SO-04
5. R-SO-05
6. R-SO-06
7. R-SO-07
8. R-SO-08
9. R-SO-09
10. R-SO-10
11. Battery board
12. Battery Probe
13. DC plug



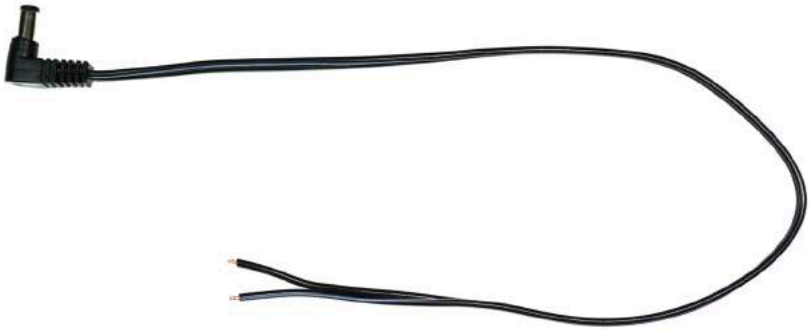
-General-purpose parts used-

62

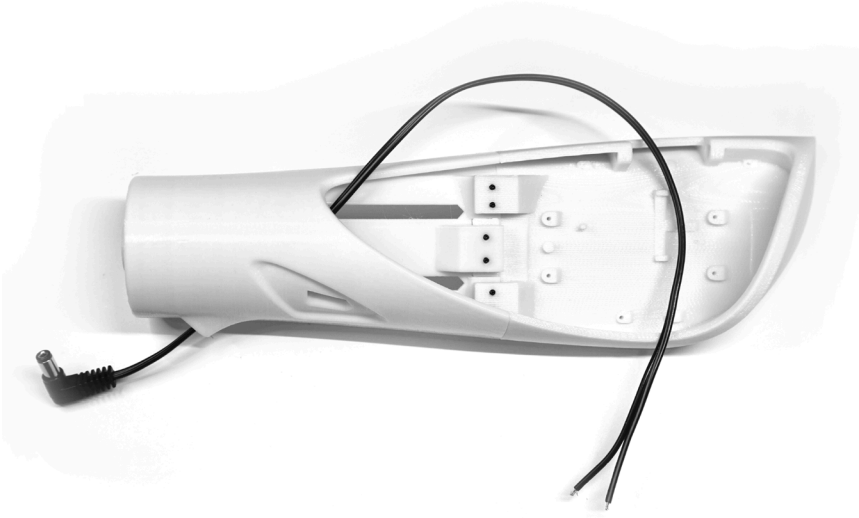
[1] Secure R-SO-01 and R-SO-02 with six screws (M2L10) as shown in the diagram below.



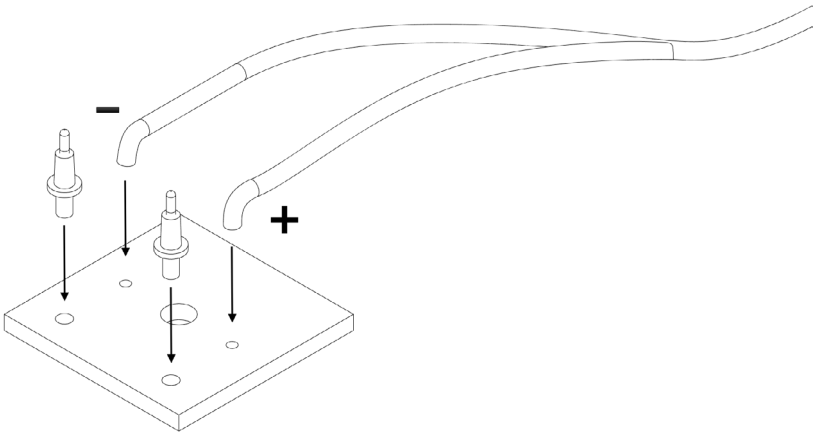
(2) Cut the DC plug into a 3.3 cm piece, strip 4 mm of the insulation from the cut end, and then twist the internal core wires together. If you don't have a wire stripper, use pliers to strip the insulation, being careful not to cut the internal core wires.

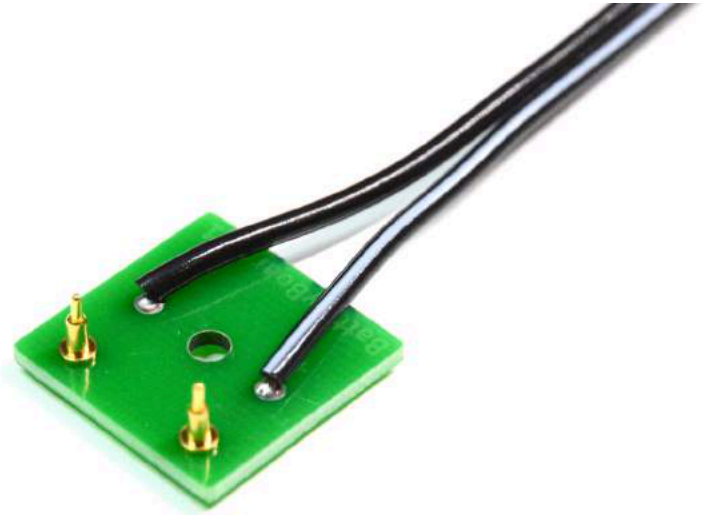


[3] Pass the wire through the hole in R-SO-02 as shown in the diagram below.

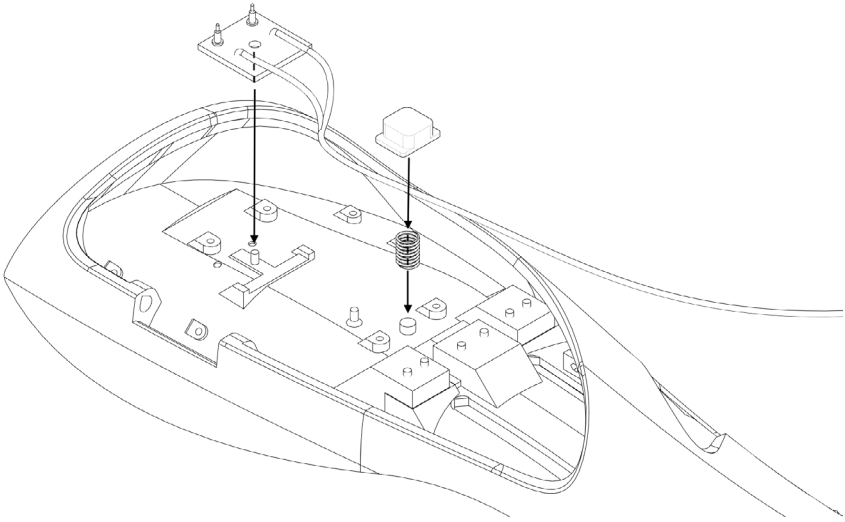


(4) Solder the battery probe and DC plug to the battery board as shown in the image. When doing so, make sure to solder the cable with the white line to the + terminal.

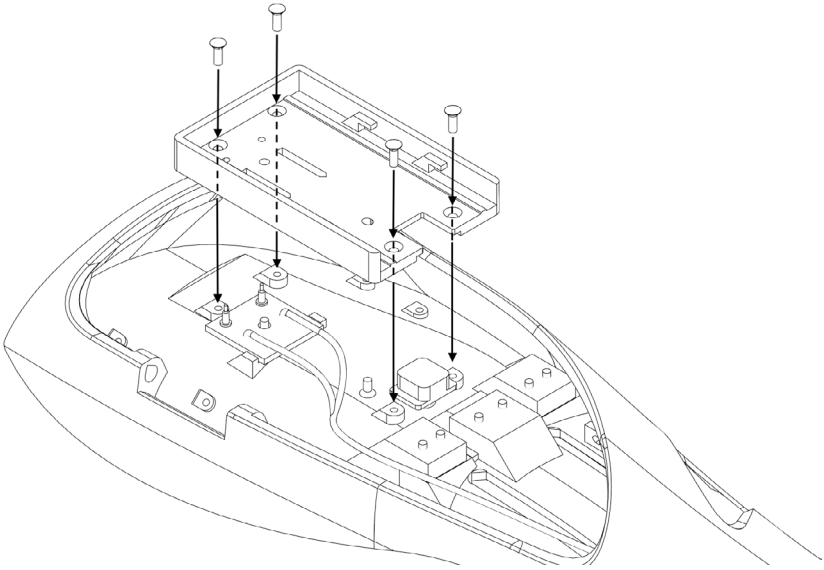




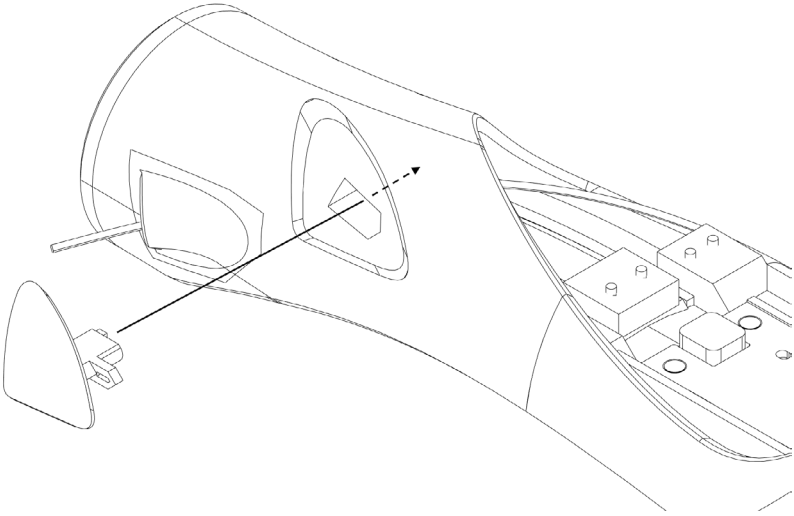
(5) Place the coil spring and R-SO-03 as shown in the figure below.



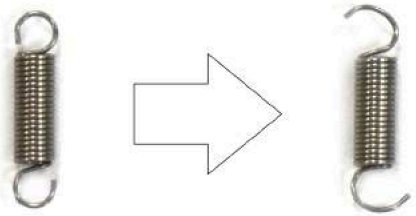
[6] Secure R-SO-04 with four screws (M2L6) as shown in the diagram below.

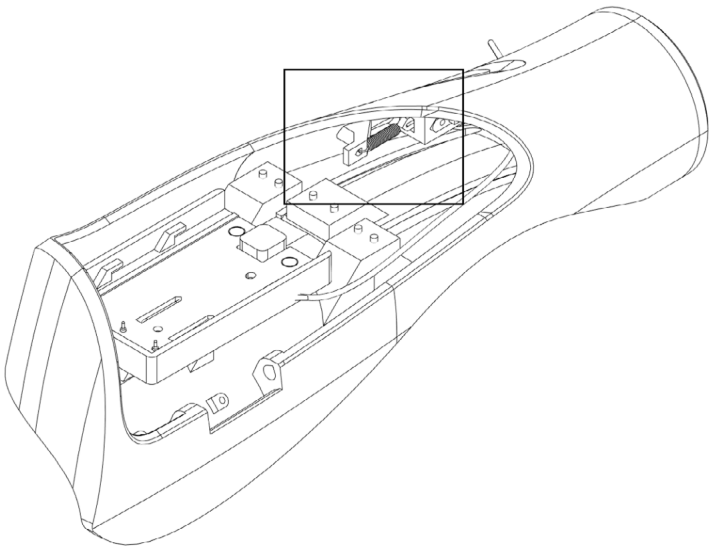


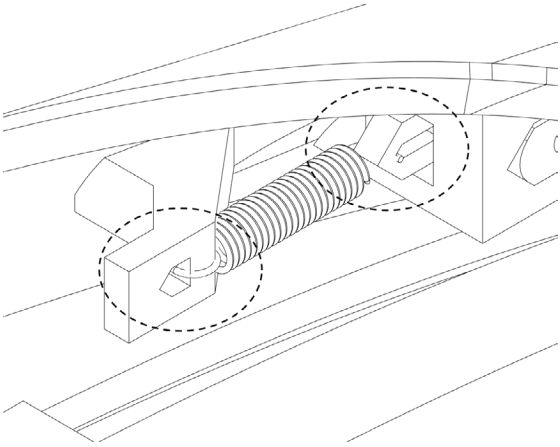
[7] Insert R-SO-05 into R-SO-02 as shown in the figure.



(8) Spread both ends of the tension spring as shown in the diagram below and install it.

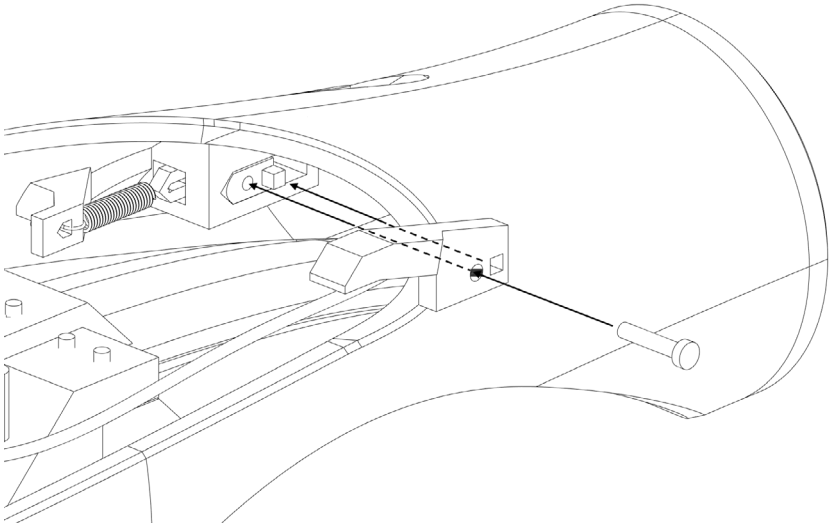




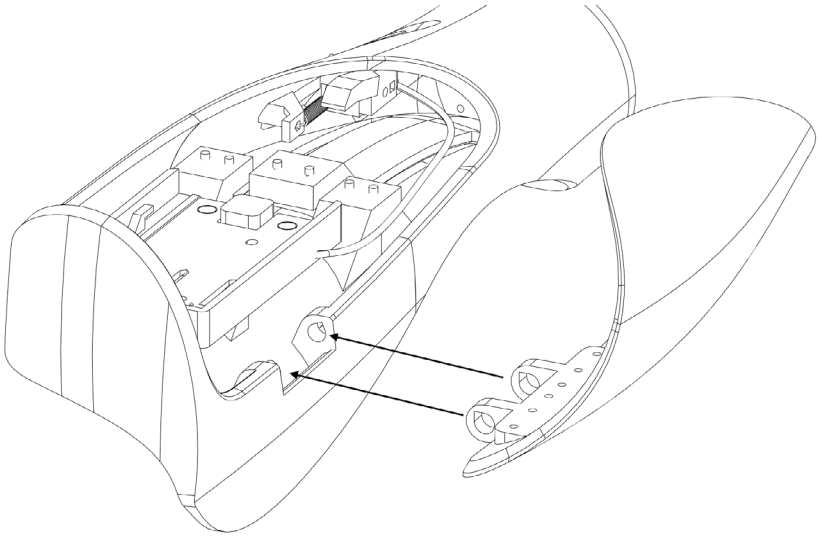


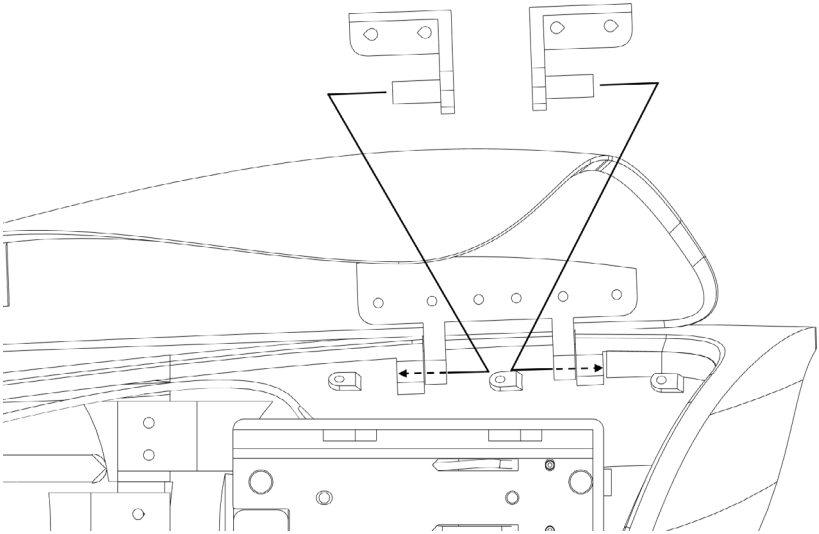
Enlarged view

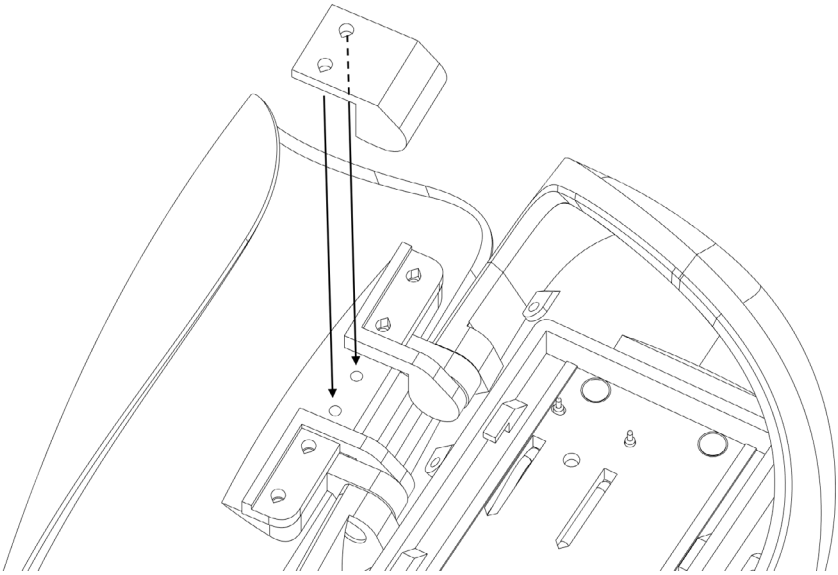
[9] Use screws (M2L10) to secure R-SO-06 as shown in the illustration.

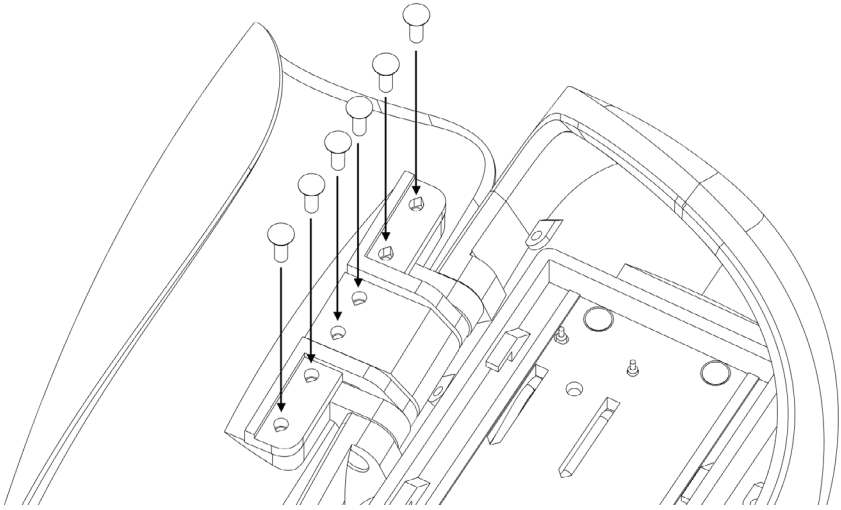


[1 0] Pass R-SO-08 and R-SO-09 through R-SO-01 and R-SO-07, and secure with six screws (M2L6).

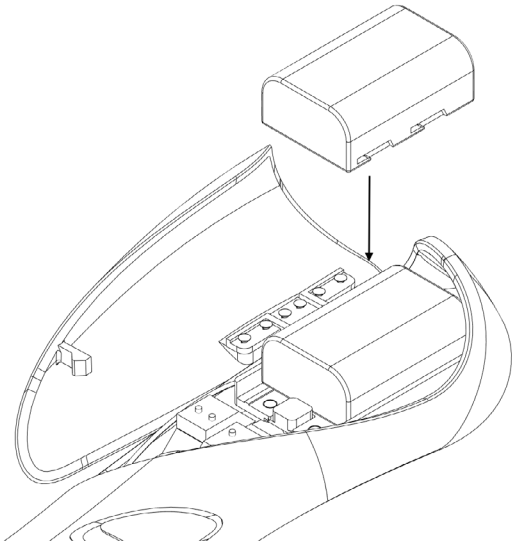




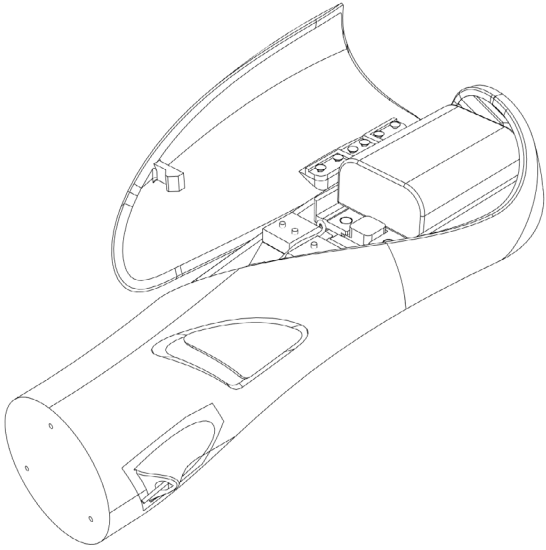




[1 1] Install the battery.



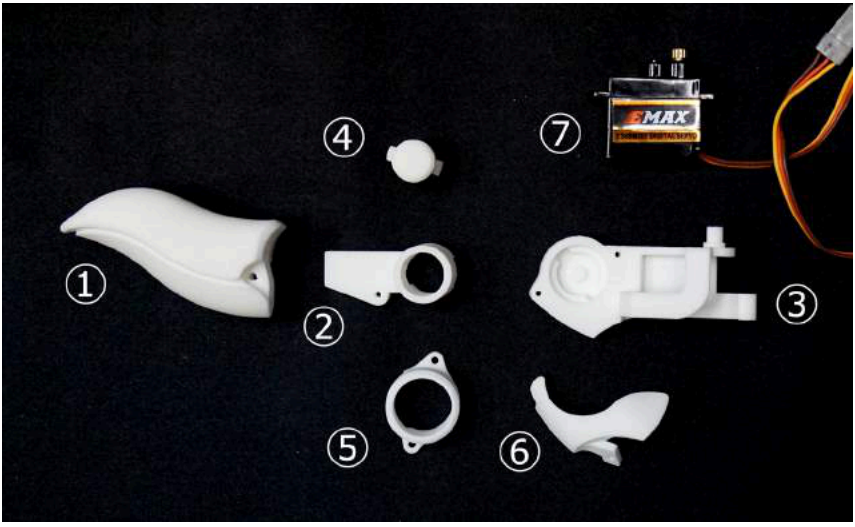
[1 2] This completes the socket assembly.



Assembling
07 Thumb assembly

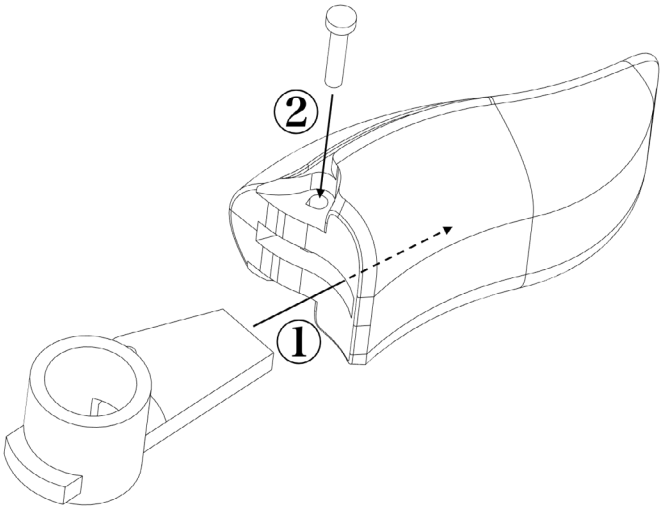
-Parts used-

1. R-T-01
2. R-T-02
3. R-T-03
4. R-T-04
5. R-T-05
6. R-T-06
7. Small servo motor



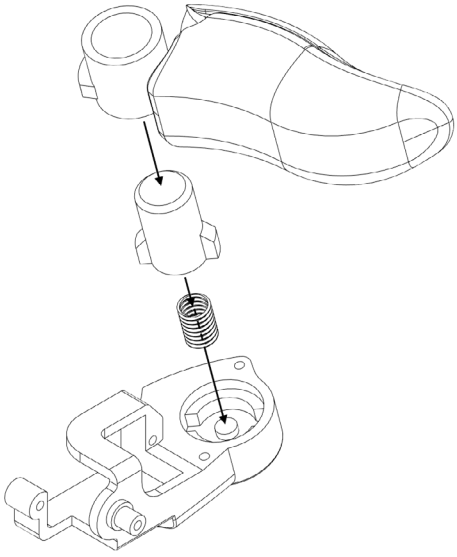
-General-purpose parts used-

(1) Pass R-T-02 through R-T-01 and secure it with a screw (M2L10).

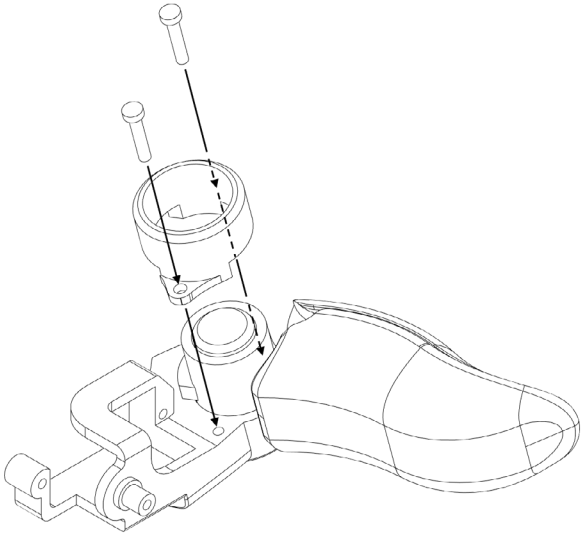


(2) Place the coil spring on R-T-03 as shown in the diagram below, align R-T-04 with the groove, and place it over the coil spring.

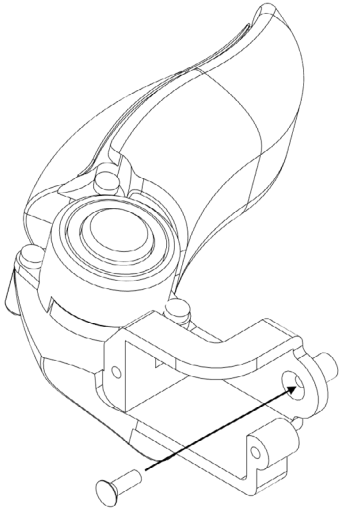
Pass it through R-T-02.



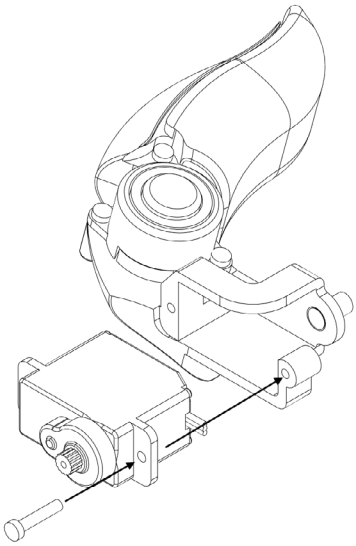
(3) Place R-T-05 over the cover and secure with two screws (M2L10).



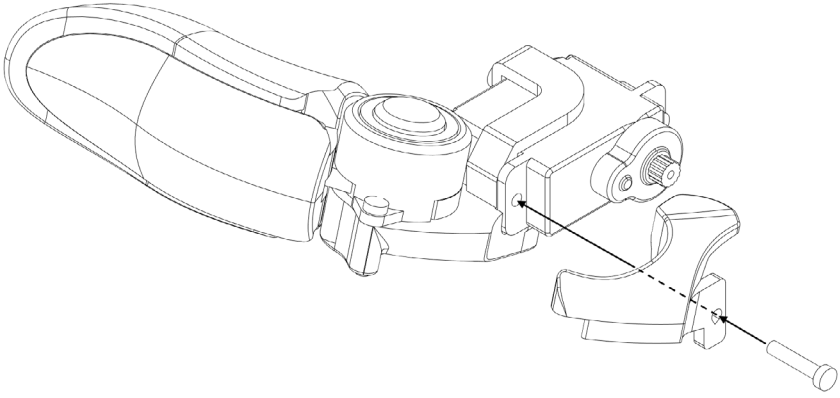
(4) Attach the screw (M2L6) to R-T-03 as shown in the diagram below.



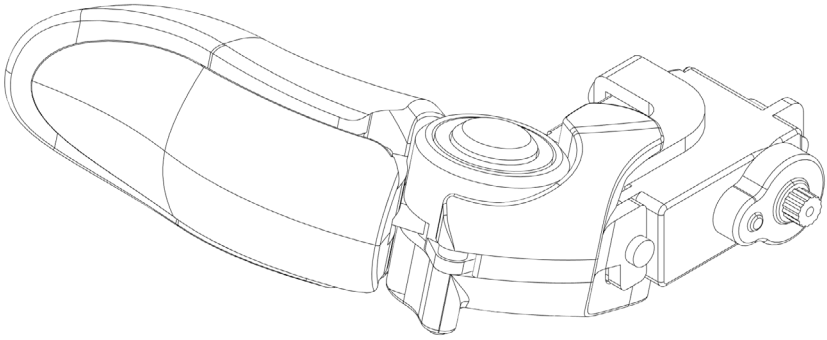
[5] Secure the small servo motor with screws (M2L10).



[6] Then, secure R-T-06 with the screw (M2L10).



[7] This completes the assembly of the thumb.



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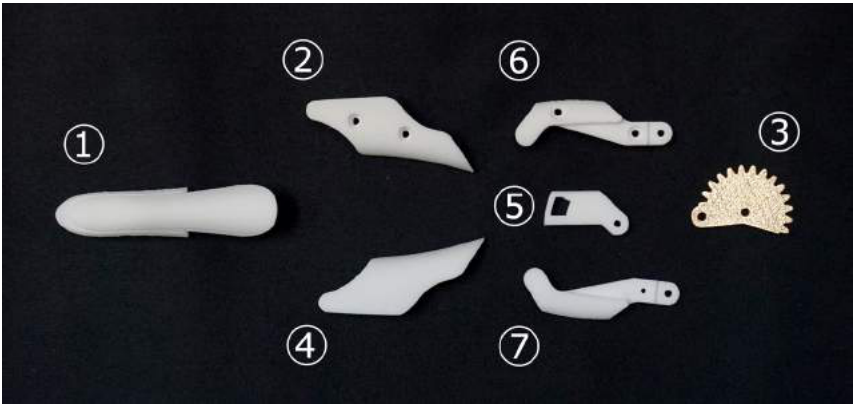
Assembling
08 Index finger assembly

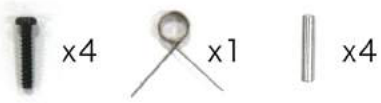
-Parts used-

1. R-I-01
2. R-I-02
3. R-I-03
4. R-I-04
5. R-I-05
6. R-I-06
7. R-I-07



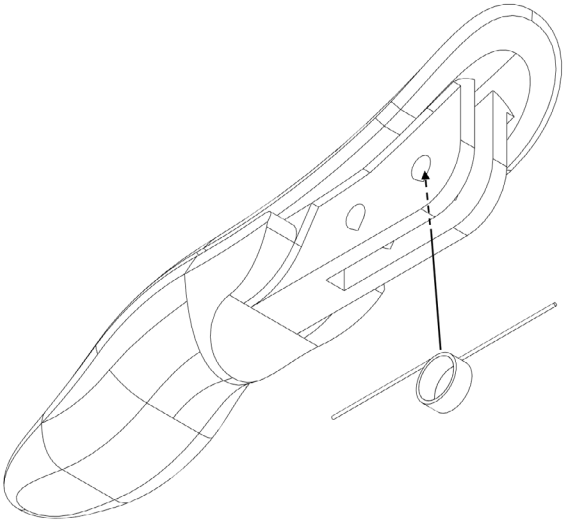
*The R-I-03 in the photo is 3D printed using metal material to increase strength. Although it is less strong and durable, you can also use resin material like other parts.

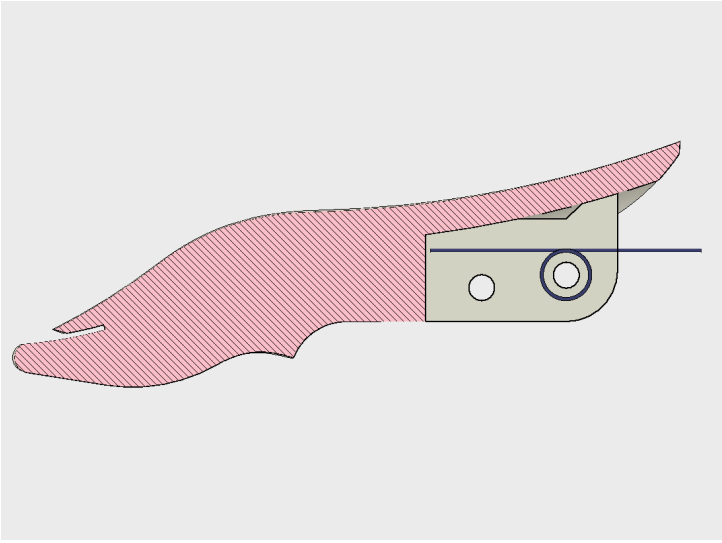
-General-purpose parts used-



80

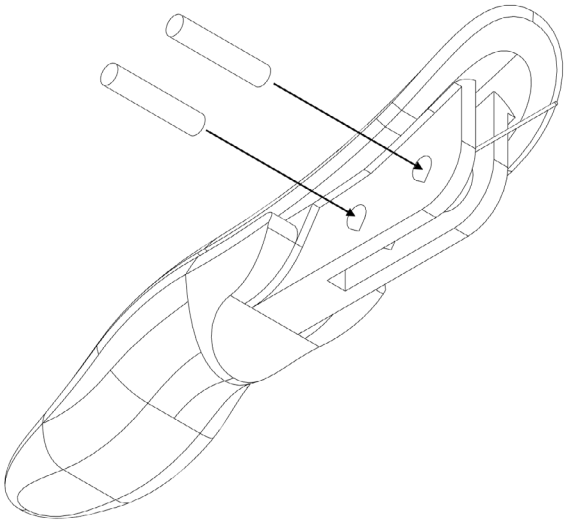
(1) Insert the torsion spring into R-I-01 as shown in the diagram below.





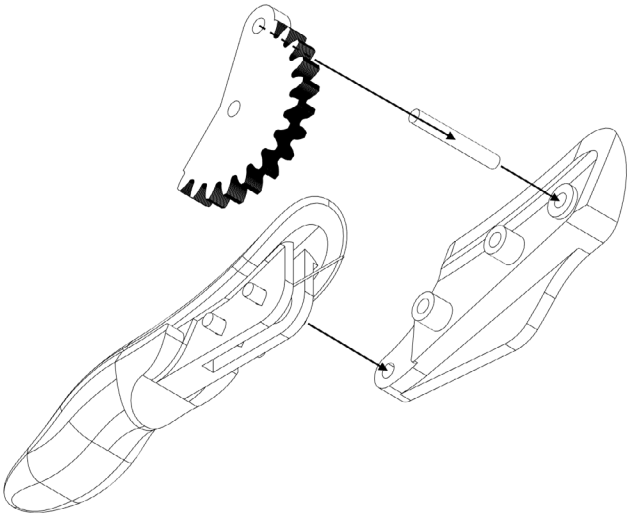
cross section

[2] Insert two shafts (10 mm) into the holes as shown in the figure below.At this time, be careful to make sure that the shaft on the right side of the figure passes through the circle of the torsion spring.

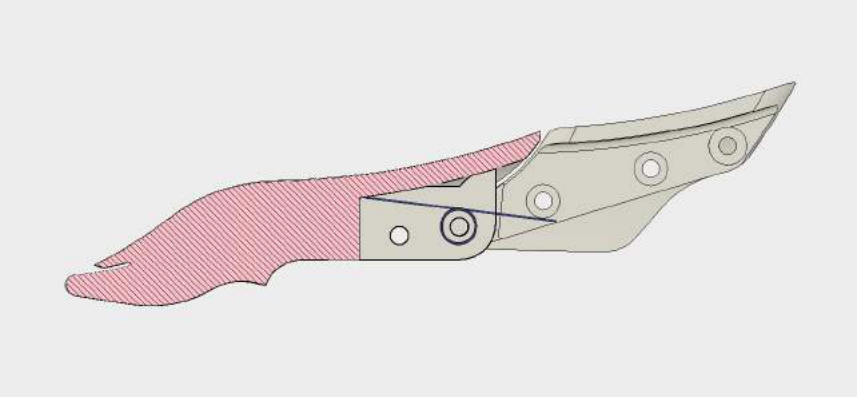


(3) Install R-I-02 and R-I-03. Install R-I-03 by passing the shaft (10 mm) through it as shown in the illustration.

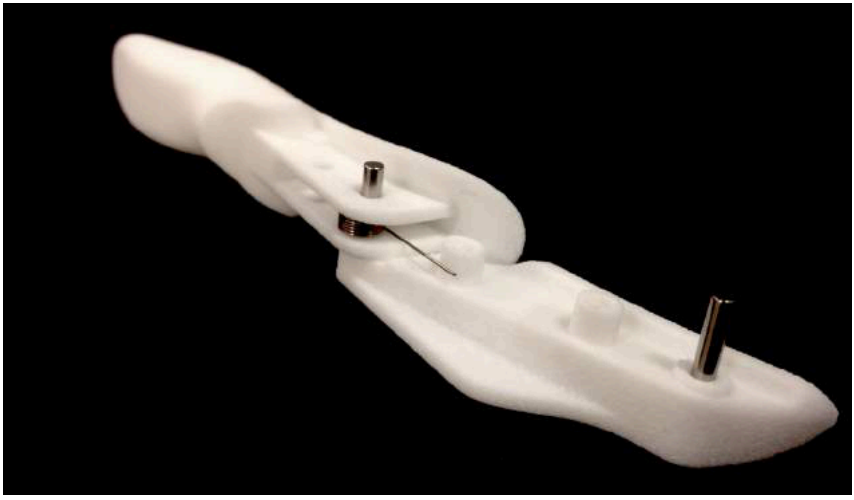
*Please pay attention to the position of the torsion spring at this time.



82

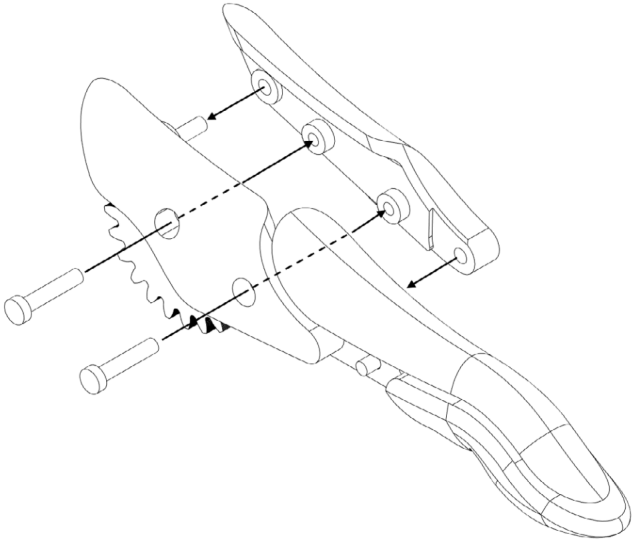


Torsion spring position cross section

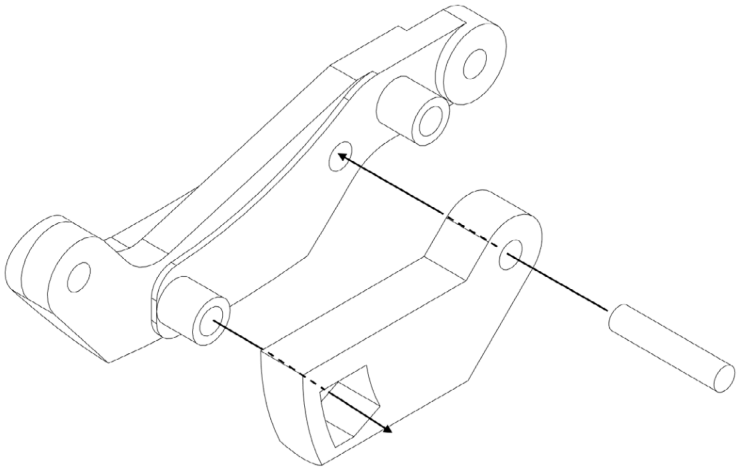


Torsion spring position photo

(4) Place R-I-04 over the cover and secure it in place with two screws (M2L10).

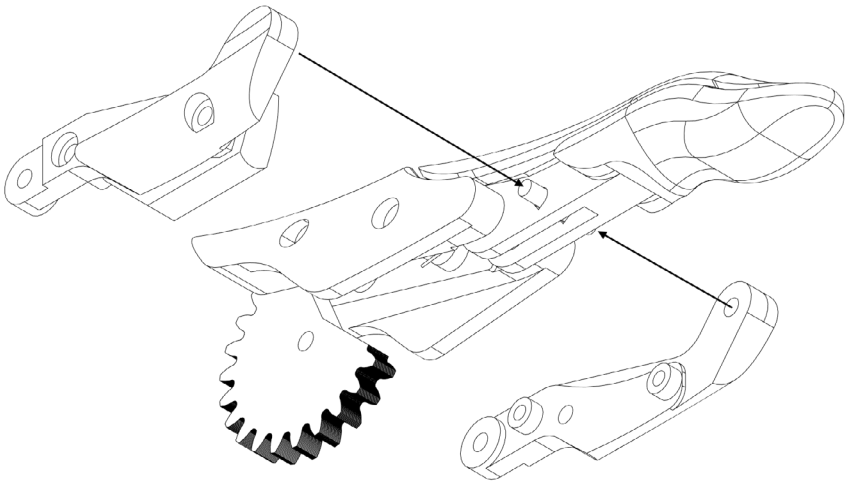


(5) After passing the shaft (10 mm) through R-I-05, attach it to R-I-06.

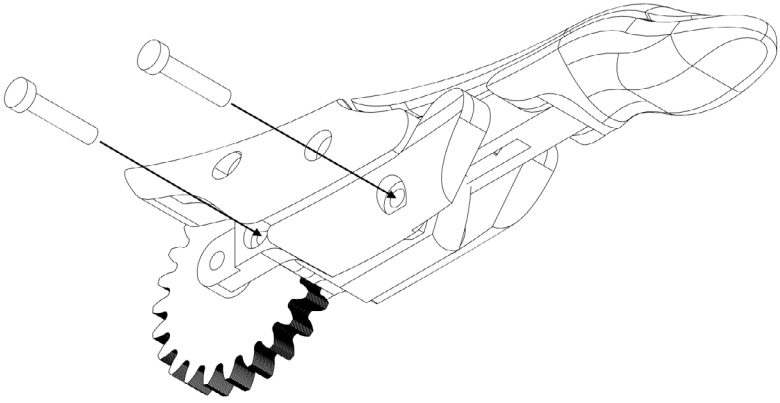


[6] Using R-I-07 and the parts assembled in [5], sandwich the parts assembled in [4] from both sides and assemble them.

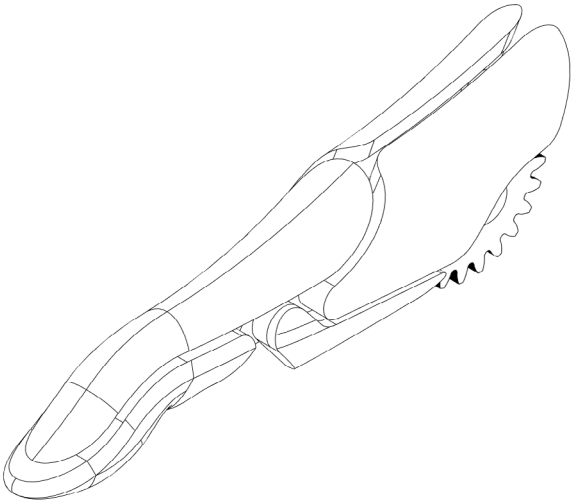
vinegar.



Secure the parts assembled in steps [7] and [6] in two places with screws (M2L10) as shown in the illustration.



[8] The index finger is now complete.

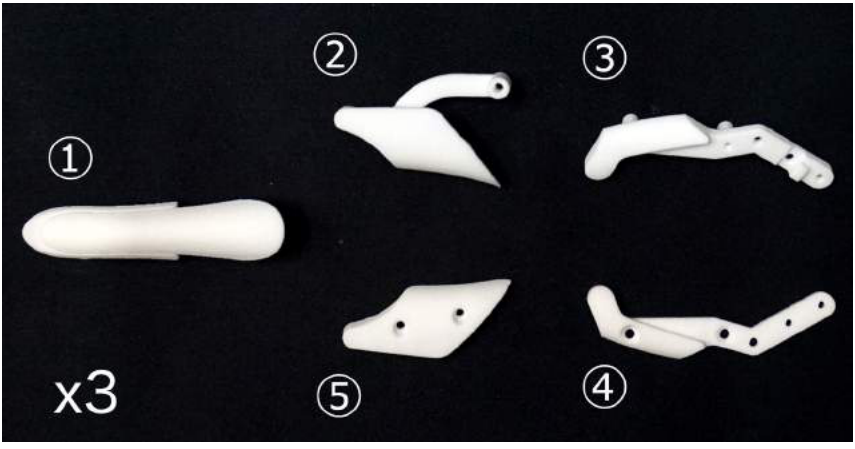


Assembling
09 Three-finger assembly

* Assemble the middle finger, ring finger, and little finger. These use the same parts.

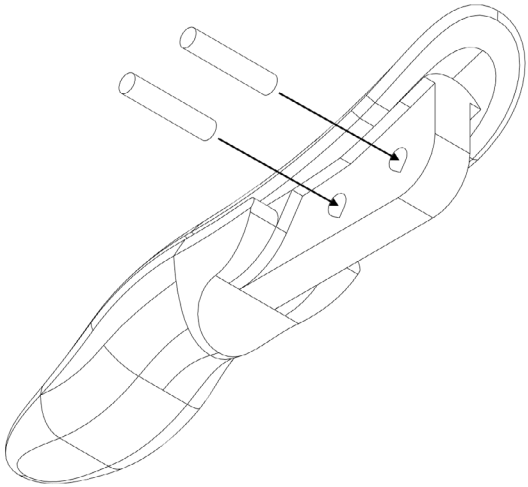
-Parts used-

1. R-O-01
2. R-O-02
3. R-O-03
4. R-O-04
5. R-O-05

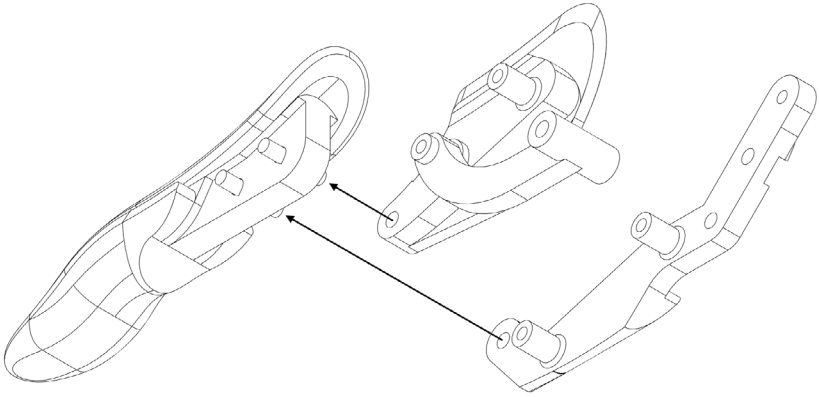


-General-purpose parts used-

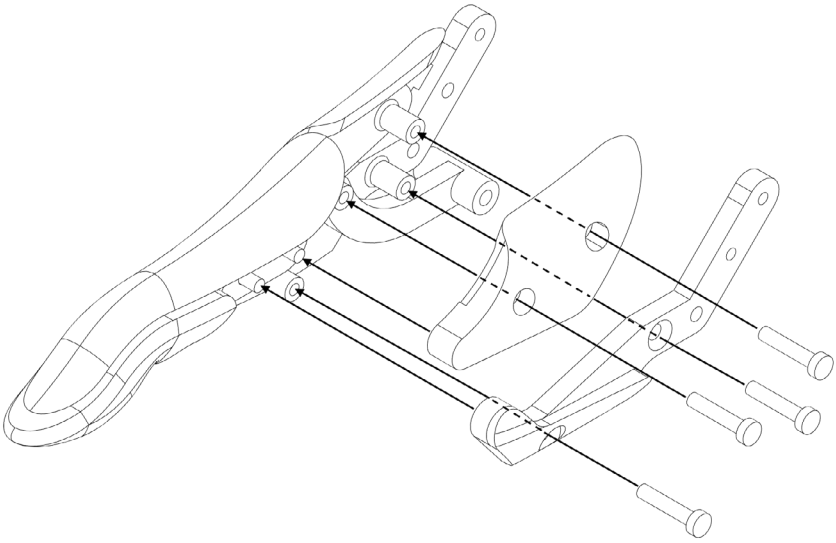
[1] Insert two shafts (10 mm) into R-O-01.

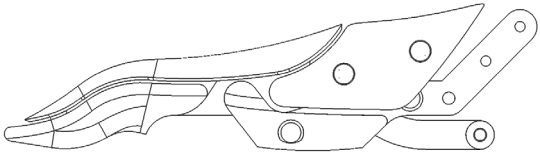


(2) Attach R-O-02 and R-O-03 as shown in the diagram below.

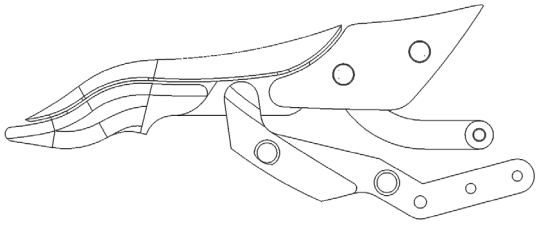


[3] Secure R-O-04 and R-O-05 together with four screws (M2L10).



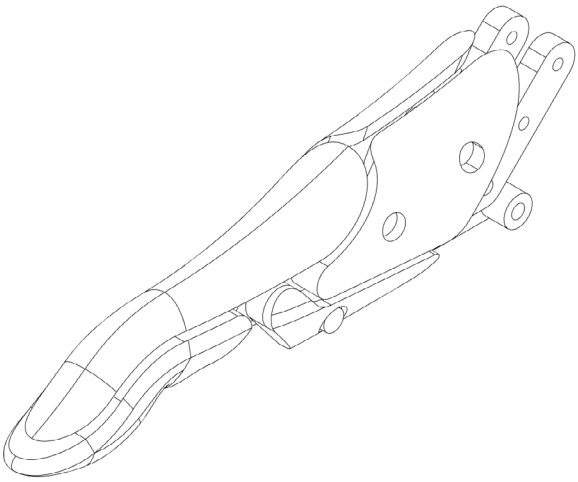


*Good example



*Bad example

[4] Assemble three of the same pieces. This completes the middle finger, ring finger, and little finger.



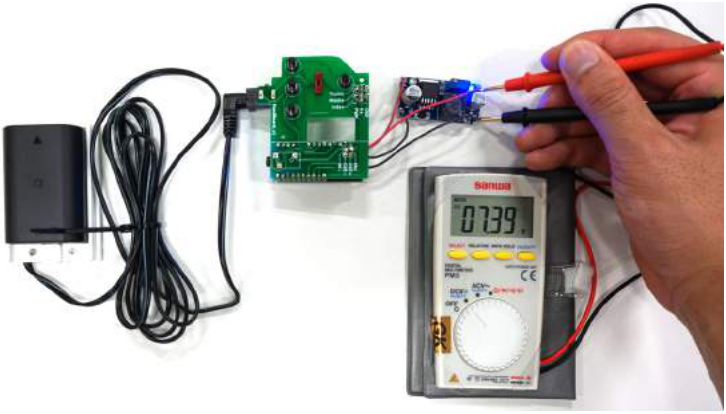
Assembling

10 Adjusting the voltage and setting the initial position of the servo motor

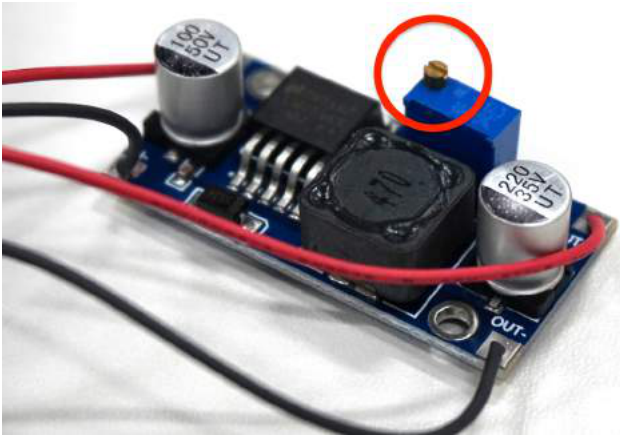
Before assembling the hand, set the servo motor angle. When using the HACKberry hand board Mk2

The combinations [1] and [2] are not necessary.

(1) Insert the DC plug into the HACKberry hand board Mk1 and flip the slide switch up to turn it on. Measure the voltage by placing a tester between the OUT+ and OUT- terminals of the DC/DC converter. When unadjusted, the battery voltage is output directly.



[2] ThisOutput voltage 5VTo step down the voltage, use a flat-head screwdriver to turn the gold part on the DC/DC converter counterclockwise. After turning it five or six times, the voltage will gradually decrease. Once you have finished adjusting it, turn off the power.



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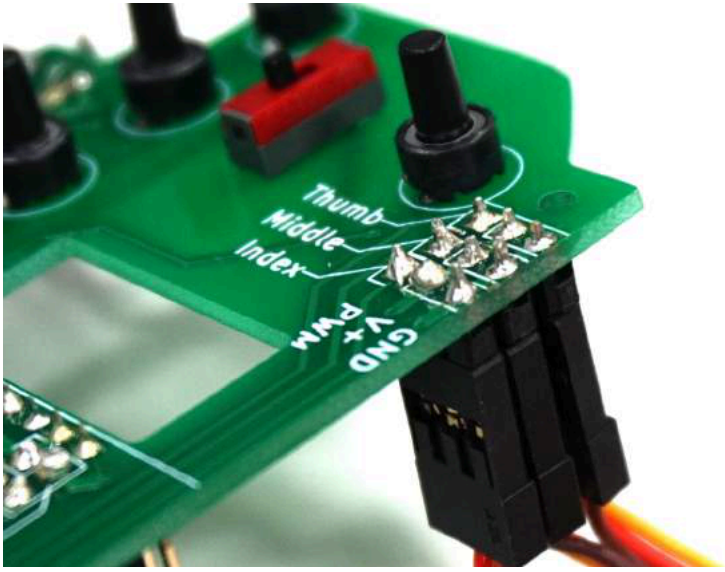
[3] Insert the servo motor connector into the pin header on the board.Make sure the black (or brown) cable is on the outside.Insert the large servo motor connector into the pin header labeled Index, and the small servo motor connector into the pin headers labeled Middle and Thumb.

For the Mk1 board, black (or brown) connects to GND, red to V+, and white (or yellow) connects to PWM. For the Mk2 board, black (or brown) connects to GND, red to VCC, and white (or yellow) connects to SIG.



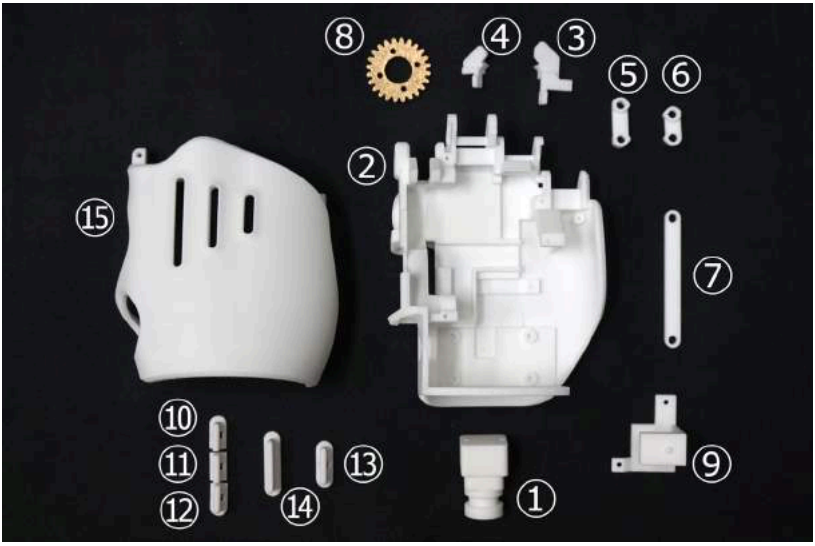
After connecting, turn on the power to the board and if the motor angle changes, the adjustment is complete.

The servo types are the sameMark the Middle and Thumb servo motors., to avoid making mistakes when reconnecting later.

Assembling
11 Assembling the palm

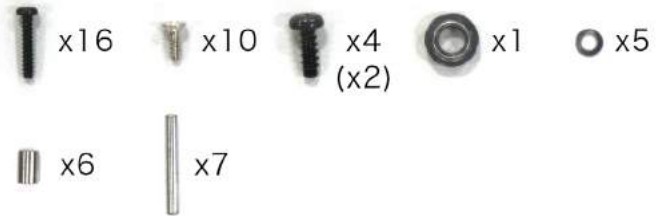
-Parts used-

1. R-H-01	2. R-H-02	3. R-H-03
4. R-H-04	5. R-H-05	6. R-H-06
7. R-H-07	8. R-H-08	9. R-H-09
10.R-H-10	11.R-H-11	12.R-H-12
13.R-H-13	14.R-H-14	15.R-H-15

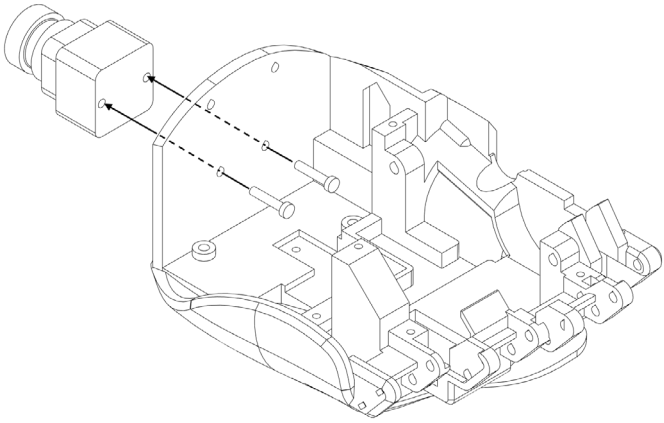


*The R-I-03 in the photo is 3D printed using metal materials to increase strength. Although it is less strong and durable, you can also use resin materials like the other parts.

-General-purpose parts used-



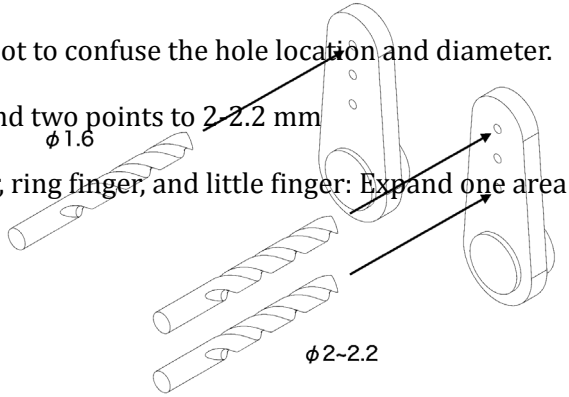
(1) Secure R-H-01 to R-H-02 with two screws (M2L10).



[2] Use a pin vise to enlarge the hole in the servo horn (shaped as shown in the figure below) that comes with the small servo motor.

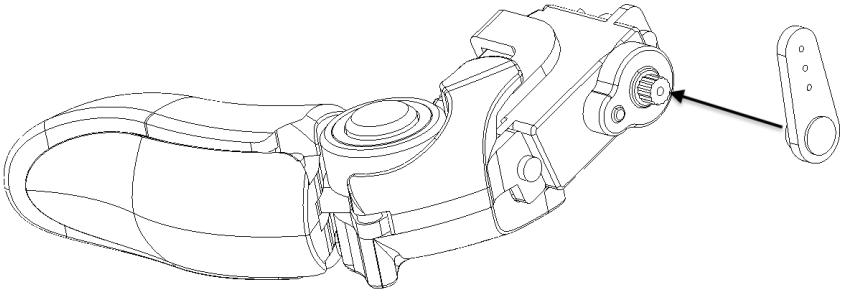
I will.Be careful not to confuse the hole location and diameter.

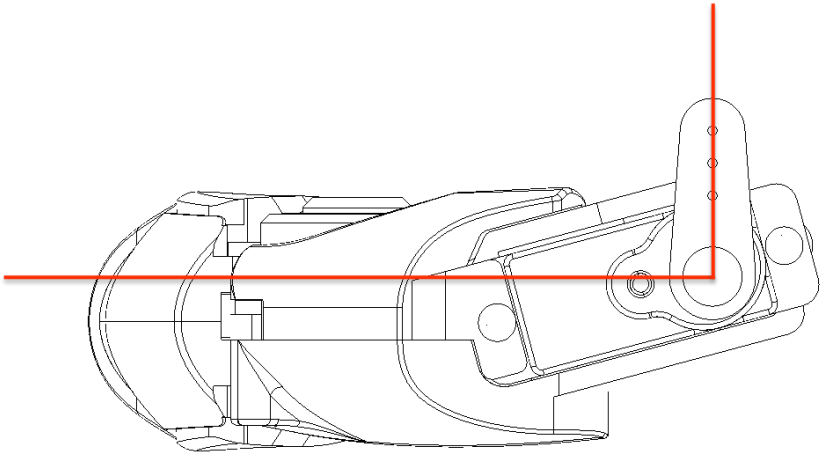
For thumb: Expand two points to 2~2.2 mm



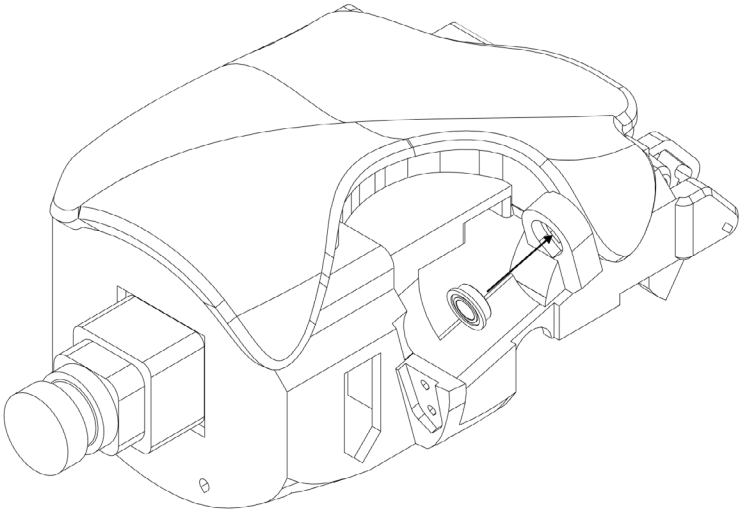
For middle finger, ring finger, and little finger: Expand one area to 1.6 mm

[3] The servo motor shaft is "10 Adjusting the voltage and setting the initial position of the servo motor"(page 92)Attach the thumb unit and servo horn at a right angle, being careful not to rotate them from the position set in step.

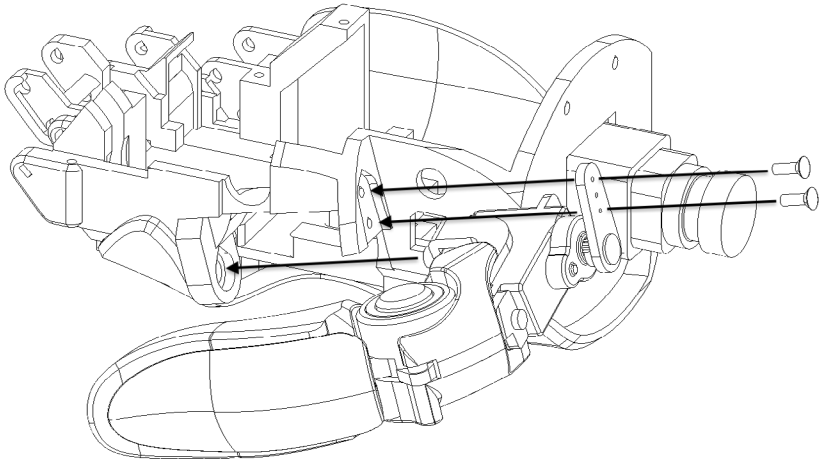




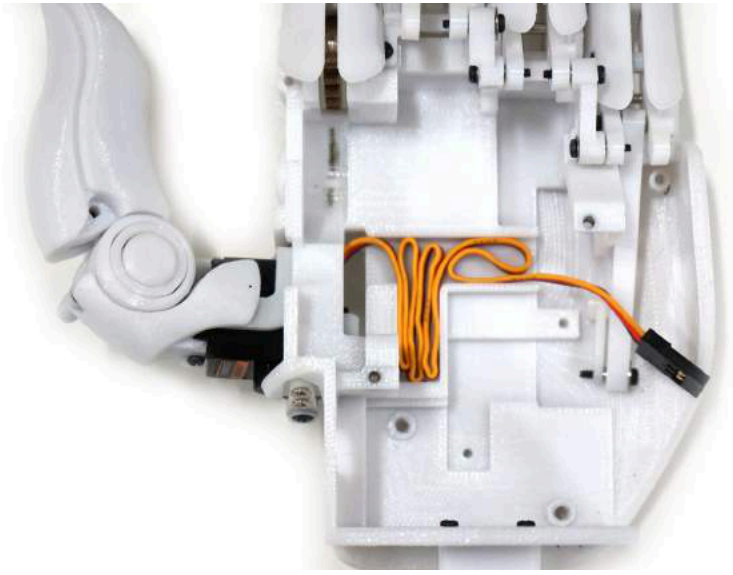
(4) Push the bearing into R-H-02.



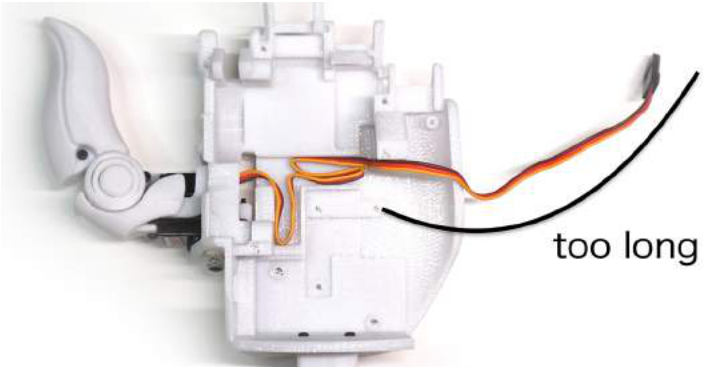
[5] Secure the thumb in place using two screws (M2L6).



[6] Insert the servo motor cable into R-H-02 as shown in the figure below and then fold it.

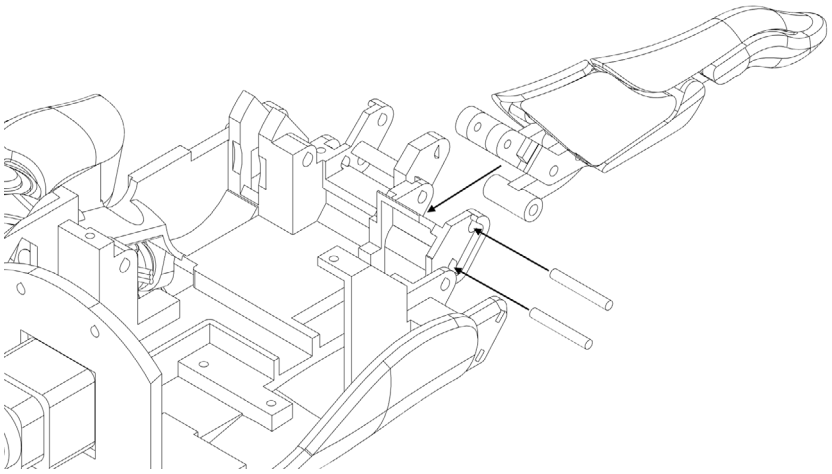


*Good example

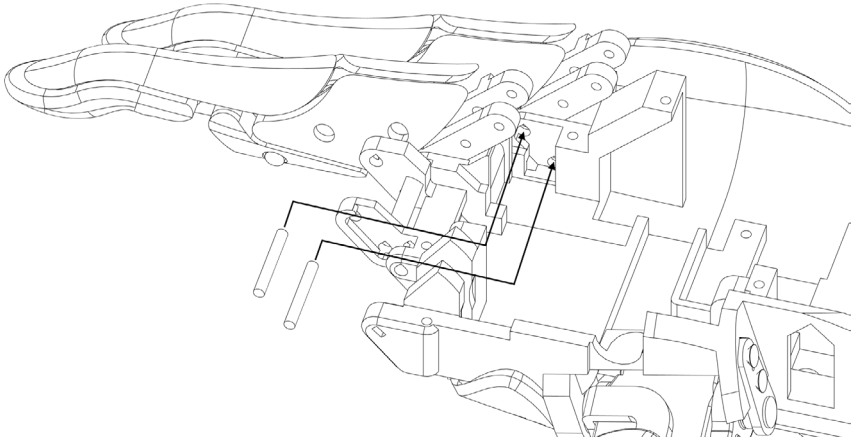


*Bad example

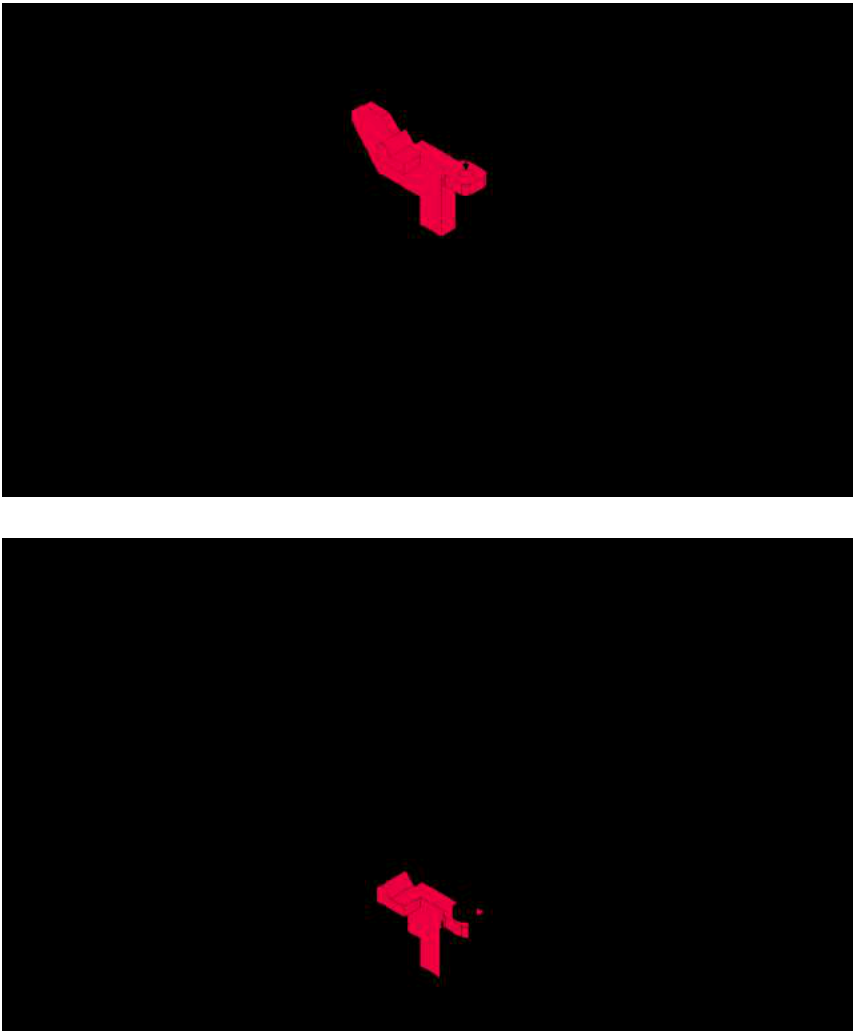
[7] Next, attach the ring finger unit through two shafts (15 mm) as shown in the figure below.



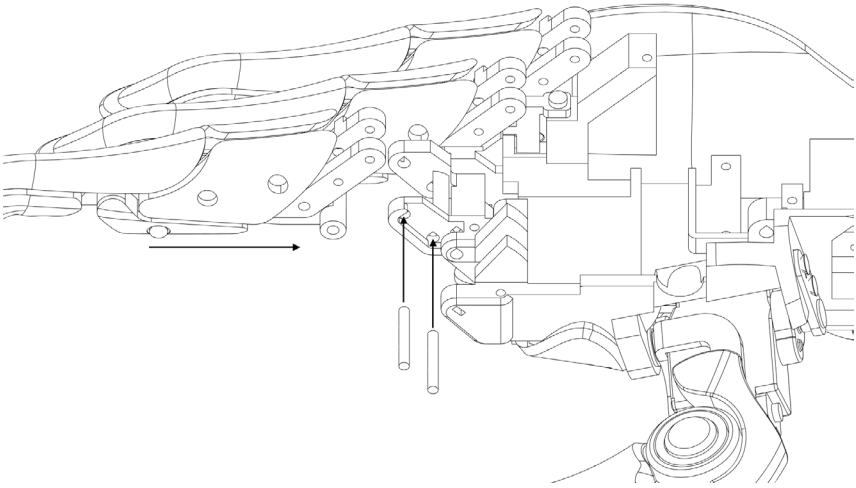
[8] In the same way, attach the little finger unit using two shafts (15 mm).



[9] Attach R-H-03 with screws (M2L10) as shown in the diagram below.

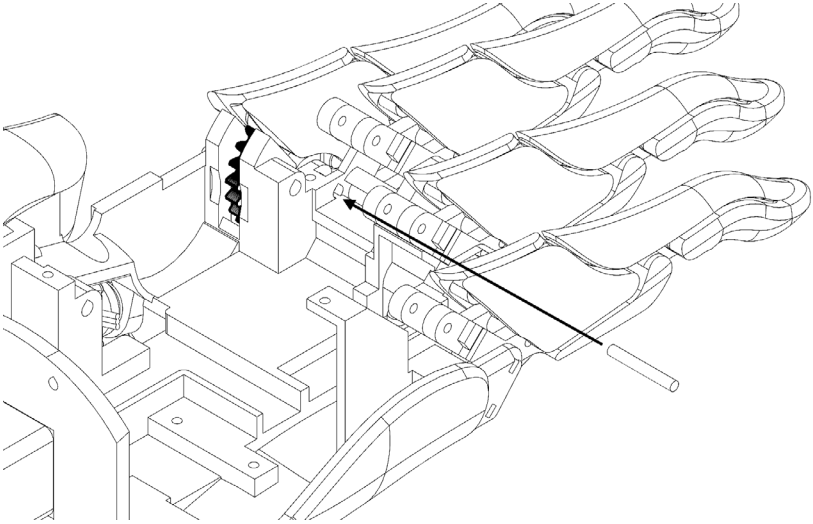


[1 0] Attach the middle finger using two shafts (15 mm) as shown in the figure below.

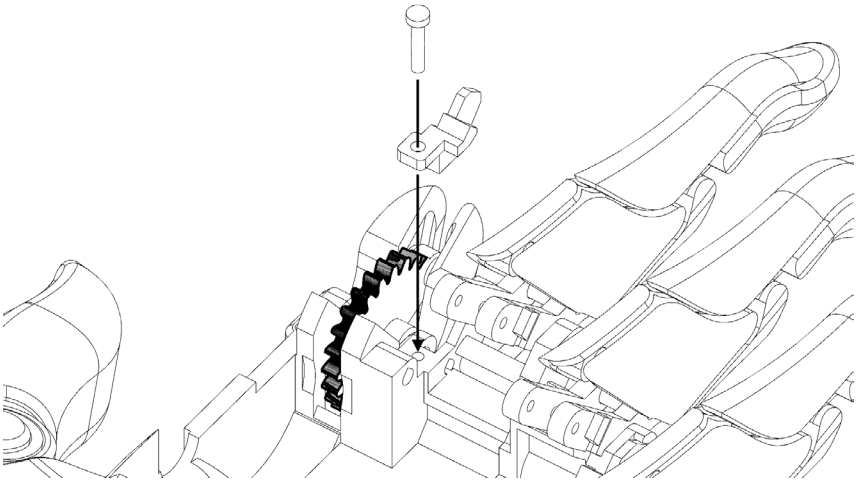


[1 1] Attach the index finger unit using the shaft (15 mm) as shown in the figure below. At this time, make sure the shaft is aligned with the gear.

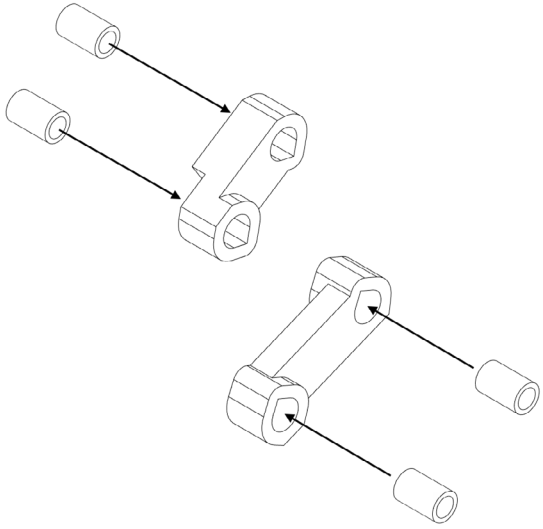
Make sure it goes through the hole.



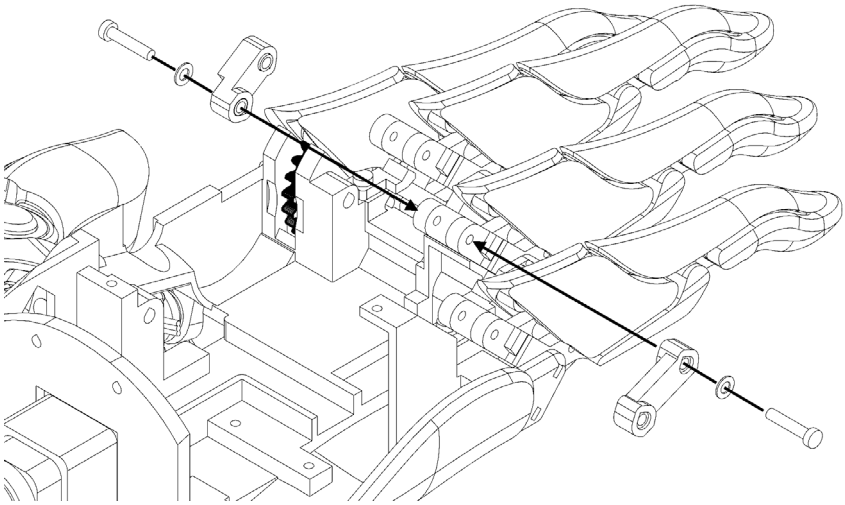
[1 2] Attach R-H-04 using screws (M2L10) as shown in the diagram below.

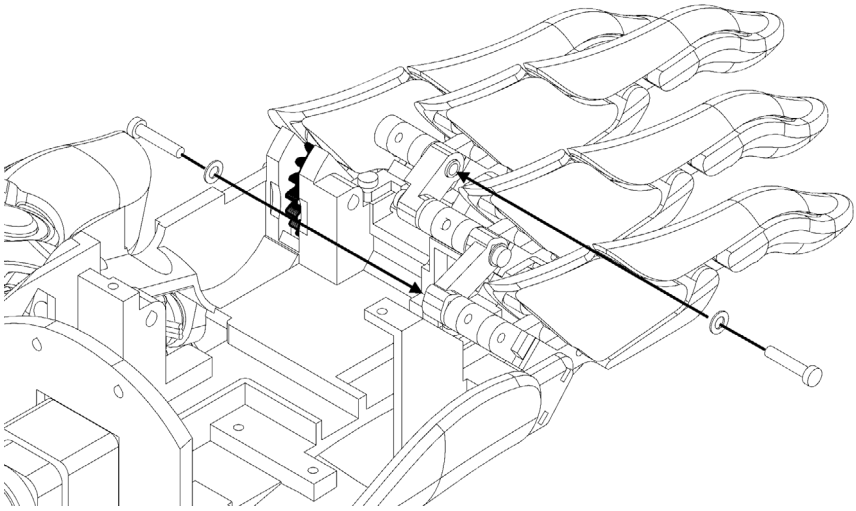


[1 3] Insert two spacers into R-H-05 and R-H-06.

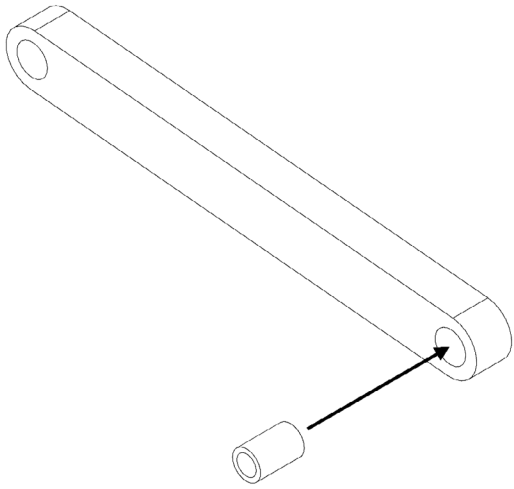


[1 4] R-H-05 and R-H-06 in two locations(4 locations in total)Pass the washer through and secure with a screw (M2L10). *Make sure that R-H-05 (the longer part) is attached between your little finger and ring finger.

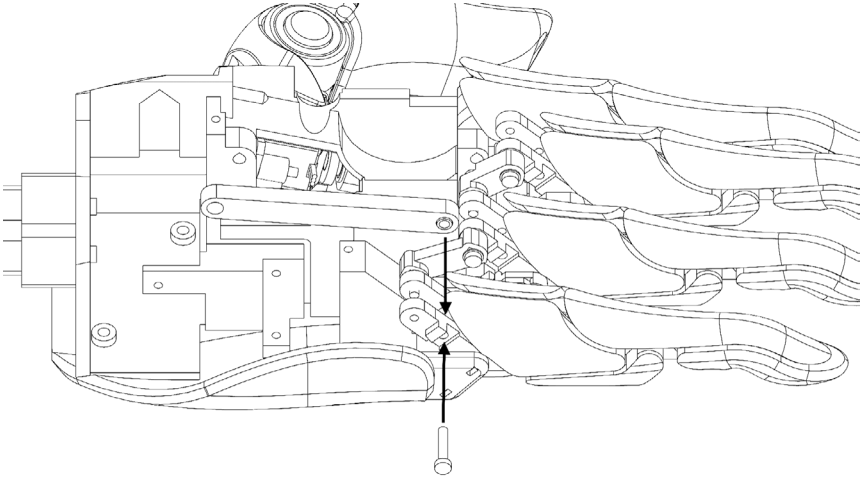




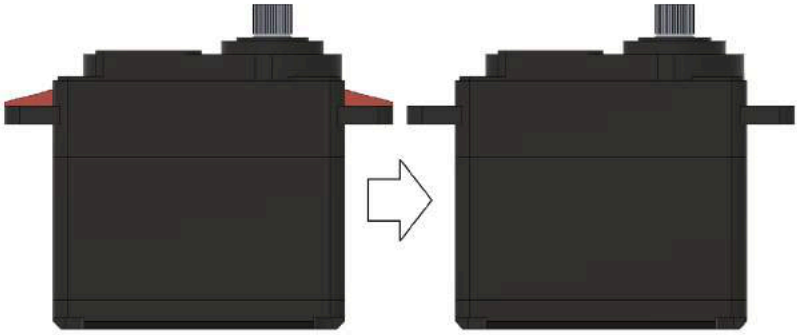
[1 5] Insert the spacer into one of the holes on R-H-07.



[1 6] Attach R-H-07 to the little finger unit using screw M (2L10).



[1 7] Cut out the part of the servo motor shown in red in the diagram below using pliers or similar.

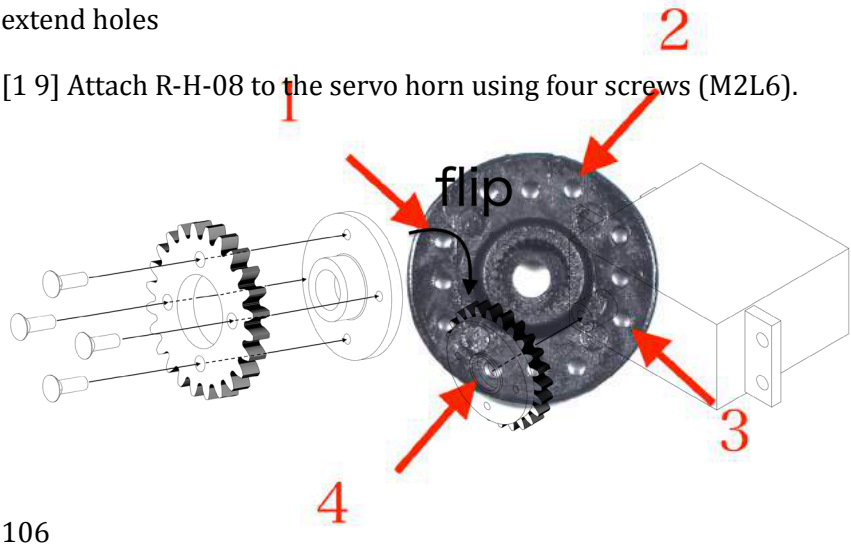


Side view

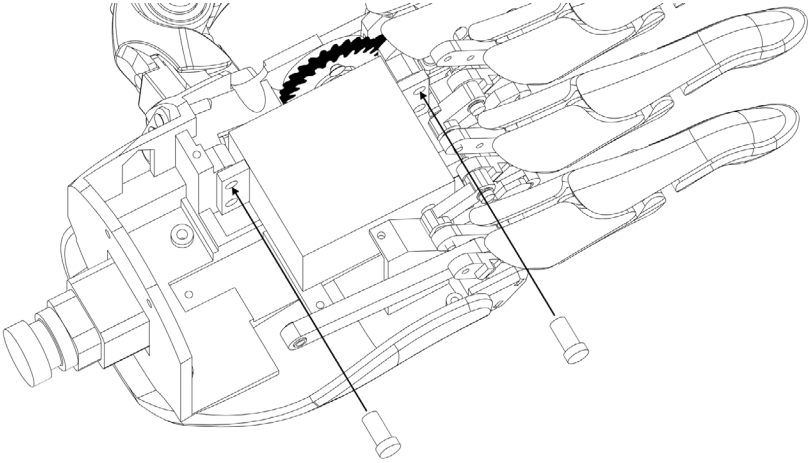
[1 8] Insert the servo horn hole (Right end of the triple hole) in four places and expand it to 1.6 mm using a pin vise.

extend holes

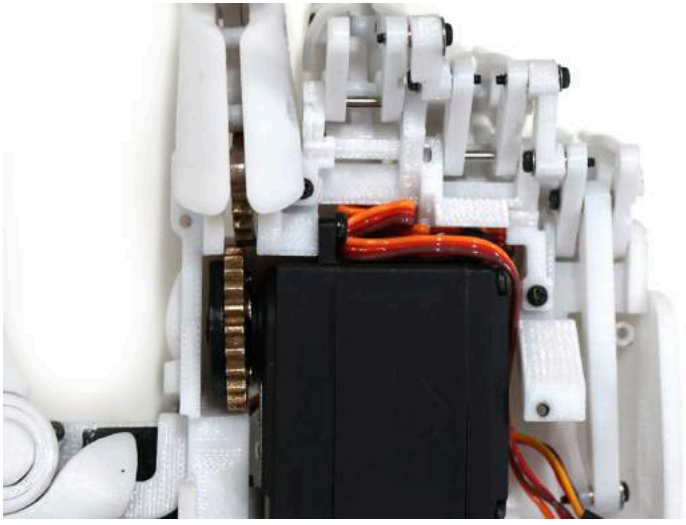
[1 9] Attach R-H-08 to the servo horn using four screws (M2L6).



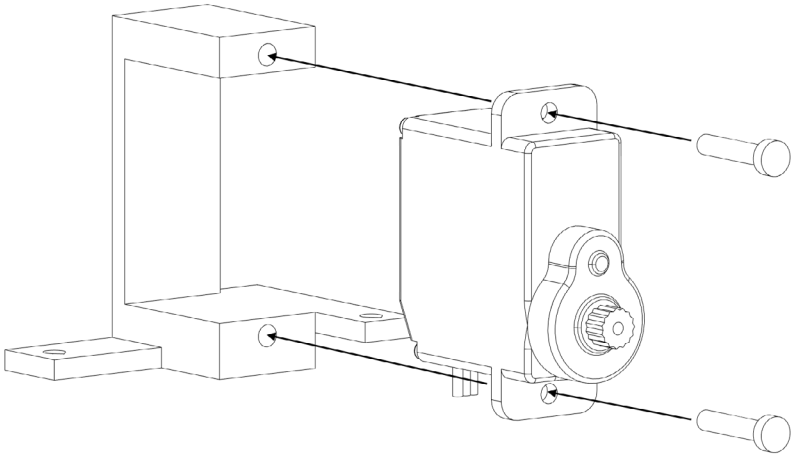
【2 0 】While being careful not to rotate the servo motor shaft, hold the index finger extended.Attach the large servo motor to R-H-02 with two screws (M3L8).



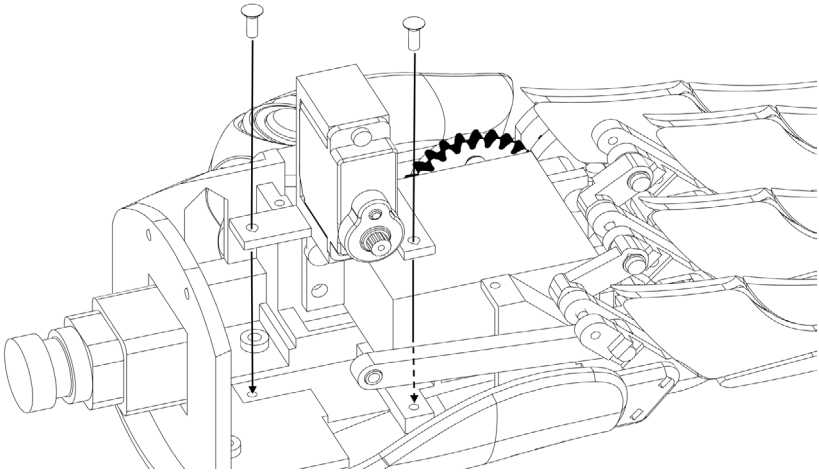
[2 1] Bundle the cables as shown in the figure below.

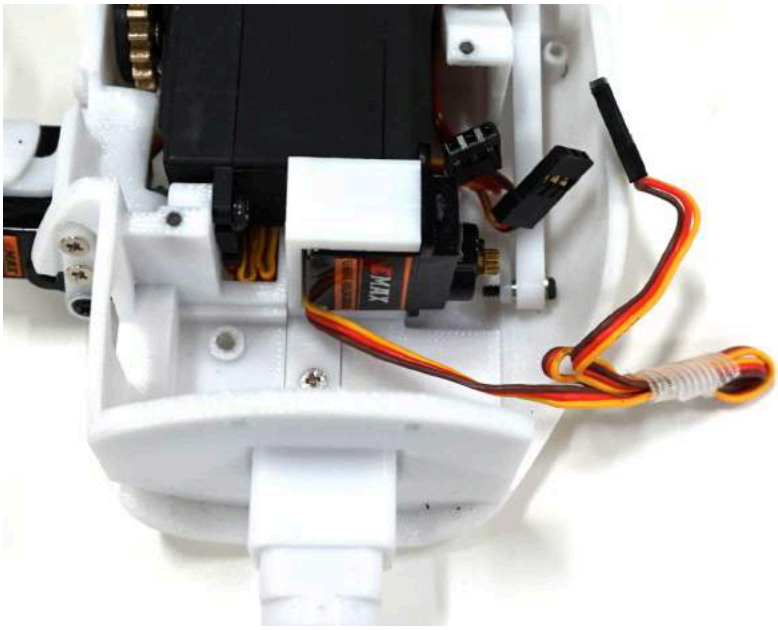


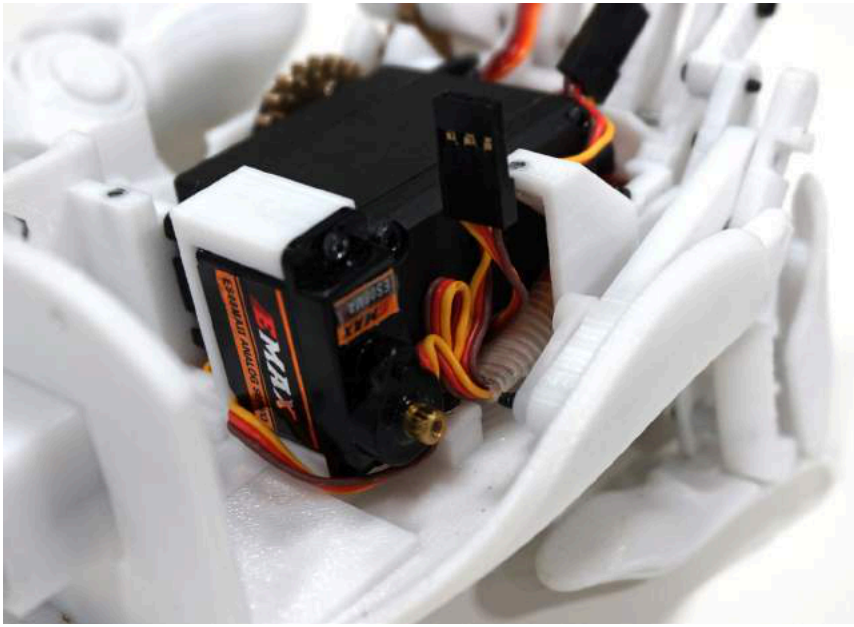
[2 2] Secure the small servo motor to R-H-09 with two screws (M2L10).At this time, be sure to pull the cable out to the left.



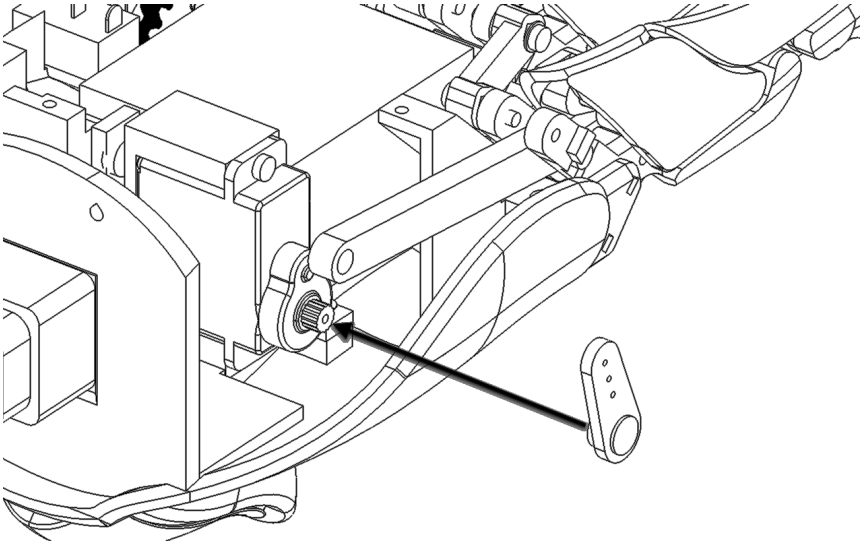
[2 3] Secure this with two screws (M2L6) and route the cable as shown in the second photo.If you accidentally use a screw (M2L10) here, the screw will stick out from your palm, so be careful.

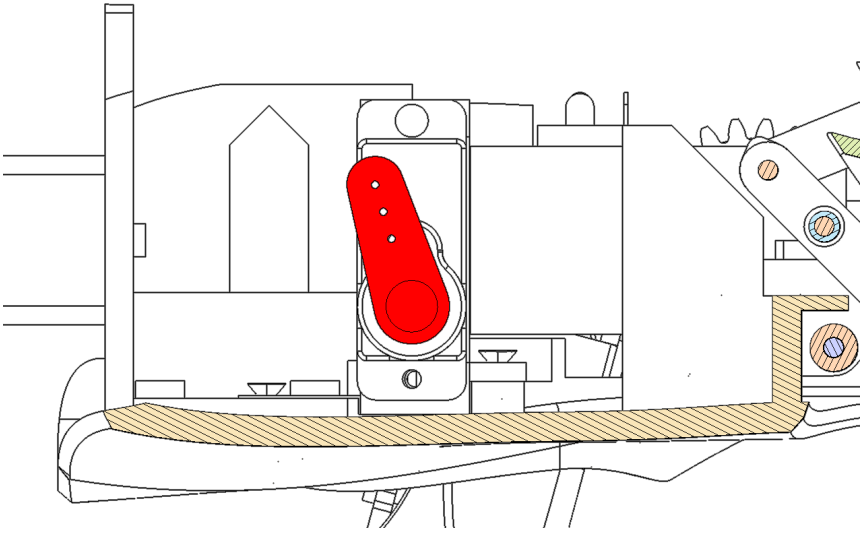




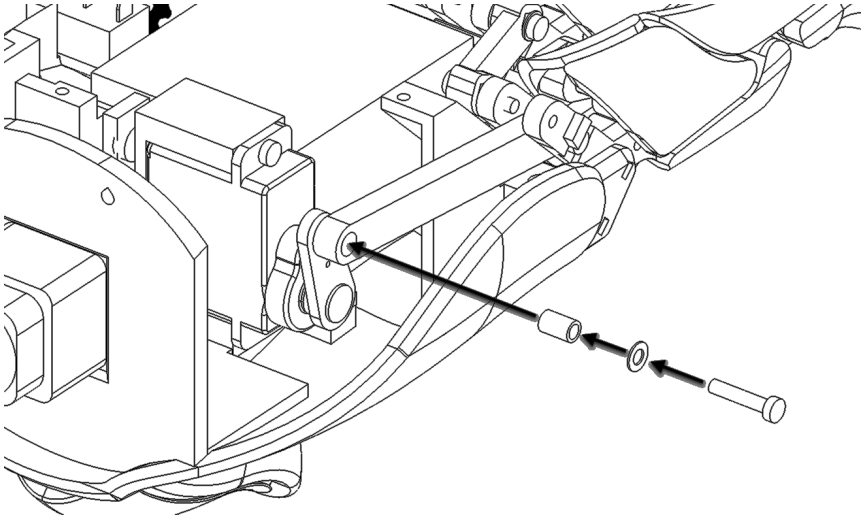

EMAX

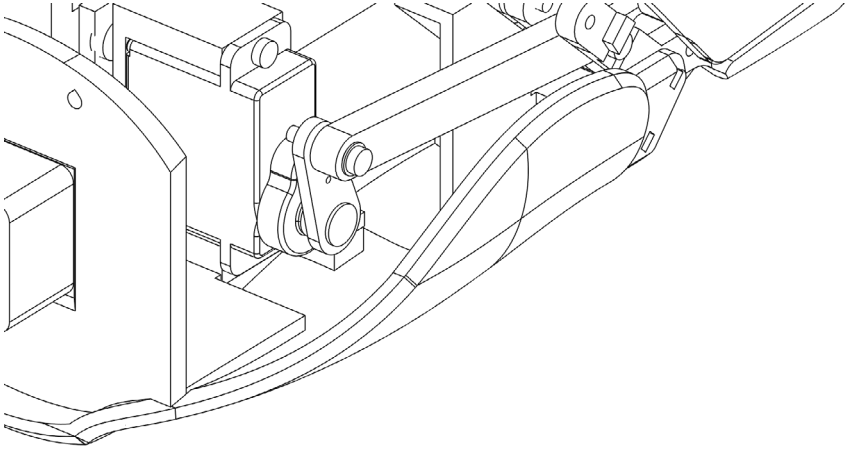
【2 4】The servo motor shaft is "10 Adjusting the voltage and setting the initial position of the servo motor"(page 92)Attach the servo horn to the servo motor at the angle shown in the diagram, being careful not to rotate it from the position set in step.





[2 5] Attach R-H-07 to the servo horn using a washer, spacer, and screw (M2L10).



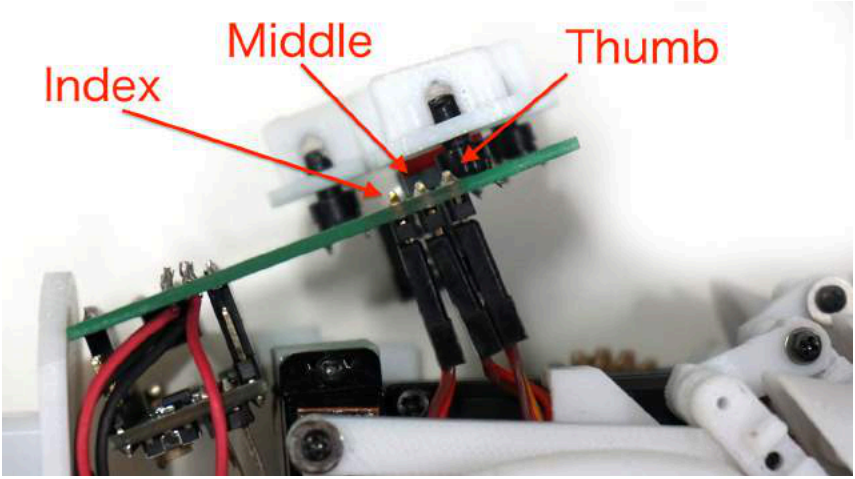


[2 6] Pass the servo motor cable through as shown in the photo.

[2 7] Insert the servo motor connector into the pin header on the board.Make sure the black (or brown) cable is on the outside.Insert the large servo motor connector into the pin header labeled Index, and the small servo motor connector into the pin headers labeled Middle and Thumb.

On the Mk1 board, black (or brown) connects to GND, red to V+, and white (or yellow) to PWM.

For the Mk2 board, black (or brown) connects to GND, red to VCC, and white (or yellow) to SIG.



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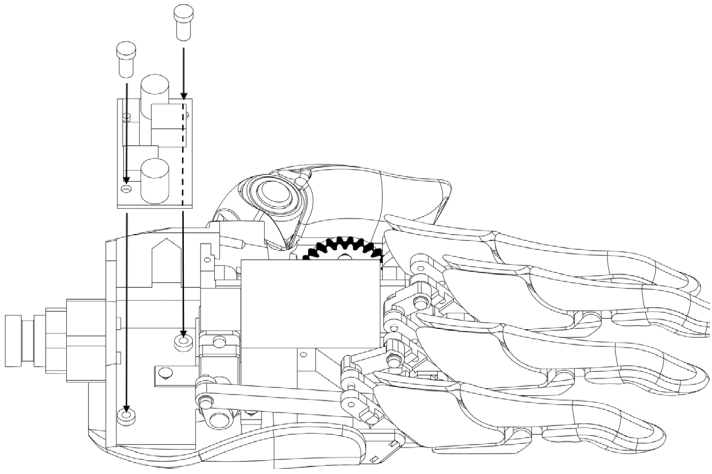
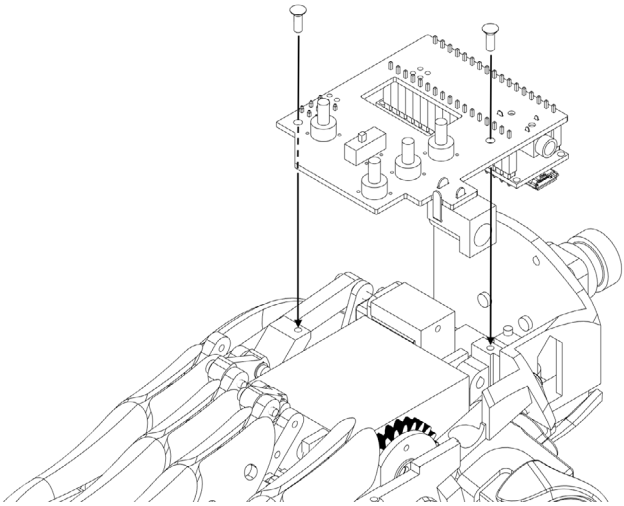
≫Next, install the circuit board.

For Mk1 boards, please refer to [28] and for Mk2 boards, please refer to [33] and onwards.

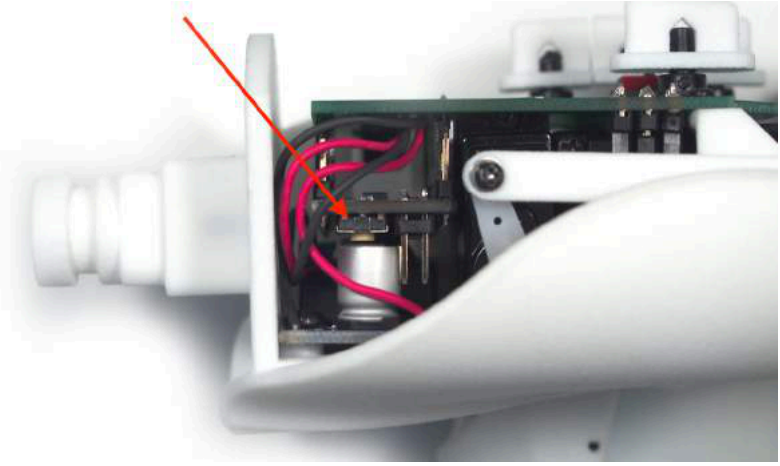
[2 8] Secure the DC-DC converter with two screws (M3L8).



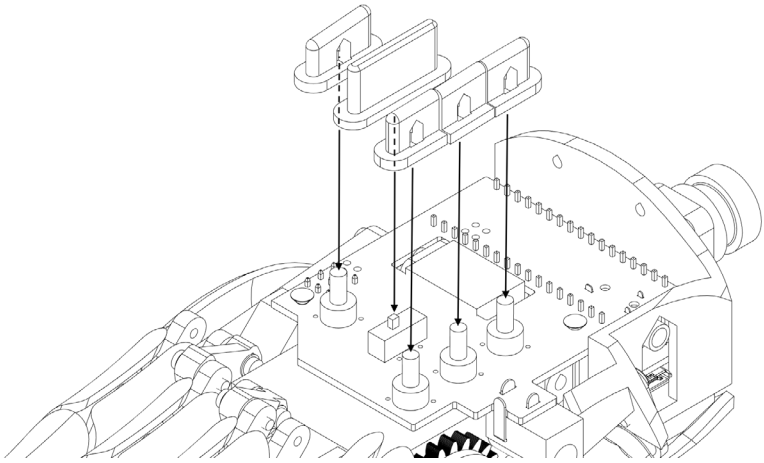
[2 9] Secure the board using two screws (M2L6).At this time, please be careful that if the Arduino Micro is not inserted all the way or the screws for fixing the board are tightened too much, the reset button on the Arduino Micro will be pressed continuously and the HACKberry will not work properly.



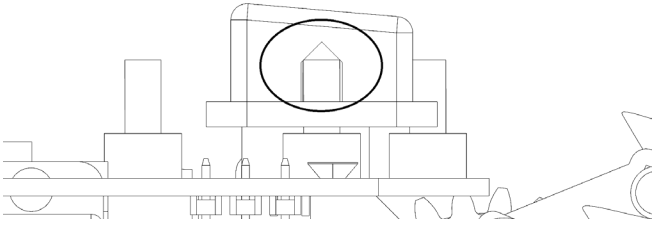
Reset Button



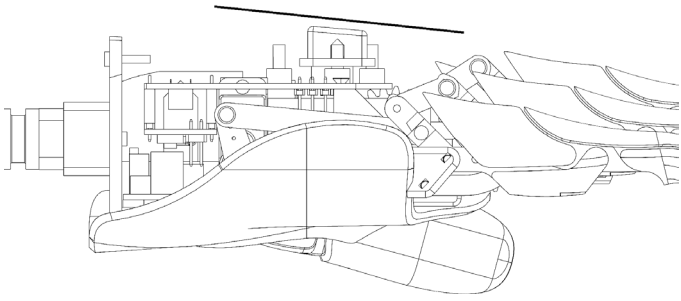
[30] Install R-H-10, R-H-11, R-H-012, R-H-13, and R-H-14 as shown in the diagram below.



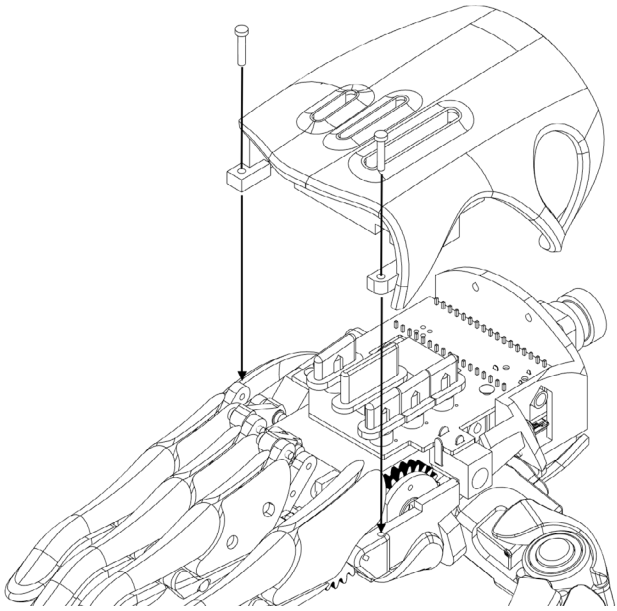
*Make sure to insert all the way in.

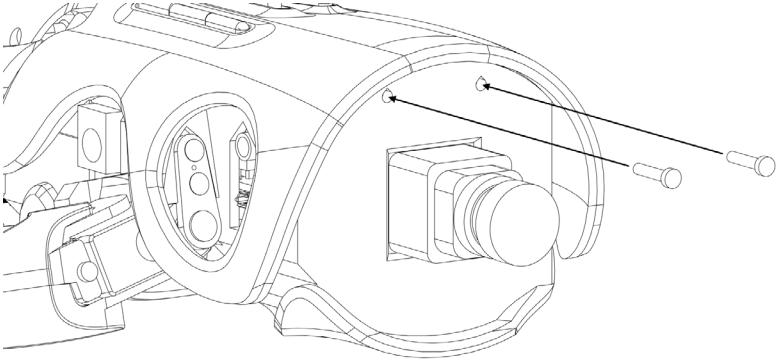


*For R-H-13, make sure the shorter side faces your fingertips.

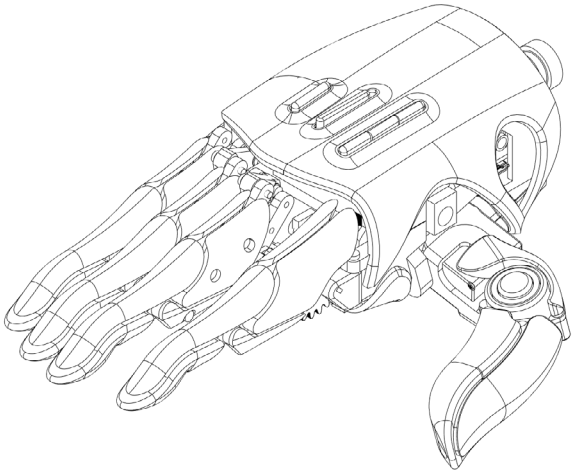


【3 1 】Fix R-H-15 using four screws (M2L10). It will be easier to install if you bend your fingers.



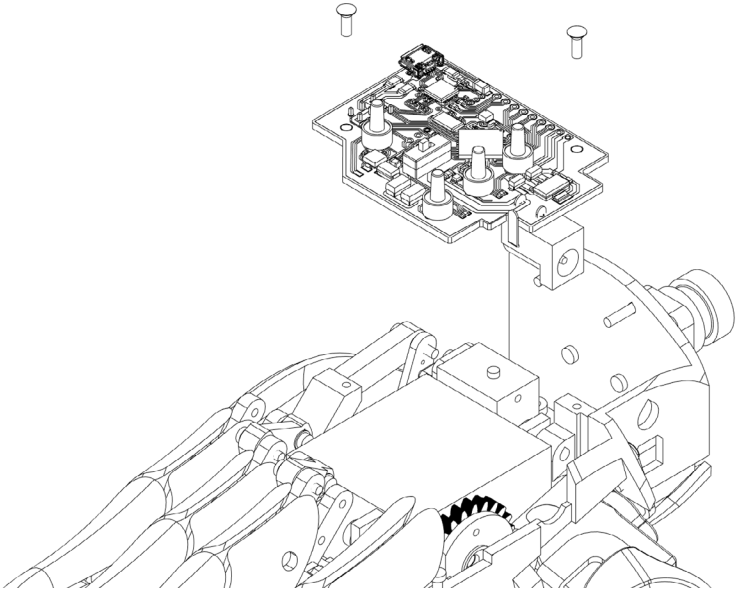


[3 2] This completes the assembly of the palm. Proceed to the next chapter.

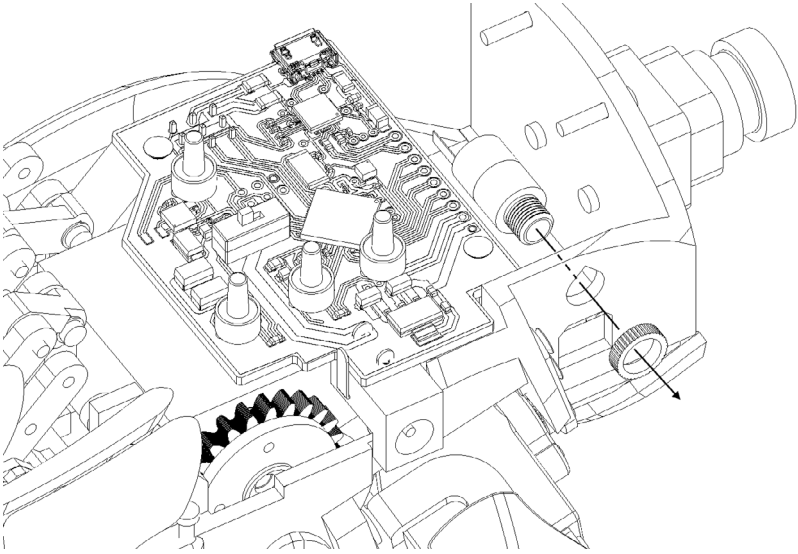


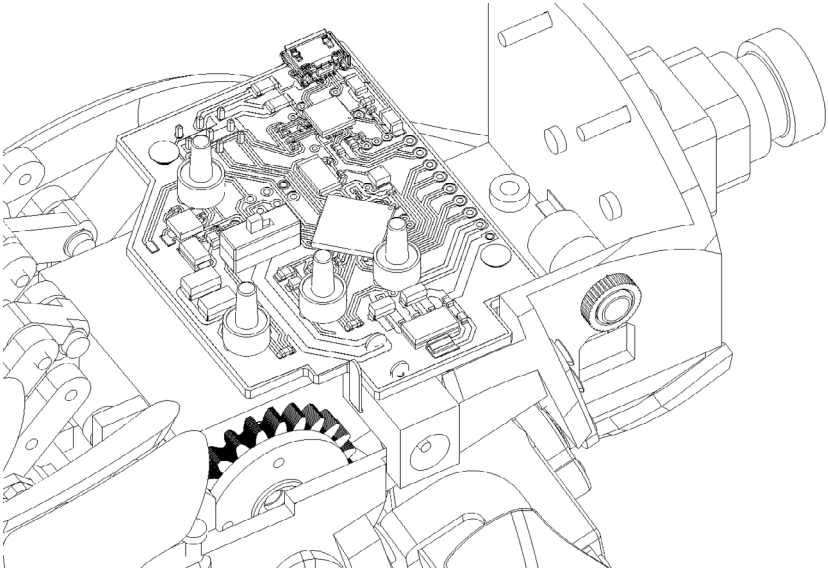
≫For Mk2 boards, start here.

[3 3] Secure the HACKberry hand board Mk2 using two screws (M2L6).

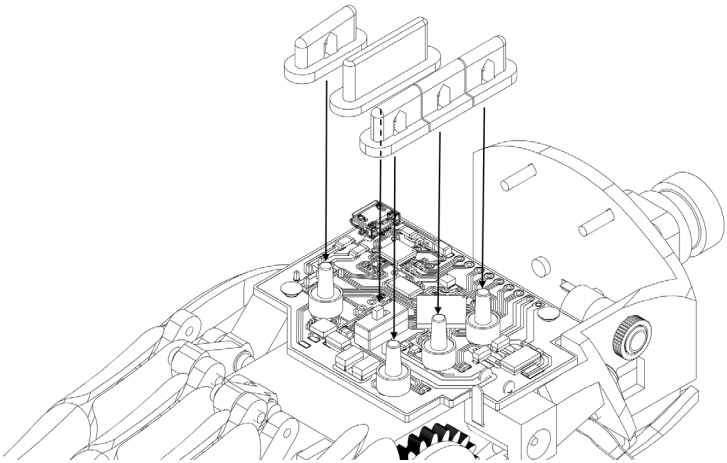


[3 4] Remove the nut from the stereo jack, then reattach it by clamping the R-H-01.

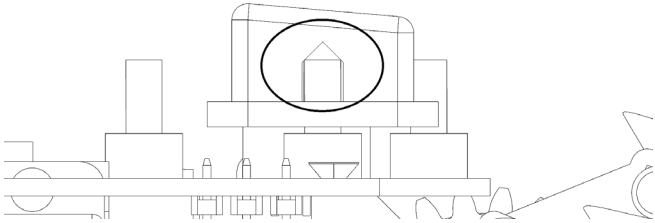




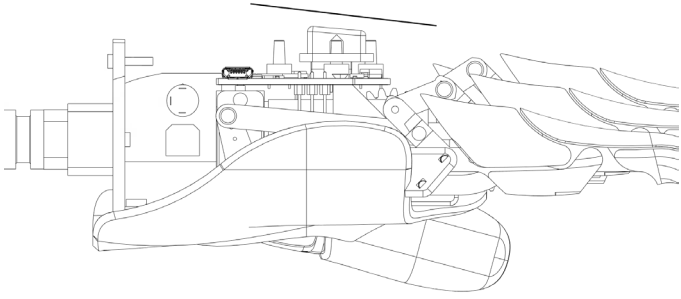
[3 5] Install R-H-10, R-H-11, R-H-012, R-H-13, and R-H-14 as shown in the diagram below.



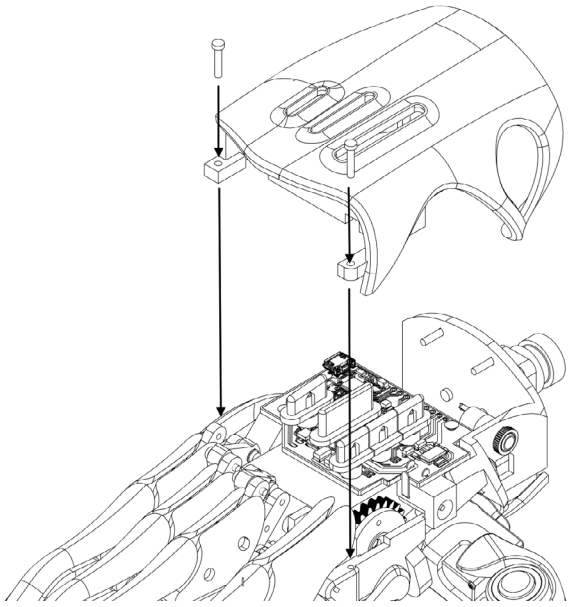
*Make sure to insert all the way in.

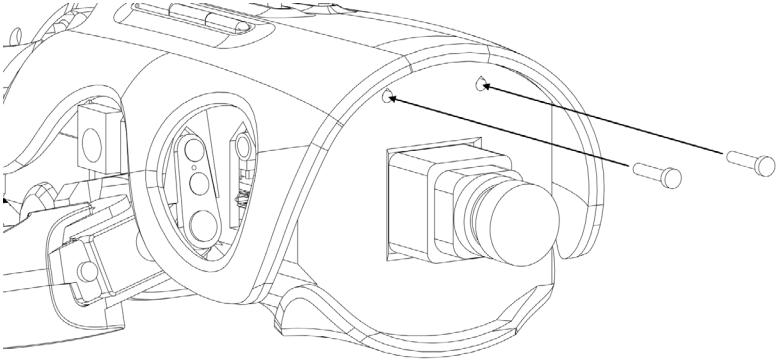


*For R-H-13, make sure the shorter side faces your fingertips.

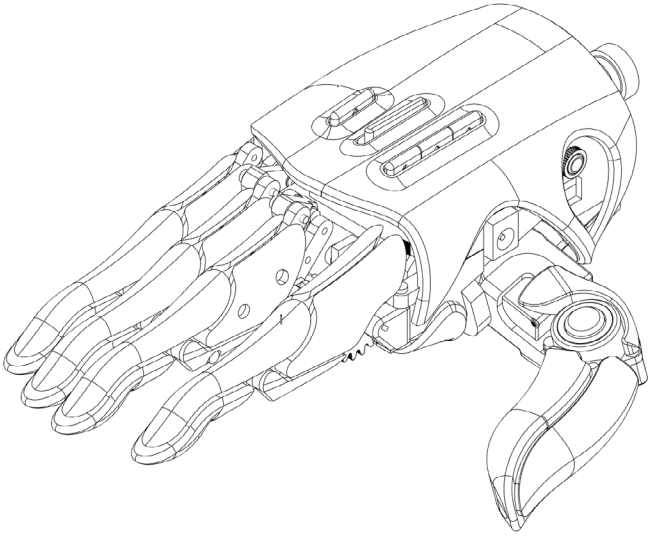


【3 6 】Fix R-H-15 using four screws (M2L10). It will be easier to install if you bend your fingers.





[4 7] This completes the assembly of the palm.



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Assembling
11Wrist assembly

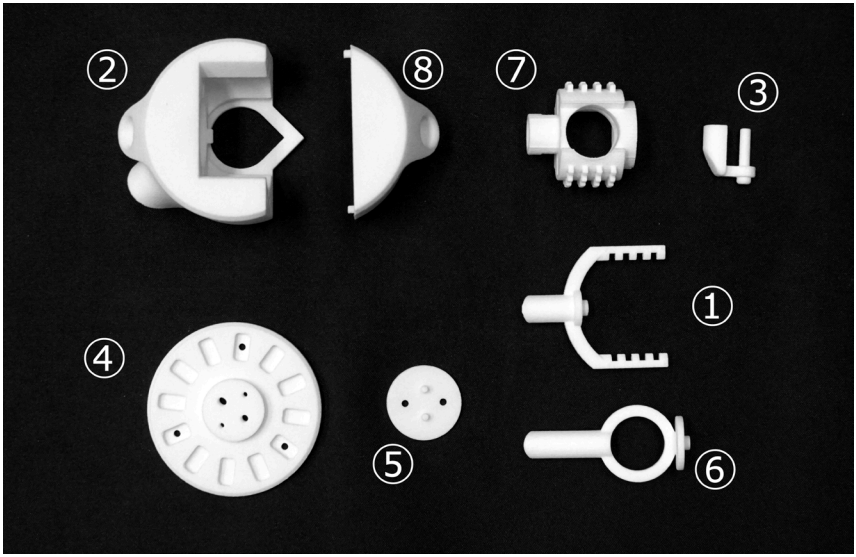
* In this section, we will assemble the wrist unit. Please note that the parts connected to the wrist unit are different depending on whether there is a dummy socket or not.

Without dummy socket

With dummy socket

-Parts used-

1. R-W-01
2. R-W-02
3. R-W-03
4. R-W-04
5. R-W-05
6. R-W-06
7. R-W-07
8. R-W-08



-General-purpose parts used-

*The number in parentheses is the quantity without dummy sockets.

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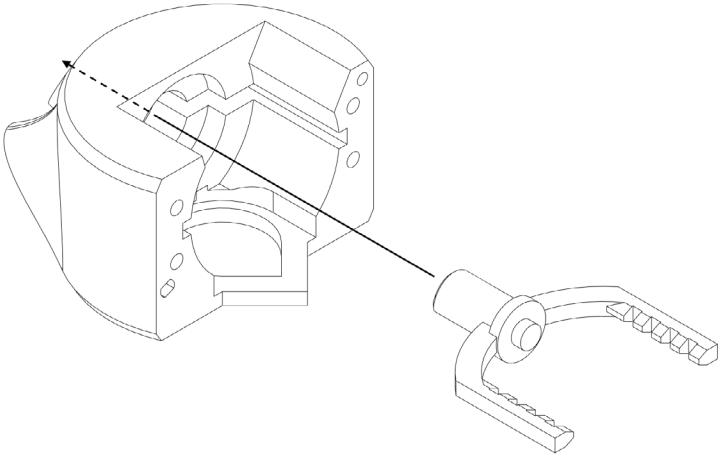
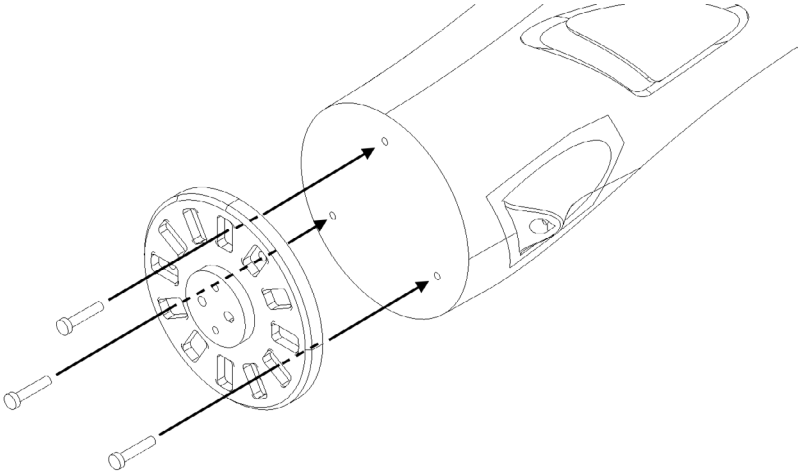
[1] Insert R-W-01 into R-W-02.The button surface of the R-W-01 is slanted, with the longer side facing downwards.

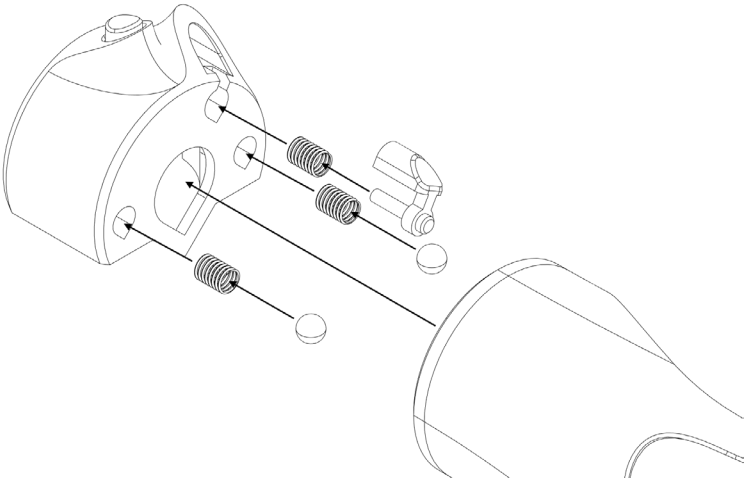


【2】(If there is no dummy socket, this step is not necessary) Use three screws (M2L10) to attach the dummy socket to R-W-04.

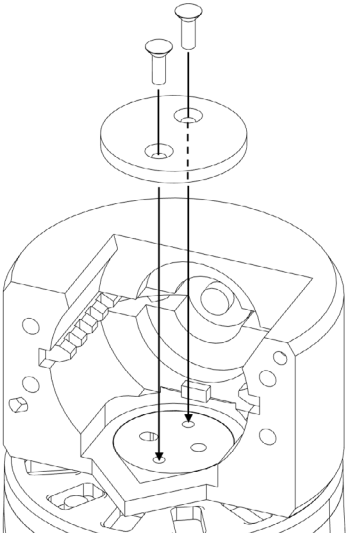
Install the socket.



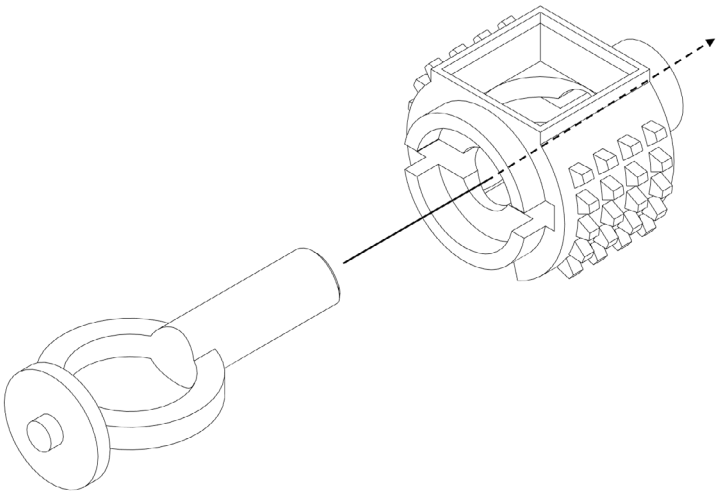
[3] R-W-02 and R-W-04 sandwich three coil springs, two steel balls, and R-W-03.



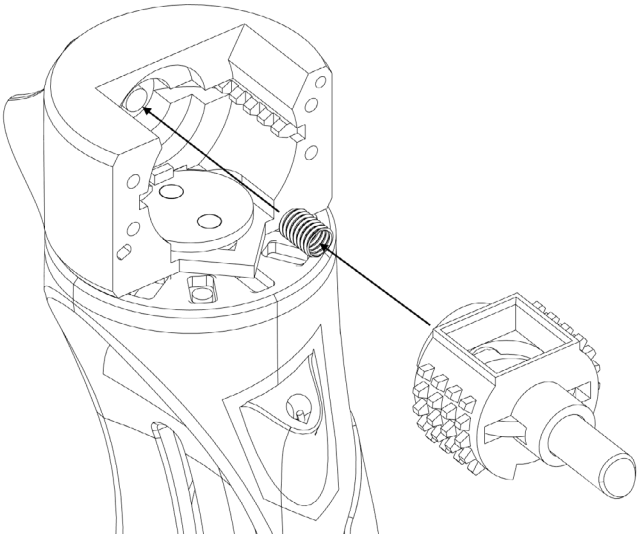
[4] Secure R-W-04 and R-W-05 with two screws (M2L6).



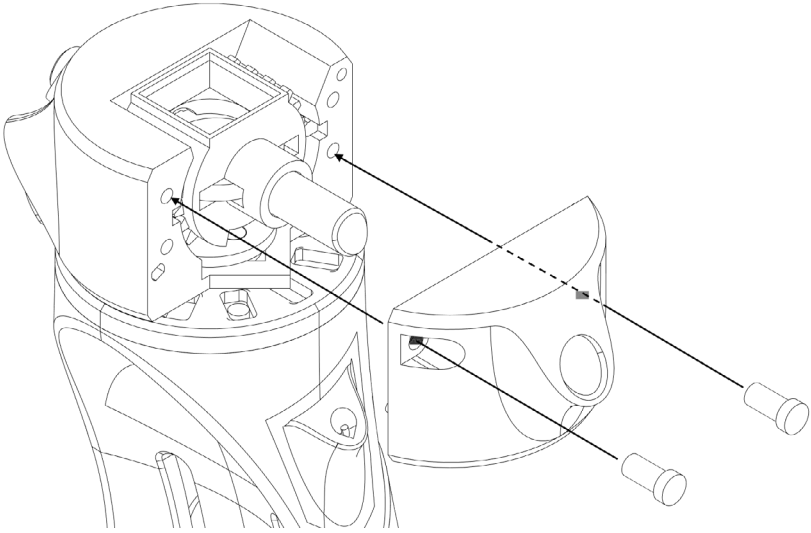
[5] Insert R-W-06 into R-W-07.The button surface of the R-W-06 is slanted, with the longer side facing downwards.

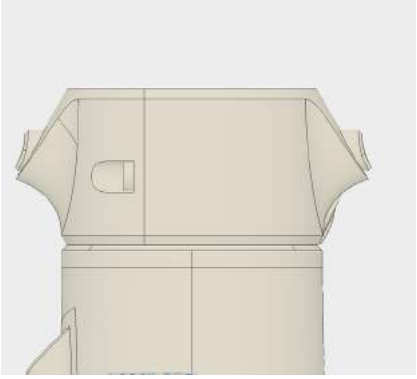


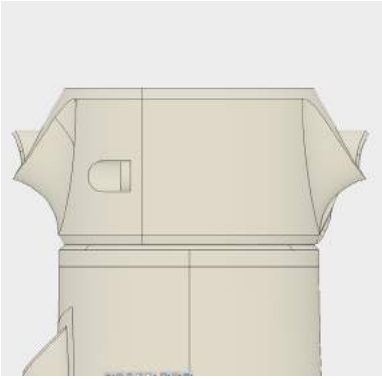
[6] Insert the coil spring and the parts you just assembled as shown in the figure.



[7] Attach R-W-08 and secure it with two screws (M3L8).



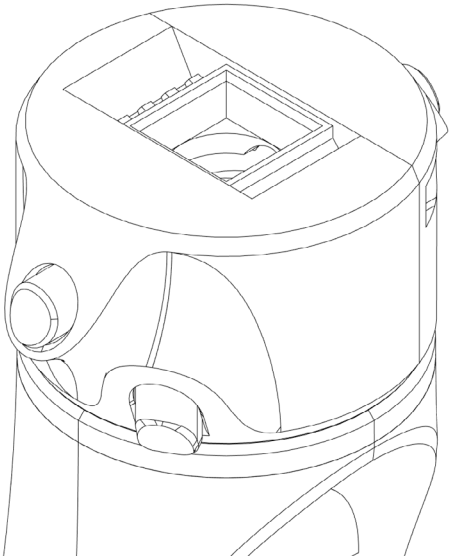




*Good example

*Bad example

[8] This completes the assembly of the wrist unit.



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Assembling

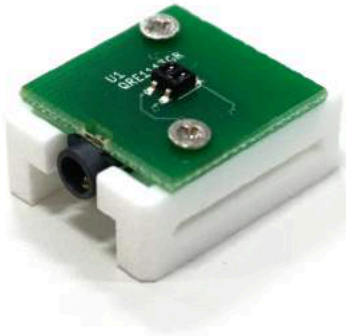
13 Assembling the Sensor

-Parts used-

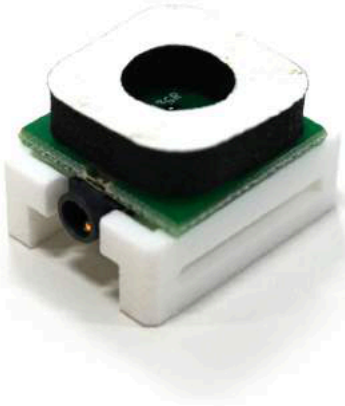
1. C-SE-01	2. C-SE-02	3. Sensor board
4. Velcro straps	5. Stereo cable	6.Sensor Cushion

-General-purpose parts used-

(1) Assemble the sensor board into C-SE-01 in the orientation shown in the photo and secure it with two screws (M2L6).



(2) Peel off the release paper from the sensor cushion and attach it to the sensor board as shown in the photo.



(3) Peel off the release paper on the other side and attach C-SE-02 to the sensor cushion as shown in the photo.

(2) Pass the Velcro strap through C-SE-01 as shown in the photo. This completes the sensor assembly.

Assembling
14 How to set the initial position

*In this section, we will set the initial position of the motor. The angle of the fingers may be slightly off due to individual differences in the motor or misalignment during installation, so we will adjust it by changing the value in the program.

If you are right-handed, please start from this page; if you are left-handed, please start from page 139.

For the right hand

(1) Insert the DC plug of the battery case or dummy socket into the DC jack on the palm unit.

(2) Press the button in the middle of the back of your hand to turn on the HACKberry. Immediately after turning it on, your thumb will be spread out to the side of your hand, and your other fingers will be spread apart, as shown in the image.However, due to individual differences in the motor or misalignment during installation, there may be cases where the finger is not fully extended or, conversely, the finger is overextended, causing the buzzing sound to continue. In this case, immediately turn off the power.

*Compared to the Mk1 board, the time it takes for the servo motor to start moving after power is turned on is slightly longer for the Mk2 board.

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As an example, in this section we will consider the case where the index finger is bent too much and the thumb is spread too far apart, as shown in the figure below.

[3] "5. Uploading the program"(Page 5 2)Open the program used in Step 1. Adjust the finger angle by changing the six numbers shown in the figure.

```
Hackberryv3.1
/*
 *  Arduino micro code for HACKberry.
 *  Origially created by exiii Inc.
 *  edited by Genta Kondo on 2017/6/11
 */
#include <Servo.h>

//Settings
const boolean isRight = 0;//right:1, left:0

//For right hand, find optimal values of ThumbMin, IndexMax and OtherMax first.
//For left hand, find optimal values of ThumbMax, IndexMin and OtherMin first.
//Then, calculate the remaining values by following rules.
//Difference of ThumbMin and ThumbMax is 86
//Difference of IndexMin and IndexMax is 117
//Difference of OtherMin and OtherMax is 55

const int outThumbMax = 150;//right:open, left:close
const int outIndexMax = 150;//right:open, left:close
const int outOtherMax = 150;//right:open, left:close

const int outThumbMin = 30;//right:close, left:open
const int outIndexMin = 30;//right:close, left:open
const int outOtherMin = 30;//right:close, left:open

const int speedMax = 6;
const int speedMin = 0;
const int speedReverse = -3;
const int thSpeedReverse = 15;//0-100
```

[4] The value marked with Max indicates the angle at which the fingers are open, and the value marked with Min indicates the angle at which the fingers are closed. In this case, the index finger was bent when it should have been open, so gradually increase the Max value to open it more.

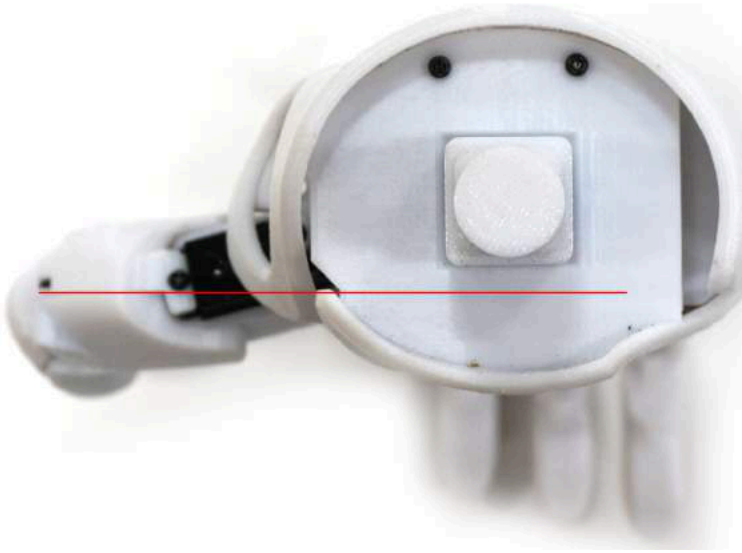
In this case,
c o n s t i n t o u t I n d e x M a x = 1 5 0 ;
What was
c o n s t i n t o u t I n d e x M a x = 1 6 1 ;
When I did this and uploaded the program, the fingers were at the correct angle.

```
17
18 const int outThumbMax = 150;//right:open, left:close
19 const int outIndexMax = 161;//right:open, left:close
20 const int outOtherMax = 150;//right:open, left:close
21
22 const int outThumbMin = 30;//right:close, left:open
23 const int outIndexMin = 30;//right:close, left:open
24 const int outOtherMin = 30;//right:close, left:open
25
```

[5] Next, adjust the thumb. The thumb is too open, so gradually decrease the Max value.

c o n s t i n t o u t T h u m b M a x = 1 5 0 ;
What was
c o n s t i n t o u t T h u m b M a x = 1 3 6 ;
When I did this and uploaded the program, the fingers were at the correct angle.

```
17
18 const int outThumbMax = 136;//right:open, left:close
19 const int outIndexMax = 161;//right:open, left:close
20 const int outOtherMax = 150;//right:open, left:close
21
22 const int outThumbMin = 30;//right:close, left:open
23 const int outIndexMin = 30;//right:close, left:open
24 const int outOtherMin = 30;//right:close, left:open
25
```



[6] Next, set the value for when the fingers are closed. For the angle difference between the closed and open states, refer to the following values for each motor.

- t h u m b : 8 6
- i n d e x : 1 1 7
- o t h e r : 5 5

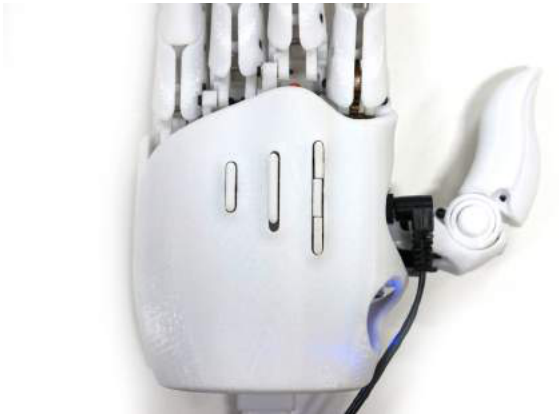
For example, since we previously set the angle of the thumb open to 136, enter 136-86. Set the other values in the same way and upload the program.

```
21
22 const int outThumbMin = 136-86;//right:close, left:open
23 const int outIndexMin = 161-117;//right:close, left:open
24 const int outOtherMin = 150-55;//right:close, left:open
25
```

For the left hand

(1) Insert the DC plug of the battery case or socket containing the battery into the DC jack on the palm unit.

(2) Press the button in the middle of the back of your hand to turn on the HACKberry. Immediately after turning it on, your thumb will be spread out to the side of your hand, and your other fingers will be spread apart, as shown in the image.However, due to individual differences in the motor or misalignment during installation, there may be cases where the finger is not fully extended or, conversely, the finger is overextended, causing the buzzing sound to continue. In this case, turn off the power immediately.



As an example, in this section we will consider the case where the index finger is bent too much and the thumb is spread too far apart, as shown in the figure below.

[3] "5. Uploading the program"(Page 5 2)Open the program used in Step 1. Adjust the finger angle by changing the six numbers shown in the figure.

[4] The value marked with Min indicates the angle at which the fingers are open, and the value marked with Max indicates the angle at which the fingers are closed. In this example, the index finger was bent when it should have been open, so gradually decrease the Min value to open it more.

In this case,
c o n s t i n t o u t I n d e x M i n = 3 0 ;
What was
c o n s t i n t o u t I n d e x M i n = 1 2 ;
When I uploaded the program, the fingers were now at the correct angle.

```
17
18 const int outThumbMax = 150;//right:open, left:close
19 const int outIndexMax = 150;//right:open, left:close
20 const int outOtherMax = 150;//right:open, left:close
21
22 const int outThumbMin = 30;//right:close, left:open
23 const int outIndexMin = 12;//right:close, left:open
24 const int outOtherMin = 30;//right:close, left:open
25
```

[5] Next, adjust the thumb. The thumb is too open, so gradually increase the Min value.

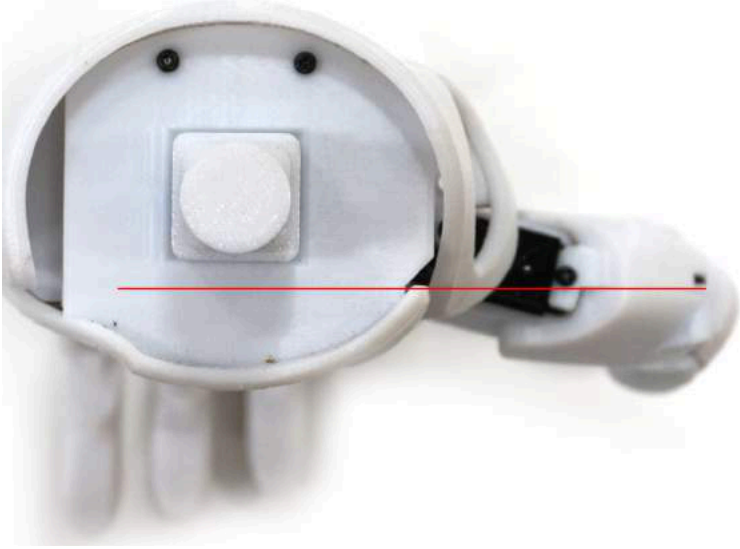
```
const int outThumbMin = 30;
```

What was

```
const int outThumbMin = 55;
```

When I did this and uploaded the program, the fingers were at the correct angle.

```
17
18 const int outThumbMax = 150;//right:open, left:close
19 const int outIndexMax = 150;//right:open, left:close
20 const int outOtherMax = 150;//right:open, left:close
21
22 const int outThumbMin = 55;//right:close, left:open
23 const int outIndexMin = 12;//right:close, left:open
24 const int outOtherMin = 30;//right:close, left:open
25
```



[6] Next, set the value for when the fingers are closed. For the angle difference between the closed and open states, refer to the following values for each motor.

- t h u m b : 8 6
- i n d e x : 1 1 7
- o t h e r : 5 5

For example, since we previously set the angle of the thumb open to 55, enter 55 + 86. Set the other values in the same way and upload the program.

```
17
18 const int outThumbMax = 55+86;//right:open, left:close
19 const int outIndexMax = 12+117;//right:open, left:close
20 const int outOtherMax = 30+55;//right:open, left:close
21
```

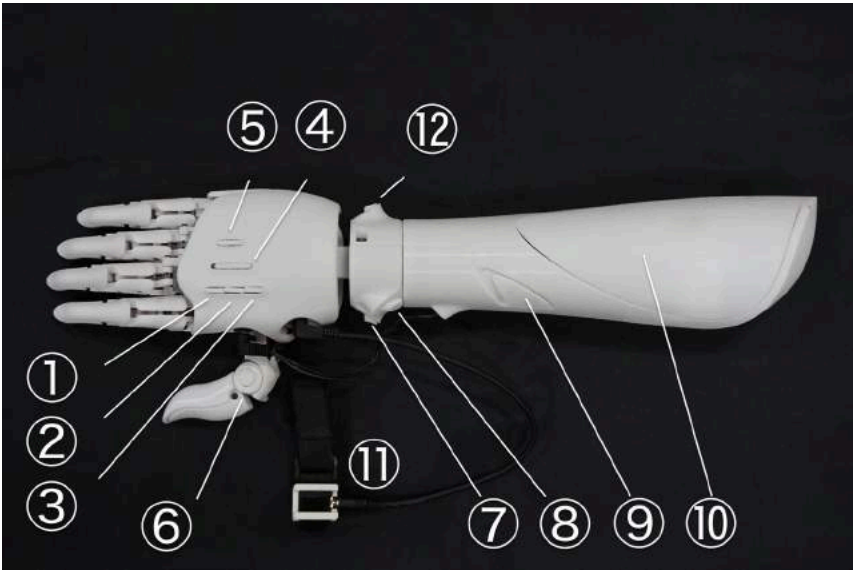
3. Let's try it out

01 Names of Each Department ‥

02 Battery installation ‥‥‥

06 Wrist abduction ··P151

How to Use HACKberry
01Department Names



1. Calibration button
2. Extension button
3. Thumb Rotation Button
4. Power switch
5. Three-finger fixed button
6. Thumb lock button
7. Wrist flexion/ulnar flexion/palmar/dorsiflexion button
8. Wrist rotation button
9. Battery door lever

10. Battery door

11. Simple pressure sensor

12. Wrist detachable button

How to Use HACKberry
02 Battery installation

Slide the battery door lever to open the battery door, then insert the battery by sliding it along the grooves until you hear a click.
(To remove it, press C-B-02 and slide it in the opposite direction.)

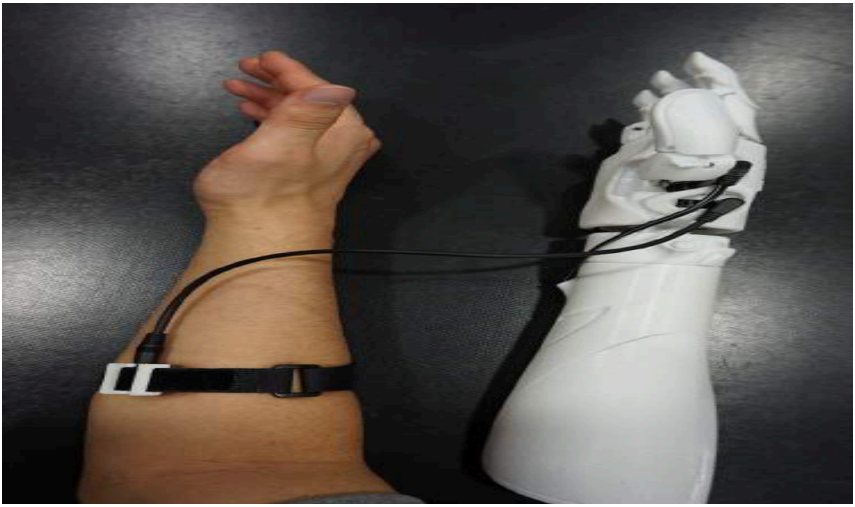
How to Use HACKberry
03 calibration

Insert the DC plug into the DC jack as shown.

Place the flat part of the sensor on the skin (preferably on a muscled area) and tighten it lightly with the Velcro strap.

Then connect the HACKberry and the sensor with an audio cable.Firmly, all the way to the back

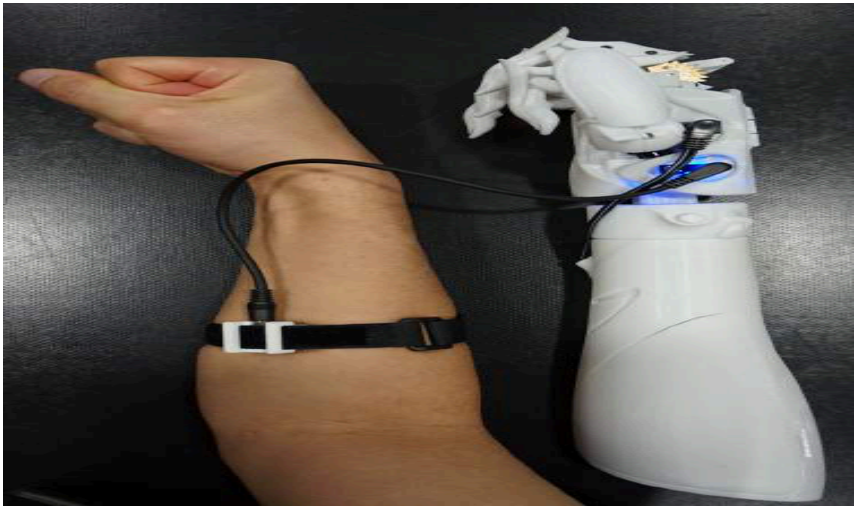
Be careful to insert it properly.



Next, perform calibration.

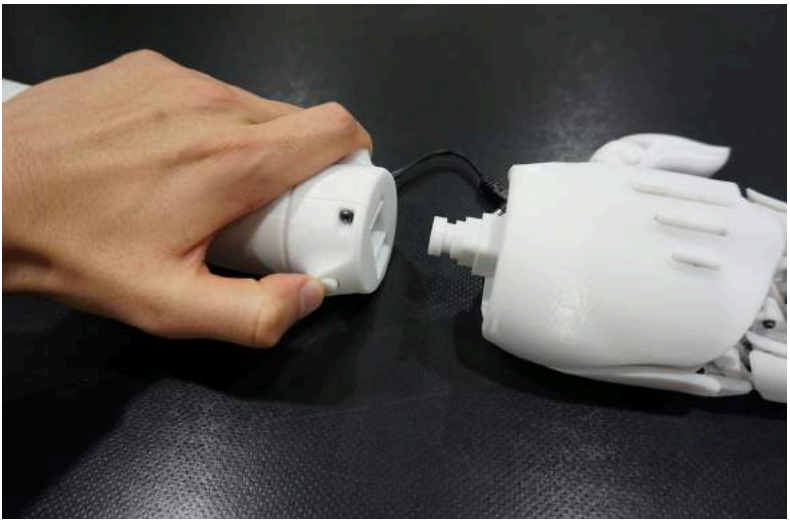
Calibration teaches HACKberry the amount of muscle bulge that varies from user to user.

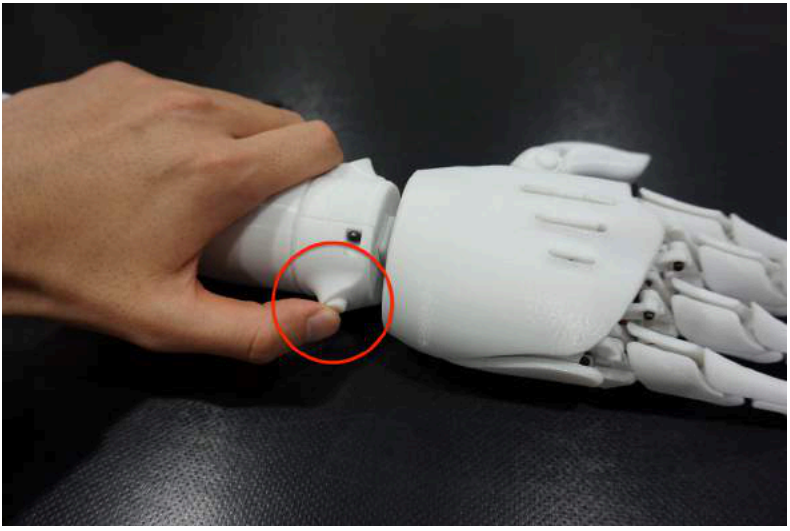
With the power on, press the calibration button and repeat the action of "clenching your fist" and then "relaxing" several times for 4 seconds.



Once calibration is complete, your index finger, index finger, and pinky finger will move in unison.

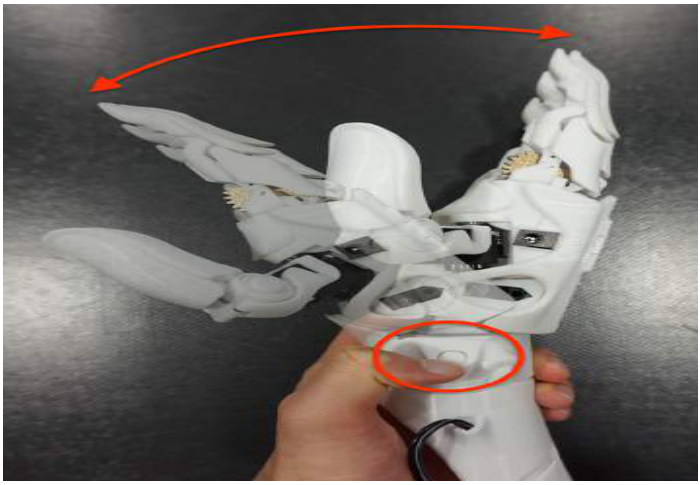
Press the hand detachment button to detach the hand and wrist.
The hand and wrist position can be adjusted in 90 degree increments.





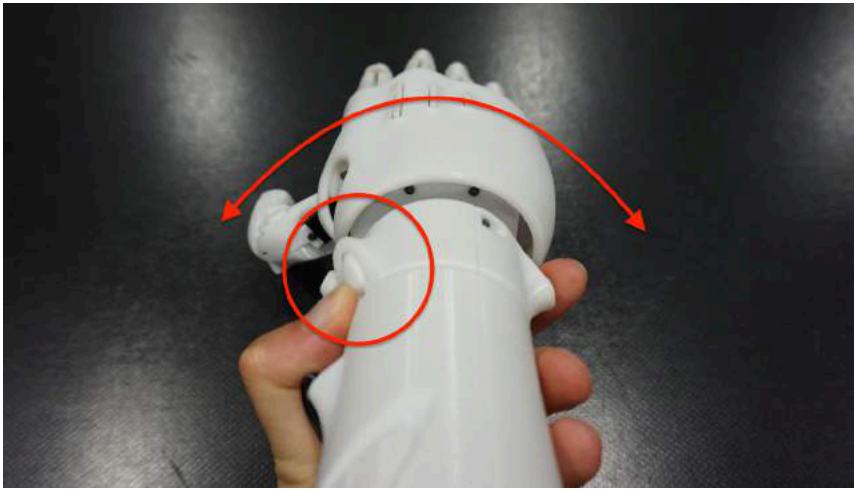
How to Use HACKberry
05 Palmar and dorsiflexion and radial and ulnar flexion of the wrist

Pressing the wrist flexion/palmar/dorsiflexion button will tilt the wrist in palmar/dorsiflexion or radial/ulnar flexion.



How to Use HACKberry
06 Wrist abduction

Pressing the wrist abduction/abduction button will rotate the hand in the direction shown.



4.HACKberryEnjoy more

01 Introduction ··P154

02 HACKberry Handboard Mk2 Specifications ·····P155

03 Controlling HACKberry using an electromyography sensor ·····P158

04 Acetone surface treatment ·····P162

05 Nylon dyeing·····P168

Customize
01 Introduction

This chapter introduces examples of customization of HACKbery.

In Section 1, customizability has been improved compared to the M k 1 board. This section explains the features and specifications of the H A C K ber ry hand boa r d M k 2 board.

In Section 2, we will introduce how to actually operate HACKberry as a myoelectric prosthetic hand using the Mk2 board and a commercially available electromyography sensor (2ch).

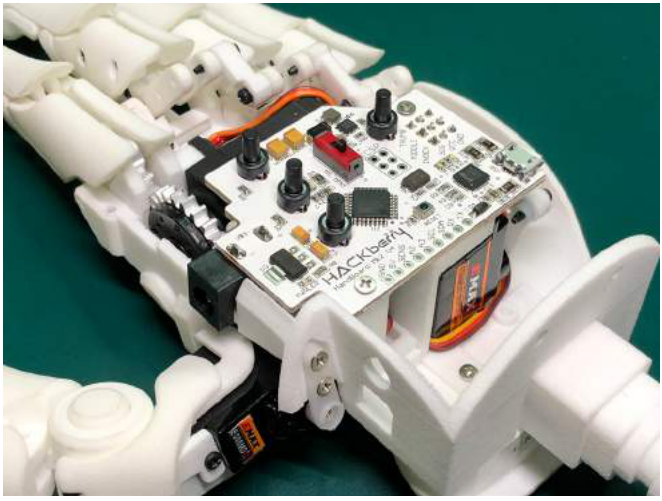
In Section 3, we introduce an acetone surface treatment method to improve the aesthetic quality of home 3D printed parts.

In Section 4, we will explain how to print using an external 3D printing service. Learn how to dye 3D printed nylon parts.

Customize
02 HACKberry hand board Mk2 specification

The HACK berry hand boa rd Mk 2 was developed primarily to improve customizability from the Mk 1. The following two points are the main improvements.

1. Space saved by integrating the board
2. Simplified customization with expansion ports

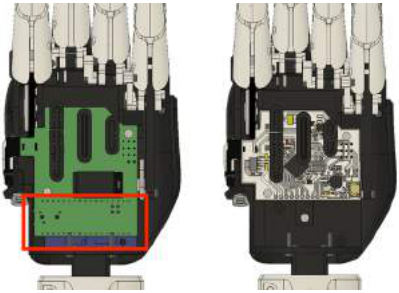


[1] Space saved by integrating the board

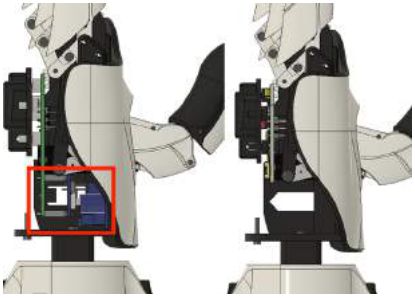
The functions of the Arduino Micro and DC/DC converter, which were separate in the Mk1, are now combined into one board.

This reduces the space required for the electrical circuitry in the housing, and in turn reduces the need for additional sensors.

There is now more space to install sensors and modules.



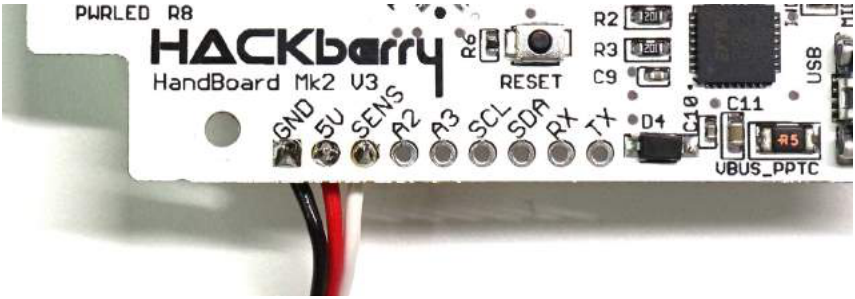
*Space comparison top view (left: Mk1, right: Mk2)



*Side view of space comparison (left: Mk1, right: Mk2)

[2] Simplified customization by adding expansion ports

Customizable input and communication ports at the bottom of the board(Analog input, I2C communication, serial communication)This allows for easy connection to other sensors and modules.



*Input and communication ports for customization at the bottom of the board

The detailed specifications are shown below. The MK2 board is an A r d u i n o compatible machine and uses the A T m e g a 3 2 8 p - A U processor, so the pin assignment is the same as that of the A T m e g a 3 2 8 P.

※ HACKberry hand board Mk2 detailed specifications

ピンアサイン	入力ピン	キャリブレーションボタン	A6
		拇指内侧触发器	A0
		三指駆動ボタン	D10
		エクストラボタン	A7
		センサー入力	A1
	出力ピン	人差し指用サーボモータ	D5
		三指用サーボモータ	D6
		親指用サーボモータ	D9
	拡張ピン	5V	5V
		GND	GND
		アナログ入力ピン	SENS(A1)
			A2
			A3
		I2C通信ピン	SCL
			SDA
		シリアル通信ピン	RX
			TX
書き込み時の設定	ボード	Arduino Nano	
	プロセッサ	Atmega328P-AU	
	書き込み装置	AVRISP mk II	
入力電圧	推奨入力電圧	7V~12V	
	最大入力電圧	20V	
	DCDCコンバータ定格入力電圧12V, 最大24V		
	三端子レギュレーター最大入力電圧20V より		
サーボピン	PPTCはthumbには入っていません		
	IndexとMiddleは500mAが限度です。		

Customize
03 Using electromyography sensorsHACKberryOperation

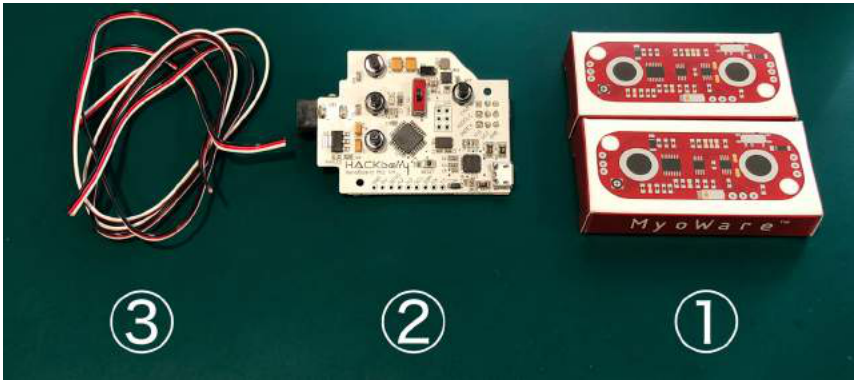
[1] Introduction

While many commercially available prosthetic hands use electromyography sensors, HACKberry uses a simple pressure sensor. Simple pressure sensors have the advantages of being low noise, easy to operate, and extremely low cost. However, they have the disadvantage of being difficult to make multi-channel and only being able to measure simple signals from one channel. If you would like to operate HACKberry using an electromyography sensor like commercially available prosthetic hands, please refer to this section for customization.

[2] What is myoelectricity?

When a person moves a muscle, an electrical signal is sent from the brain and transmitted to the muscle via the nerves. This electrical signal leaks slightly onto the surface of the skin. The signal observed by electrodes attached to the surface of the skin is called myoelectricity.。

[3] Parts used



1. MyoWare (2 items)
This electromyography sensor is manufactured by Advancer Technologies. It has built-in circuits for amplifying and smoothing myoelectric potential, making it relatively easy to measure electromyography signals.

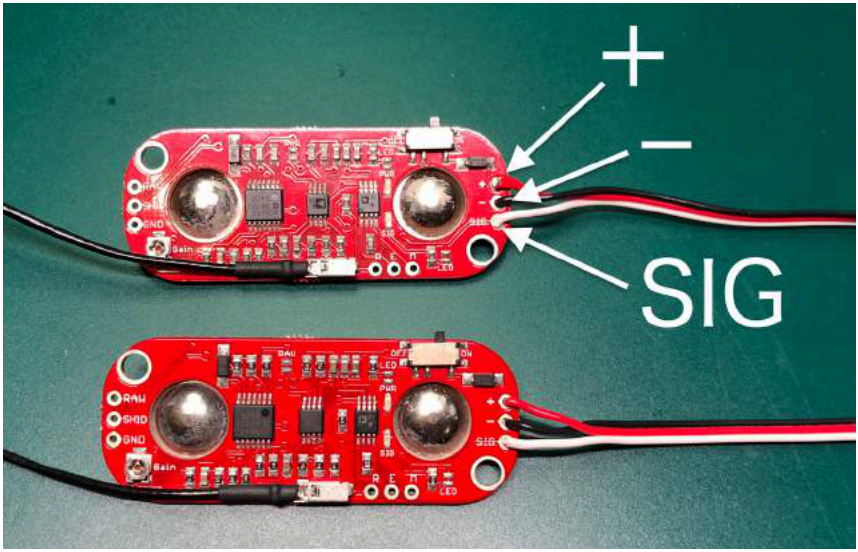
2. HACKberry hand board Mk2
The audio jack is not connected.

3. 2 x 50cm triple-row flat cable

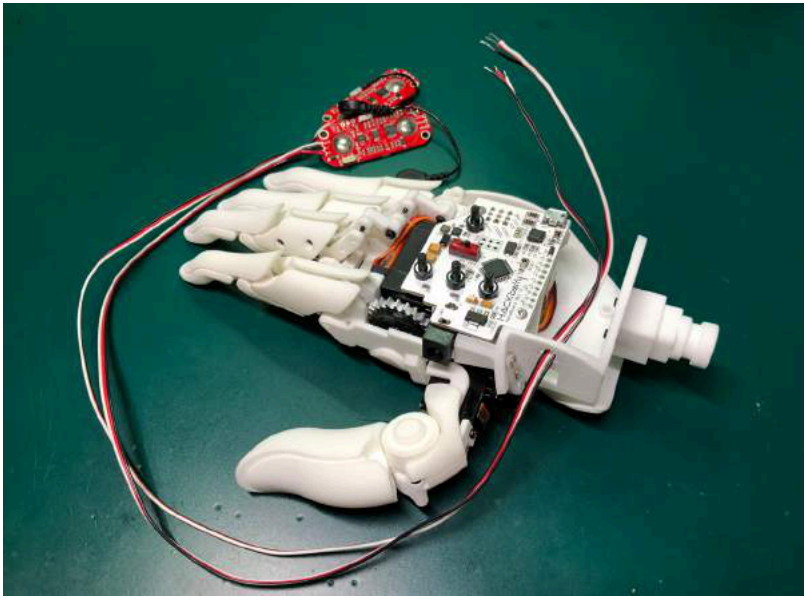
In this section, two My O Ware sensors are used to detect the palmar flexion (bending the wrist toward the palm) and dorsiflexion (bending the wrist toward the back of the hand) movements of the arm, respectively, to realize two-channel operation of the H A C K be r r y.

[4] Implementation method

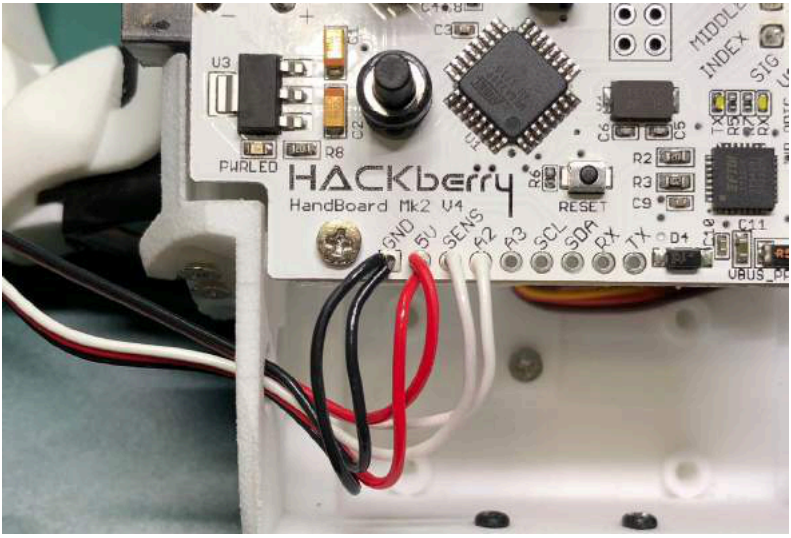
Solder the red, black, and white wires to the +, -, and SIG points on the MyoWare.



Attach the HACKberry hand board Mk2 to the HACKberry and pass the flat cable through the audio jack hole as shown in the figure.



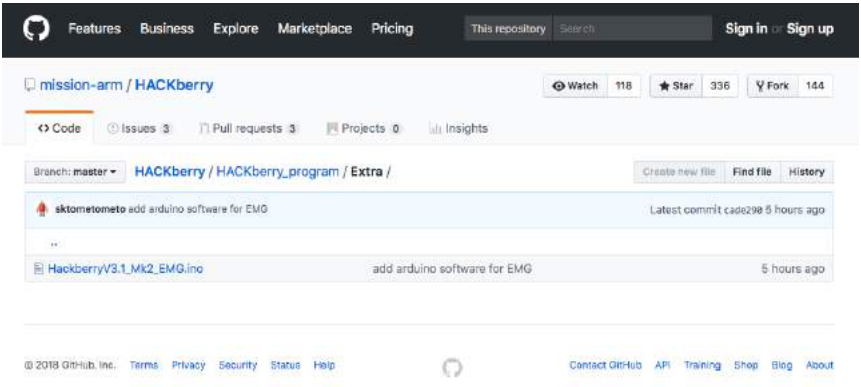
Solder the flat cable to the expansion port of the HACKberry hand board Mk2. As shown in the figure, connect the two black wires to GND, the two red wires to 5V, one white wire to SENS, and the other white wire to A2.



This completes the hardware work.

Next, download the program for EMG control from Github.

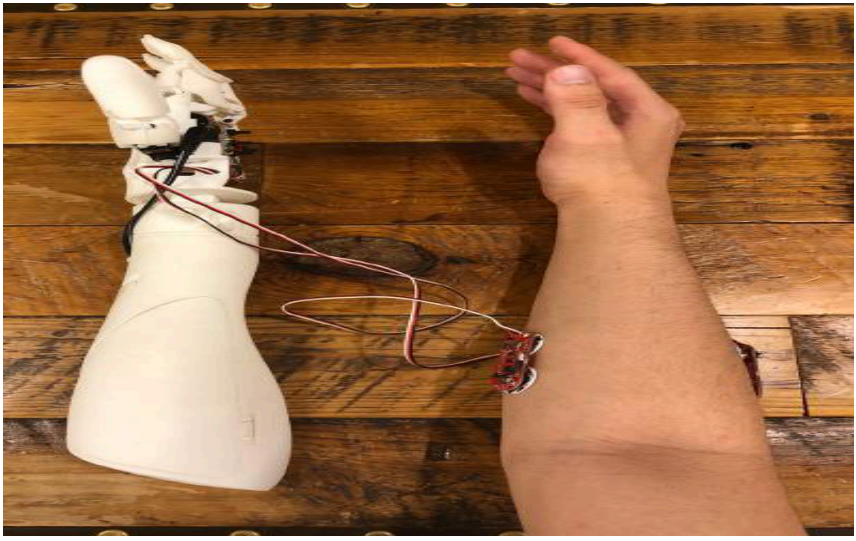
[02 How to obtain parts](Page 19)Please refer to the link below to access the HACKberry Github page and download "HackberryV3.1_Mk2_EMG_180412" under Program.



Then, click [0 5 Upload Program](Page 52)Using this as a reference, write the program you just downloaded to the HACKberry hand board Mk2.

【5】Operating Method

Attach the MyoWare with the white cord connected to the SENS external port on the HACKberry hand board Mk2 to the flexor muscle (the muscle that rises during palmar flexion), and the other MyoWare to the extensor muscle (the muscle that rises during dorsiflexion). For details on sensor attachment positions, please refer to the MyoWare instruction manual.



Turn on the HACKberry and press the calibration button. Calibration will take place for four seconds after pressing the button, during which time you apply force alternately to the flexor and extensor muscles to preset the width of the electromyographic signal to the HACKberry. After calibration is complete, applying force to the flexor muscles will flex your fingers, and applying force to the extensor muscles will extend your fingers.

YouTube electromyography video

Customize
03 Surface treatment with vaporized acetone

[1] Introduction

FDM (Fused Deposition Modeling), commonly used in home 3D printers, allows for inexpensive printing of three-dimensional shapes, but due to the nature of the process of building up layers one by one, the roughness of the surface becomes noticeable. In such cases, the surface can be smoothed by surface treatment with vaporized acetone. The advantages and disadvantages of this method are listed below.

merit
- It takes less time (about 1/10 of the time required compared to using sandpaper to smooth the surface). - It can also handle fine details that are difficult to handle with sandpaper.

-It is effective even when the layer pitch is large during 3D printing, so printing time can be significantly reduced. -The surface is melted and leveled, creating a kind of molten state, which reduces the weak bonding strength between layers and improves overall strength.

Disadvantages
-Since organic solvents are used, care must be taken when using it.

-Due to the nature of melting and smoothing the entire surface, there is a possibility that details may be crushed, making it unsuitable for precision fitting or press-fitting.

Countermeasures for these disadvantages include separating the exposed parts of the casing from the other structural and mechanical parts and performing surface treatment only on these, or placing the parts to be surface treated in a position where they are more likely to be exposed to acetone (see below).

[2] Points to note

Acetone is an organic solvent. It is designated as a Class 4 hazardous material (Petroleum Class 1, Hazard Level 2, Water-soluble) under the Fire Service Act. The

characteristics of acetone that require particular attention are:Highly flammable (ignites at room temperature)"and"High volatilityPlease note the following points:

- To prevent ignition and excessive inhalationHandle in a well-ventilated area.、Don't let steam build uplet's.

-If any spills occur during use, wipe them up quickly with a cloth.

-If acetone vapor gets into your eyes, rinse them quickly with plenty of water. Even if you don't feel any pain or discomfort, we recommend seeking medical attention immediately.

-If you accidentally swallow or inhale acetone, we recommend that you seek medical advice.Wash your hands and garglePlease make sure to follow these rules thoroughly.

[3] Equipment

●An airtight container (made of a material that does not dissolve in acetone. We used a 5L polyethylene container.) ●Aluminum foil
●Kitchen paper
Acetone (Hotoku Pure Acetone 1L)
Pipette

[4] Procedure

[1] Place aluminum foil on the bottom of an airtight container. Cut a piece of aluminum foil large enough to hang over the outside of the container. This is so that when you remove the parts from the container later, you can hold the edge of the aluminum foil without touching the parts.

[2] Place the parts on the aluminum foil.Place the surface to be treated facing upThis is because the vaporized acetone falls onto the parts from above, making the top parts melt more easily.

[3] Fold three sheets of kitchen paper in half and place them as shown in the photo. Use a pipette to soak them in 15 ml of acetone.

[4] Immediately after that, close the lid and start the timer. If the print is at 0.3 mm pitch, it will take about an hour.

Please use the degree as a guide.

(5) After the specified time has passed, open the lid and remove the parts. At this point, the parts will be soft.Remove the aluminum foil by grasping its edges without touching the components.

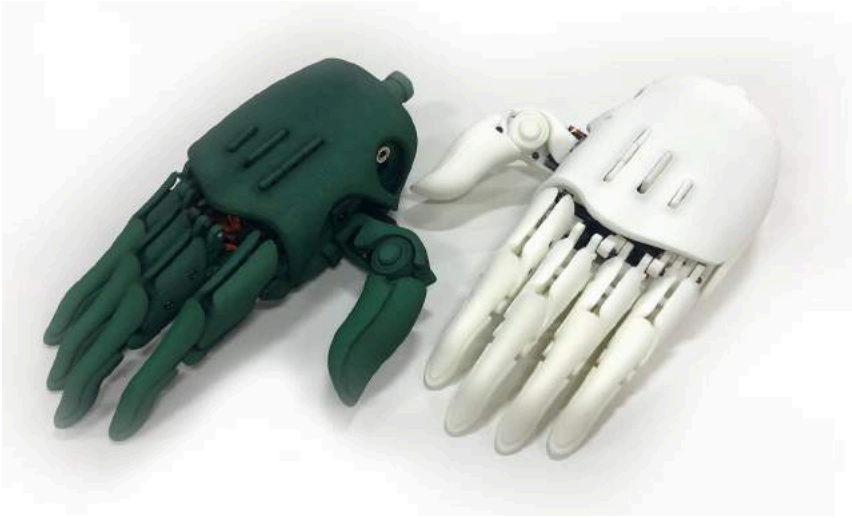
[6] You can see that the surface has melted and become smooth.Leave in a well-ventilated place for 4 to 5 hours.

[7] If the purpose is surface treatment, the surface will be fairly smooth at this point, but if you are concerned about the shine, you can spray it with a matte spray to restore the original texture.

※Image before matte finish

※Image after matting

Customize
04 Nylon dyeing



[1] Introduction

In this section, we will demonstrate one method of coloring HACKberry: dyeing 3D printed nylon parts using commercially available resin dyes.

3D printed parts made of nylon are printed white, making them easy to dye. (Not only that, nylon is low-cost and strong, which is why we use it as our standard material at HACKberry.) Please note that nylon is not compatible with home 3D printers that use the FDM (fused deposition modeling) method. (Home 3D printers generally use ABS or PLA resins, which are not suitable for dyeing.) Nylon is compatible with 3D printers that use the Selective Laser Sintering (SLS) method, which involves layering powder and burning it together using a laser or other device. Because SLS 3D printers are not widely used for home use, we recommend using an external 3D printing service (using an external service allows you to choose from a variety of materials in addition to nylon). We use the 3D printing service provided by DMM.com.

DMM.com 3D printing service:https://make.dmm.com/print/

Some 3D printing services that use nylon also offer dyeing services. In this case, you can dye the material yourself using commercially available resin dyes to adjust the color to your liking.

[2] Equipment

1. Diluted dye for resin
2. Stainless steel colander
3. Kitchen paper
4. Rubber gloves
5. Measuring cup
6. Funnel
7. Chopsticks
8. Thermometer
9. IH heater (This is for heating the dye solution, so a gas stove will also work.)
10. Stainless steel pot (large enough to fit a colander)

[3] Dyeing process

[1] First, make the dye solution. This time, we used Osaka Kasei Co., Ltd.'s "Resin Dye SDN." Using a measuring cup, dilute the dye 20 times, using the amount that will cover the parts you want to dye. Please note that the dilution concentration varies depending on the dye.

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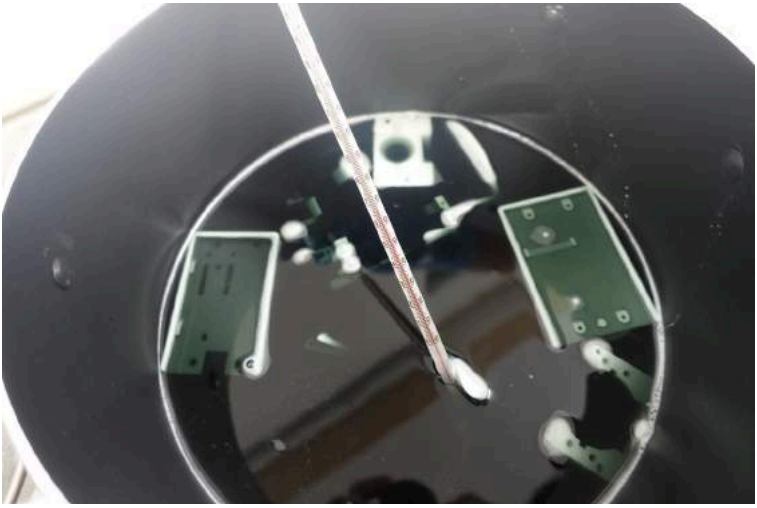
[2] Pour the dye solution into a stainless steel pot and heat it.

[3] The standard dyeing conditions for nylon with "Resin Dye SDN" are 70±10°C for 3 to 5 minutes.

Once the temperature has risen to within ±10°C, immerse the 3D printed object in the dye solution and start the timer.

The dye solution is kept at the specified temperature of 70±10°C during the dyeing process.



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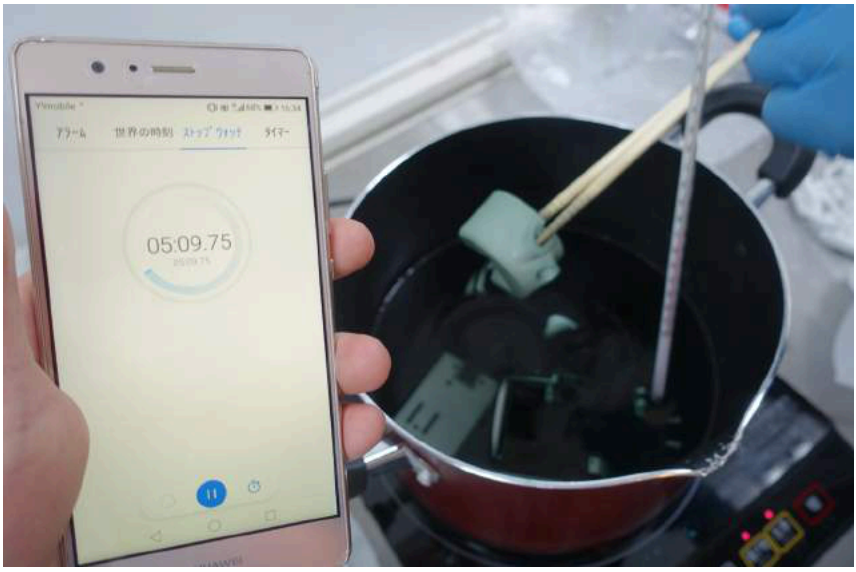
[4] The manufacturer recommends a staining time of 3 to 5 minutes, but in our environment, even if we removed the stain after 5 minutes,

It didn't seem to have dyed properly, so I dyed it for about 20 minutes.



* 5 minutes later

* 20 minutes later

[5] Once the desired color has been achieved, remove the 3D printed object from the dye solution and arrange it on a piece of kitchen paper.

Let it dry. This completes the dyeing of the nylon 3D printed object.

[6] The dye contains substances that are harmful to the human body.Just throw it down the sink

It is not possible.The dye solution can be reused, so store it in a plastic bottle using a funnel.

If you want to do this, dilute the dye solution by 200 times or more and discard it in the sink.

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5. If you think something is broken

Troubleshooting
If you think it might be broken

No power

◆Check that the battery is installed correctly.

◆ Make sure the DC plug is firmly inserted into the DC jack on the board.

◆ Check that the soldering position is correct.

➡Are the DC plug wires soldered correctly to the positive and negative terminals?

→Is VCC (or V+) and GND on the board shorted?

◆Use a tester to check that the voltage is displayed correctly on the battery terminals.

→If the circuit is shorted, the protection circuit within the battery may have been activated, preventing voltage from being generated. Please contact the battery manufacturer for information on how to restore this state.

Calibration is not possible

◆When you press the calibration button, make sure you feel a click.

→ The button may have been soldered at an angle, causing it to remain pressed down. Either re-solder it or repair it by scraping off the housing part.

*Please check that you have written the correct version of the program (left or right, mk1 or mk2). *Please check that the initial position settings of the servo motors are correct (see page 134 for details).
◆Please check that the servo motor pin socket is connected to the board in the correct orientation. (For details, See page 112.)
◆ Check that the sensor is connected correctly.

→Is the audio cable inserted all the way in?

→ If there is no reaction even when connected correctly, the sensor or the audio cable itself may be damaged. I will.

My fingers don't bend properly

◆ Check that the initial position settings of the servo motors are correct. (For details, see page 134.)
Please.)

The servo motor makes a strange noise

◆ Check if the initial position setting of the servo motor is correct. If you try to move in a direction that you cannot turn, an abnormal noise will occur. (For details, please refer to page 134.) ◆ Check that the voltage adjustment of the DC-DC converter is correct. If the voltage is not adjusted, the servo Excessive voltage will be applied to the motor and the servo motor will be damaged. (See page 92 for details.)

The servo motor suddenly stopped working

◆(For mk1) Check that the polyswitch is inserted correctly.

If the motor is not inserted, it may overheat and be damaged. Replace it, insert the PolySwitch and reassemble.

Even if the motor is inserted, the circuit inside the servo motor may short out due to poor soldering or insufficient insulation. If this is the case, replace the motor.

◆Please check that the voltage of the DC-DC converter is adjusted correctly. If the voltage is not adjusted, excessive voltage will be applied to the servo motor, causing it to be damaged. (For details, see page 92.)

The sensor does not respond when pressed

◆ Check that the initial position settings of the servo motors are correct. (For details, see page 134.)
Please.)
◆ Check that the sensor is connected correctly.

→Is the audio cable inserted firmly and all the way in?

→ If there is no reaction even when connected correctly, the sensor or the audio cable itself may be damaged. I will.

◆Try adjusting the position where the sensor is installed.

➡Please install it in a position where the ridges are most likely to protrude.

The program cannot be written

◆Please make sure you are using a USB cable with communication functions. (USB cables that are only for charging will not function.)
I can't trust it.)
* Check that the COM port is correctly recognized. (See page 54.) * Check that the correct board is selected. (See page 56.) * Check that the driver is correctly installed.

The driver is generally installed automatically, but this may not be possible depending on your environment. For driver installation, please refer to the following website:

My fingers are running wild

◆ Check that the sensor is connected correctly.

→Is the audio cable inserted all the way in?

→ If there is no reaction even when connected correctly, the sensor or the audio cable itself may be damaged. I will.

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